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Progress, Challenges, and Opportunities in Oxide Semiconductor Devices: A Key Building Block for Applications Ranging from Display Backplanes to 3D Integrated Semiconductor Chips

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As Si has faced physical limits on further scaling down, novel semiconducting materials such as 2D transition metal dichalcogenides and oxide semiconductors (OSs) have gained tremendous attention to continue the ever-demanding downscaling represented by Moore's law. Among them, OS is considered to be the most promising alternative material because it has intriguing features such as modest mobility, extremely low off-current, great uniformity, and low-temperature processibility with conventional complementary-metal–oxide–semiconductor-compatible methods. In practice, OS has successfully replaced hydrogenated amorphous Si in high-end liquid crystal display devices and has now become a standard backplane electronic for organic light-emitting diode displays despite the short time since their invention in 2004. For OS to be implemented in next-generation electronics such as back-end-of-line transistor applications in monolithic 3D integration beyond the display applications, however, there is still much room for further study, such as high mobility, immune short-channel effects, low electrical contact properties, etc. This study reviews the brief history of OS and recent progress in device applications from a material science and device physics point of view. Simultaneously, remaining challenges and opportunities in OS for use in next-generation electronics are discussed.

1. Introduction

1.1. Birth of Amorphous Oxide Semiconductors

Semiconducting crystalline oxide substances such as zinc oxide (ZnO), indium oxide (In_2O_3), and gallium oxide (Ga_2O_3) are representative wide-bandgap semiconductors that are promising for optoelectronics applications due to their inherent transparency and n-type dopability. The term "amorphous oxide semiconductor (AOS)" was historically defined by Hosono and co-workers in 2004 where a high-performance transistor with an AOS channel layer composed of indium (In), gallium (Ga), zinc (Zn), and oxygen (O) was demonstrated at room temperature, surprising the whole electronic materials community.[1] This ground-breaking finding began from a conceptual design of a carrier percolation pathway.[2] A carrier path in the electronic material is composed of intercalated atomic orbitals, which implies that the degree of orbital intercalation is one of the most important factors to dictate the carrier transport. For this reason, the carrier transport properties can be tailored through controlling the size, shape, and arrangement of atomic orbitals that contribute to formation of conduction band (CB)/valence band (VB) edges. Here, there is a significant disparity between AOS and conventional silicon (Si). Solid-state Si has interatomic covalent bonding, which causes both CB and VB edges to be formed by sp^3 hybridized orbitals of Si atom that have highly anisotropic, spatial directionality (**Figure 1**). Conversely, in oxide materials, electron charge transfer occurs from electropositive metals to electronegative oxygen where their electronic structure is stabilized by the Madelung potential.[3] Along with this innate ionic bonding, AOS fully benefits from an innovative design concept using heavy post-transition metal cations with an electronic configuration of $(n-1)d^{10}ns^0$ ($n \geq 5$).[4] As a result, a CB edge of AOS can be formed by vacant metal cation s-orbitals that have spherical, isotropic, and spatially spread shapes (Figure 1). This huge disparity in terms

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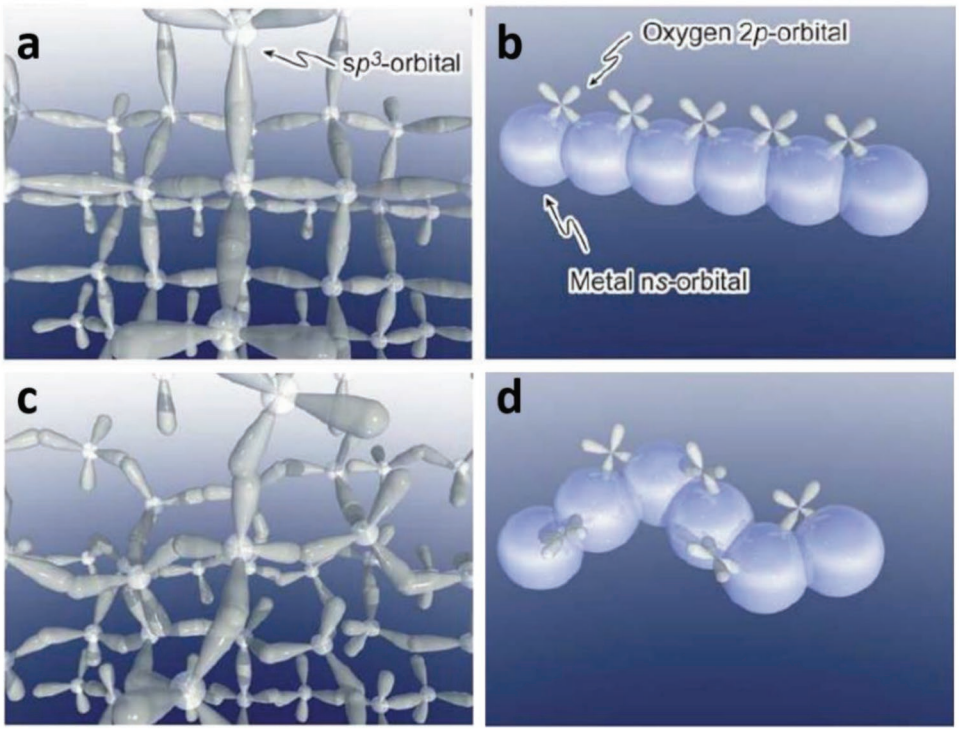


Figure 1. A schematic of orbital bonding in crystalline (top) and amorphous semiconductors (bottom): a,c) Si composed of anisotropic sp^3 orbitals; b,d) oxide semiconductor composed of metal *n*s and oxygen 2p orbitals, which shows effective overlap of spatial, isotropic metal *n*s orbitals even in amorphous structure. a–d) Reproduced with permission.[1] Copyright 2004, Springer Nature.

of contributing orbitals affects the direction-dependent carrier transport properties, leading to a noticeable difference in the electrical characteristics of amorphous Si (a-Si) and AOS field-effect transistors (FETs). The former and the latter have field-effect mobilities (μ_{FE}) around 1 and ≥10 $cm^2\ V^{-1}\ s^{-1}$, respectively. Remarkable work by Hosono and co-workers proved that a certain wide bandgap (E_g) material with an amorphous structure can have great semiconducting characteristics, and this has created significant worldwide interest for the AOS family in academia and industry.

In addition to promising electrical characteristics of a high μ_{FE} and an extremely low off-current, AOS has numerous fascinating advantages such as optical transparency, low fabrication cost/temperature, and great uniformity over a large area.[1–4] This indicates their strong potential for diverse next-generation electronic applications that require transparency, deformability, and/or long durability. Among them, AOSs have been extensively studied as channel layers in thin-film transistors (TFTs). In fact, active-matrix organic light-emitting diode (AMOLED) televisions have been produced using In–Ga–Zn–O (IGZO)-TFT-based backplanes by LG Display since 2013, successfully replacing hydrogenated a-Si TFTs despite the very short incubation period for AOS-based TFTs. Since then, AOS TFTs have been a mainstream large-sized display backplane technology. Nevertheless, AOS TFTs with device performances comparable to low temperature polycrystalline Si, especially in terms of μ_{FE}, must be developed to satisfy ever-growing demand on them for semiconductor industries as well as display industries. Several approaches to increase the μ_{FE} in AOS TFTs have been reported such as introducing a multichannel layer structure, controlling cation combinatorial composition, and crystallization. However, it is not trivial to boost the μ_{FE} without degradation of reliability on external stress in AOS TFTs. This trade-off relationship between mobility and stability will be discussed later.

1.2. Emerging Needs of Oxide Semiconductor Devices

Memory, logic, and processor chips in the semiconductor industry are developed toward high performance, low power consumption, and reduced cost. Key considerations are speed, power, density, and cost when designing new or emerging semiconductor chips, which can be represented as power–performance–area–cost (PPAC).[5] To meet this PPAC ground rule, the semiconductor industry has achieved a number of major technological innovations. Logic transistors have been developed through: 1) advanced material and process implementation such as high permittivity (κ) gate dielectrics and strain engineering and 2) structure innovations from planar to 3D transistors such as FinFET or gate-all-around (GAA), which is also known as multibridge-channel FET (MBCFET).[6,7] Volatile and nonvolatile memories represented by dynamic random access memory (DRAM) and vertical NAND flash memory (VNAND), respectively, have also been developed on the basis of PPAC. However, the semiconductor industry is currently facing huge challenges for further transistor miniaturization due to the physical limitations of Si. It is an open question as to whether the industry can continue its growth trajectory considering the increasingly difficult, incremental, and expensive technological advancements.[8]

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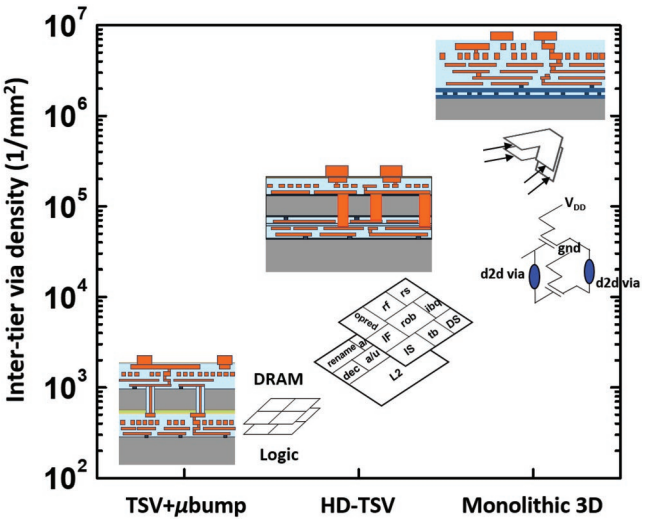


Figure 2. Achievable intertier via density as a function of the specific 3D integration scheme: die stacking, layer transfer, in situ sequential integration. Adapted with permission.[9] Copyright 2019, IEEE.

3D integration technology is being proposed as one of the most important technologies to overcome the issues of PPAC originating from traditional scaling limitations. The sequential 3D monolithic (M3D) integrated circuit can provide interlayers via density that is approximately two orders of magnitude denser than through-silicon-via (TSV) density (**Figure 2**).[9] When this monolithic technology is available, the power and performance of any semiconductor chip can be dramatically improved by reducing the wiring length, and its cost and area can be substantially reduced compared to conventional 2D technologies. To realize this state-of-the-art technology, however, upper-layer transistors in this architecture must be fabricated at a low temperature (<400 °C) to prevent the electrical performances of devices in underlying layers from being undermined.[9] For this reason, Si requiring a high temperature to exploit its exceptional electrical characteristics cannot be adopted in the upper-layer transistors, necessitating an alternative material with great electrical characteristics that can be achieved using low temperature fabrication. Here, there are several promising candidates satisfying this stringent thermal requirement including 2D materials and AOS. Previously, 2D materials such as molybdenum disulfide (MoS_2), molybdenum diselenide, and tungsten diselenide (WSe_2) gained more attention than AOS due to their extraordinary physical properties including the high mobility and dangling-bond-free surfaces. Furthermore, their atomically thin body can provide strong immunity against the notorious short channel effects (SCEs), enabling the demonstration of transistor applications with a sub-10 nm gate length (L_g).[10–12] However, the direct growth of 2D materials with high crystalline quality on a Si substrate at the low temperature is still limited, posing significant challenges for both controlling thickness of these thin films and fabricating them uniformly over a large area. For this reason, it is difficult to fully take advantage of their outstanding physical properties in practice. In this regard, AOS has recently emerged as another promising alternative for the upper layer transistors because it is stackable using conventional complementary-metal–oxide–semiconductor (CMOS)-compatible methods such as sputtering and atomic layer deposition (ALD) routes at low temperature. The extremely low leakage current (<10^{-18} A μm^{-1}) originating from the large E_g (>3 eV) and huge gap state above the VB edge in AOS is strongly attractive for low-power applications, making AOS critical for both the memory and foundry industries. However, many unknown factors in a scaled dimension, such as carrier transport, electrical contact properties, and SCEs, remain unresolved, requiring in-depth studies on miniaturized AOS TFTs. Furthermore, CMOS TFT technology should be available for versatile back-end-of-line (BEOL) applications. Unfortunately, developing p-type oxide semiconductor (OS) channels is difficult because of strong localization of the VB edge by oxygen 2p orbitals,[13] which should be solved by improving their electrical characteristics or finding another feasible candidate. As such, both 2D materials and AOS have own strengths and weaknesses, implying that these candidates can be applied in different roles in the M3D architecture depending on the purpose of the applications.

Note that there are several excellent review articles on OSs including AOS.[3,14–23] Since most studies on OS have been carried out for display applications to date, the main focus is on display applications with rather long channel lengths (≈10 μm) and thick structures (≥200 nm). However, works demonstrating the feasibility of OS for memory and synaptic devices as well as BEOL-compatible transistors have been reported recently beyond the display backplane technology. Indeed, they have been increasingly addressed at the IEEE International Electron Devices Meeting (IEDM) and Symposium on VLSI Technology and Circuits, two of the most renowned conferences in the field of semiconductor technology, where the total number of papers regarding OS devices steadily increased from 3 in 2017 to 18 in 2021. This strongly indicates increasing interest in OS from the semiconductor industry. Nevertheless, a study summarizing the potential of OS for semiconductor applications is lacking at present. Herein, the recent progress of OS devices including advanced applications as well as unit FETs are addressed. The rest of this paper is organized as follows. Section 2 describes the fundamental backgrounds of AOS, such as the role of respective constituents, conduction mechanism, and reliability issues. Anyone reading the previous review papers may skim or skip Section 2 depending on his/her understanding of AOS. Then, the recent progress of OS TFTs using diverse approaches and demonstrating promising immunity against SCE is dealt with in Section 3. Here, some of the leading studies are reviewed in detail. Furthermore, highly important and relevant works suggesting the potential of emerging OS applications such as memory and synaptic devices are introduced in Sections 4, and 5, respectively. Note that we focus on reviewing the latest studies (mainly reported after 2017). Finally, key technological obstacles and potential opportunities which can arise from the use of OS are discussed in Section 6.

2. Backgrounds of Amorphous Oxide Semiconductor Devices

2.1. Role of Respective Constituents

Electrical and optical characteristics of AOS are strongly dependent on the relative composition ratio of metal cations

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and oxygen. The degree of oxygen deficiency in AOS can be controlled to some extent by modulating oxygen partial pressure (P_O) during deposition. This influences the fraction of oxygen vacancy (V_O) and interstitial oxygen (O_i), which acts as an electron source and/or trapping sites. Therefore, it can be inferred that modulating the oxygen content is a first step in controlling the free electron concentration (N_e) in AOS. More importantly, AOS is a network of binary metal oxide polyhedrons, and the coordination number (CN) of each cation with nearby oxygen atoms and the way to share them between polyhedrons are strongly dependent on the configuration of metal cations and network packing density. Therefore, understanding the role of metal cations in the AOS system is critical as this helps predict the physical properties of various ternary and quaternary combinations of binary In_2O_3, Ga_2O_3, ZnO, and tin oxide (SnO_2).

In cation is known as a creator of V_O in In-based AOS because 1) the In−O bond strength is the weakest among M−O bonds (M = Zn, Ga, In, and Sn) and 2) the bixbyite In_2O_3 unit cell associated with an indium oxide polyhedron motif contains two V_Os because it is a derivative structure of the calcium fluorite.[24] Furthermore, spherically extended In 5s orbitals are effectively intercalated with each other and thus mainly characterize the CB edge in the In-based OS. This leads to a low electron effective mass (m_e*), enabling realization of their high μ_{FE}. It should be noted that these unique characteristics of In cations are still effective in the amorphous structure. Sn cations are also considered as an electron mobility enhancer along with In cations because they have identical isotropic 5s orbitals, which form a great electron conduction path.[4] Unlike In and Sn cations, Ga cations act as a carrier suppressor in AOS. Since the Ga−O bond strength is the strongest among M−O bonds,[4] V_O generation is effectively suppressed by incorporation of Ga cations, which hinders the free electron generation in AOS. Zn cations are known to stabilize amorphous structures because ZnO has a CN of 4 with the corner sharing a wurtzite structure.[25,26] Note that both bixbyite In_2O_3 and rutile SnO_2 have a CN of 6 with an edge sharing structure,[25] which indicates that the incorporation of Zn cations can effectively hinder the formation of an edge sharing crystalline structure. It is noteworthy that coordination structures of constituent ions in the amorphous structure are similar to those in the corresponding crystalline structure, but the average CN is generally reduced in the amorphous phase.[26] In addition to the abovementioned metal cations, there are other candidates, such as tungsten (W), aluminum (Al), and Si, which can be combined in AOS for various roles. The amount and kind of cations could be different depending on the purpose of AOS applications.

2.2. Conduction Mechanism of Amorphous Oxide Semiconductors

Carrier conduction is closely related to the electronic structure. Amorphous materials have numerous strained chemical bonds that form tail states below the CB and/or above VB edge as well as deep states. For this reason, the carrier conduction in a-Si occurs by variable range hopping (VRH) between tail states, leading to poor drift properties.[27] Note that although they possess this innate amorphous property, AOS is relatively insensitive to local strain compared to other amorphous materials because of the effective intercalation of contributing orbitals. The intriguing band-like transport property in AOS is explained by a percolation conduction (PC) mechanism, where electron mobility increases with increasing electron concentration.[27–29] Random distribution of various cations such as In^{3+}, Ga^{3+}, and Zn^{2+} cations in AOS causes nanoscale bandgap fluctuations, which are responsible for the existence of position-dependent potential barriers near CB edge (**Figure 3**a). This PC mechanism is mainly determined by the potential barrier height and the ratio of barrier width to distance, both of which are reduced with increasing N_e.[29] Thus, the electron mobility is enhanced with the N_e in AOS. However, it should be noted that the electron conduction is dictated by trap-limited conduction (TLC) before an applied gate voltage (V_G) is high enough to make the Fermi level (E_F) reach the CB edge. In the TLC region, the electrons are subjected to multiple trapping events in the localized tail states below the CB edge during the transport (Figure 3b). Once the E_F exceeds the CB edge, the electron transport is dominated by the PC, where the electrons in the nonlocalized states travel through the path with the least resistance among the potential barriers.[29] While μ_{FE}, i.e., electron transport, is dependent on the temperature in the TLC region, it shows temperature-independent behavior in the PC region. Finally, electrons are transported by the VRH at low temperature [e.g., ≤–113 °C (160 K)] in the localized tail states ($E_F < E_{CB}$) due to the lack of sufficient thermal energy.[28]

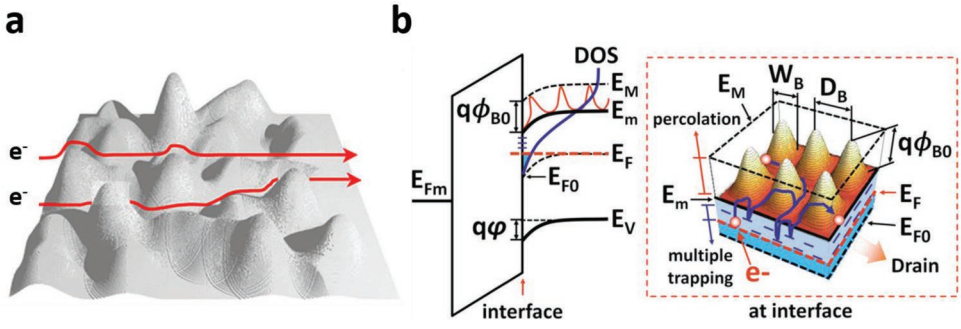


Figure 3. a) Illustration of percolation conduction over distributed potential barriers in CB edge. Reproduced with permission.[27] Copyright 2009, IEEE. b) Schematic energy band diagrams of an oxide semiconductor with tail states and potential fluctuations above CB edge. Reproduced with permission.[29] Copyright 2011, American Institute of Physics.

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We now briefly introduce how to distinguish between the dominance of TLC and PC in the electron transport using an equation. Note that gate-voltage-dependent μ_{FE} can be described by the power law equation: $\mu_{FE} = K(V_G - V_{T,P})^{\gamma}$, where V_G, V_T, and V_P are the gate, threshold, and percolation voltages, respectively, and the prefactor K and exponent γ are related to the nature of carrier transport.[29] In particular, the prefactor K includes the percolation term from the Thomas–Fermi approximation for both PC and TLC. In the PC, the contribution of the spatial characteristics of potential barriers are involved in K. Meanwhile, the exponent γ is dependent on either the spatial characteristics of the potential barriers for PC or the temperature for TLC, which implies that this is an indicator of the different transport mechanisms that dictate the electron conduction. Here, it was confirmed by Lee et al. that the values of γ are ≈0.1 and 0.7 in the PC- and TLC-dominant regions, respectively.[29]

3. Reliability Issue of Amorphous Oxide Semiconductors

As potential applications for AOS such as ultrahigh resolution displays, chip input/output, power devices have emerged, the development of AOS with high μ_{FE} comparable to that of conventional Si is required. For this reason, there have been enormous efforts to increase the μ_{FE} of AOS TFTs using various approaches. However, it should be noted that there is a trade-off relationship between mobility and reliability in AOS, which has been troublesome for developing AOS TFTs with high μ_{FE}. Therefore, it is important to understand the degradation mechanisms to affect the reliability in AOS TFTs, and they can be classified to: 1) an intrinsic cause mainly arising from point defects in AOS and 2) an extrinsic one by atmospheric adsorbates such as oxygen and water molecules. Before discussing the reliability issues, it is worth noting that point defects such as the V_O and O_i cannot be defined clearly in AOS due to their high relaxation structure, which means that these terminologies are not accurate (unlike crystalline materials). Nevertheless, it should be emphasized that the abovementioned terms will be used throughout this paper as they are commonly used in the AOS system.

In general, the threshold voltage (V_{TH}) of AOS TFTs is positively shifted under a positive gate bias stress (PBS). The PBS-induced V_{TH} instability can be attributed to: 1) trapping of accumulated electrons at the interface between a gate insulator (GI) and an AOS channel layer or 2) their injection into bulk trap states of GI (**Figure 4**).[18,30] This indicates that the N_e of AOS channel layer is reduced during the PBS condition, leading to a positive shift of the V_{TH}. Here, a V_O site existing near some undercoordinated In (In*) cation (referred to as In*–M by Nahm and Kim) can be an intrinsic acceptor-like trap, which easily captures two electrons and converts to $(In^*–M)^{2-}$ states, aggravating the PBS stability in the In-based AOS with oxygen deficiency.[31] The O_i defects in AOS that come from excess oxygen also act as an electron trap sites. These defects capture two electrons and becomes an O_i^{2-} site, which can degrade the PBS stability in AOS.[32] It is noteworthy that the formation energy of the O_i^{2-} site is so low that this chemical reaction can spontaneously occur. The ubiquitous hydrogen (H) impurities must be considered in AOS because they result in complex defects with V_O or O_i.[33] H impurities, such as interstitial H (H_i) and substitutional H at the oxygen site (H_O), are generally considered as shallow donors in the AOS system.[33–35] However, the H impurities also can negatively impact the PBS stability, especially in H-rich and oxygen deficient AOS TFTs, which would be explained by the H bistability model: 1) At low H concentration ([H]), the H impurities act as a shallow donor in AOS, as mentioned above; 2) at high [H], however, the normally positively charged H_O^+ defects can capture electrons, dissociating into V_O and H_i^+ through a large lattice relaxation, where this Frenkel defect is referred to as a H-DX$^-$ center.[33,36] This understanding was supported through density functional theory (DFT) calculations, where formation energies of four types of H impurities (H_i^+, H_O^+, H_i^-, and H-DX$^-$) were calculated in an amorphous In–Zn–Sn–O (IZTO) system.[33] Here, it was confirmed that the average thermal transition level from H_i^+ (H_O^+) to H_i^- (H-DX$^-$) was 2.79 (2.96) eV above the VB edge (**Figure 5**), which indicates that H_i^+ and H_O^+ defects are more stable than H_i^- and H-DX$^-$ defects in most of the E_F region except the critical condition in which E_F is close to the CB edge. For this reason, the electrons can be trapped by the H-related defects in the H-rich AOS TFTs under PBS conditions where

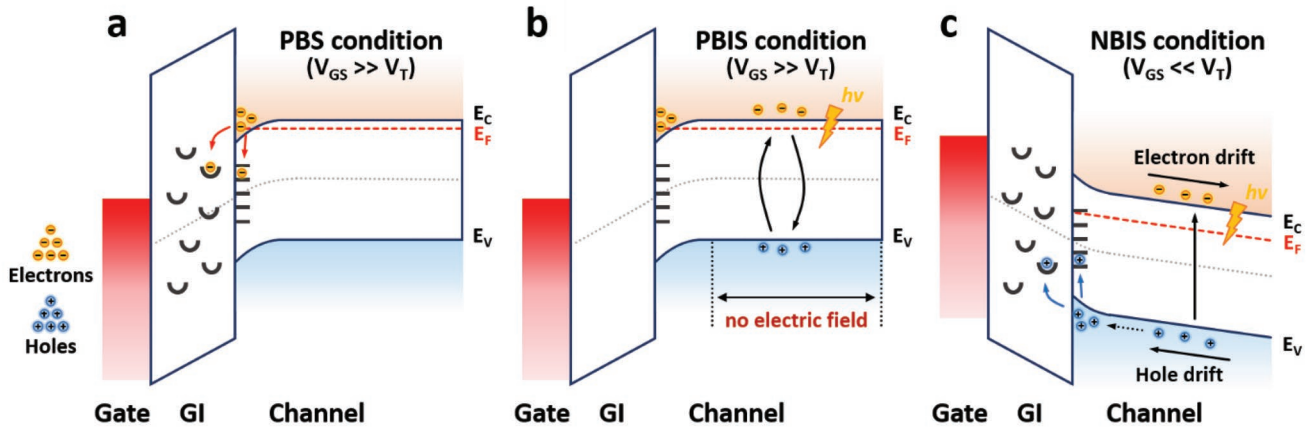


Figure 4. a–c) Energy band diagrams of metal/insulator/semiconductor structure under different external bias conditions: a) PBS, b) PBIS, c) NBIS.

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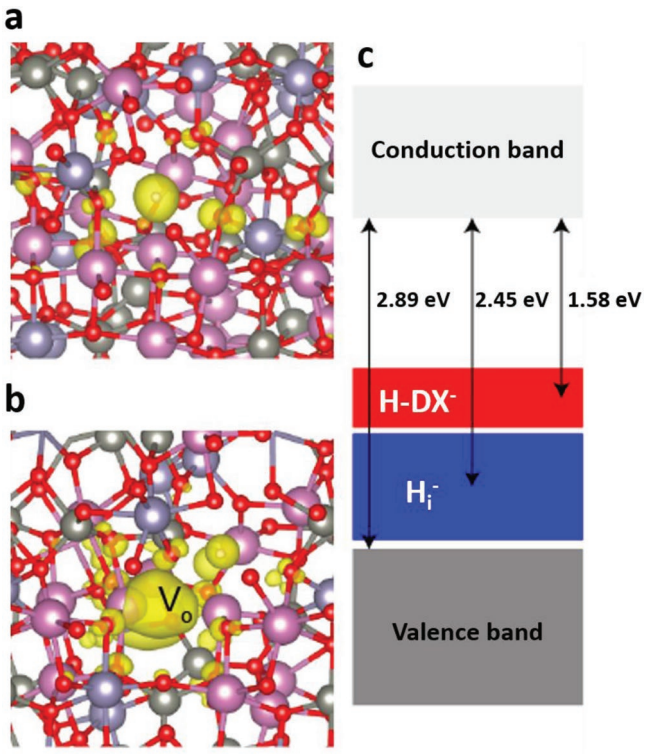


Figure 5. a,b) Partial charge densities of different electronic states: a) H_i^-, b) H-DX^-. c) A schematic diagram of energy distributions of different H levels. a–c) Reproduced with permission.[33] Copyright 2015, Wiley-VCH.

the E_F reaches the CB edge, which can lead to poor stability. In addition to these intrinsic causes, the reliability against the PBS condition can deteriorate due to atmospheric effects, which is mainly attributed to the dynamic adsorption of ambient gases. Oxygen molecules in air are adsorbed on the surface of AOS film, and this can be significantly increased by an electric field under PBS conditions (**Figure 6**).[37] These adsorbed oxygen molecules can capture electrons due to their high electronegativity, depleting the underneath AOS layer. Thus, this chemical reaction by the ambient oxygen is responsible for the positive V_{TH} shift of AOS TFTs, which can be often observed for the unpassivated AOS TFTs. Based on this degradation mechanism, it is important to 1) reduce trap states including interfacial traps (D_{it}) through optimizing the GI and AOS channel layer, 2) minimize the nonstoichiometric defects such as V_O and O_i by careful deposition and annealing processes, and 3) passivate AOS using a high-quality dielectric layer to improve the stability against the external PBS duration. It is worth mentioning that the main causes of the positive gate bias thermal stress (PBTS) are identical to those of PBS.

As opposed to the PBS, V_{TH} is negatively shifted under a negative gate bias stress (NBS) in AOS TFTs (Figure 4). It should be noted that the V_{TH} shift is negligible or small under the NBS conditions, including negative gate bias thermal stress (NBTS), as compared to the PBS conditions. This implies that the stability against the NBS conditions is generally superior to that against the PBS conditions, which can be mainly attributed to scarcity of hole carriers in most of the AOS.[30] However, electron–hole pairs can be generated from deep level subgap states above the VB edge by exposure to visible/ultraviolet (UV) light; these pairs are separated by the local electric field under NBS conditions (Figure 4).[30] In particular, photoinduced hole carriers move toward the interface between the GI and channel layer, being trapped (injected) at the interface (into the bulk of GI). This results in significant deterioration in the reliability against the NBS conditions, which is called negative gate bias illumination stress (NBIS). The subgap states creating the electron–hole pairs by the light can originate from V_O defects, which are responsible for the NBIS degradation. While some are in the shallow donor level, most of the V_O defects exist in deep level states that are electrically inactive.[38] This theoretical result was experimentally corroborated, where the improvement in the stability against the NBIS was shown by reducing the amount of V_O in the amorphous IGZO TFTs.[39] Moreover, O_i defects in AOS can degrade the stability against the NBIS when H defects exist near them.[40] The H defects, a shallow donor as mentioned above, easily donate electrons to the network, which are taken by the O_i defects, becoming O_i^{2-} states. Since two electrons in the O_i^{2-} states are excited to the CB edge

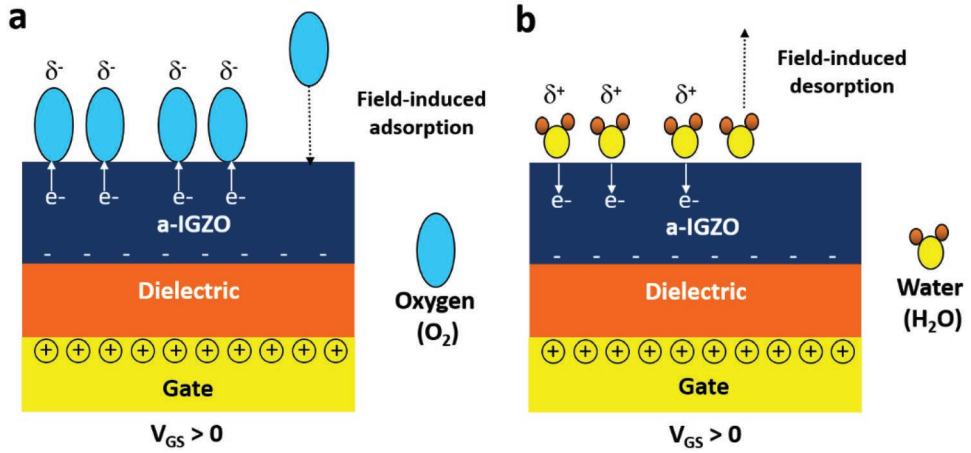


Figure 6. a,b) Illustrations of electric-field-induced behavior of atmospheric molecules under the application of positive gate bias: a) oxygen, b) water. a,b) Reproduced with permission.[37] Copyright 2008, American Institute of Physics.

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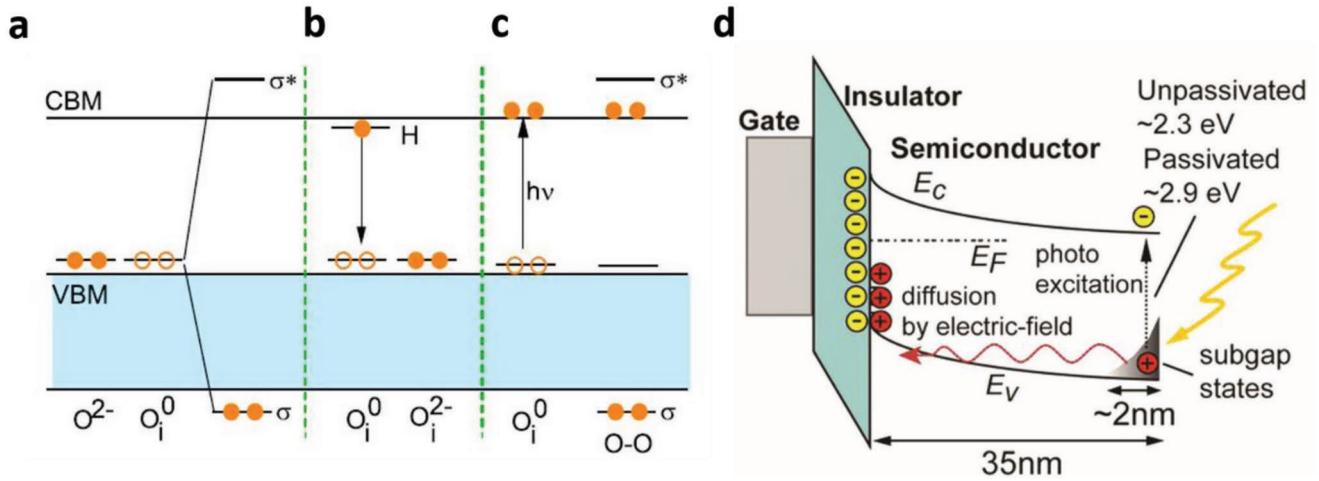


Figure 7. a–c) Schematic diagrams: a) formation of a neutral O_i site, b) electron transfer from two H to the neutral O_i state, creating an O_i^{2-} state, c) photoexcitation of electrons from the O_i^{2-} state. a–c) Reproduced with permission.[40] Copyright 2014, American Institute of Physics. d) Illustrated energy band diagram which shows accumulation of positive charges by photoexcited electrons from defect states near VB edge existing at surface. Reproduced with permission.[43] Copyright 2011, American Institute of Physics.

by the light illumination, they can be responsible for NBIS degradation in AOS (**Figure 7**a–c). The reliability on the NBIS is also affected by atmospheric factors. Adsorbed water molecules by exposure to air donate electrons to the underlying AOS layer, leading to the huge negative displacement of V_{TH}.[41,42] Note that the amount of adsorbed water molecules significantly increases due to the electric field under NBS conditions.[37] For this reason, AOS TFTs without passivation show inferior NBS stability compared to those with passivation. This trend is extended to the NBTS and NBIS as well, emphasizing again that passivating AOS thin films is an essential step to obtain highly reliable transistor applications. Furthermore, passivation can also reduce the subgap states near the VB edge existing at the surface of AOS thin films, thus improving the stability against the NBIS conditions (Figure 7d).[43]

It is noteworthy that a high mobility AOS like IZTO generally shows poor reliability even against NB(T)S conditions. Recently, an impressive study was reported by Shiah et al., where a possible origin of this trade-off relationship between the μ_{FE} and reliability was suggested involving carbon-monoxide (CO)-related impurities.[44] Note that these are readily introduced into the thin film during the universal photolithography process, which was confirmed through thermal desorption spectroscopy (TDS) (**Figure 8**a,b). These CO-related impurities were formed by the chemical reaction between the oxide and photoresist, which requires postdeposition annealing (PDA) at >350 °C to desorb these detrimental impurities. Here, it is interesting to note that IZTO TFTs annealed at >350 °C after the photolithography exhibited significant improvements in the reliabilities against the NBS conditions compared to those of the TFTs annealed before the photolithography (Figure 8c–g). This result demonstrates the correlation between CO-related impurities and reliability. The proposed degradation mechanism is as follows. Deep states formed by the CO-related impurities are transformed to shallow states by the external NBS stress shifting down the Fermi level, and thereby donate electrons to AOS, leading to a negative shift in V_{TH}. However, it is surprising that IGZO TFTs barely showed instabilities by the CO-related impurities even when having almost the same number of impurities. This noticeable difference between IZTO and IGZO TFT can be derived from the electronic structure, especially the CB edge. Note that AOS with a high μ_{FE} including IZTO has effective spatial overlap of larger metal s orbitals and therefore a deeper CB edge compared to AOS like IGZO. Here, the externally created shallow states have a lower activation energy in AOS with a deep CB edge, which easily donate electrons, leading to severe instabilities against the NBS conditions in the high mobility AOS TFTs. As such, the reliability of AOS TFTs is closely related to numerous defects and cations that contribute to the formation of the CB edge. This trade-off relationship between the μ_{FE} and reliability emphasizes that controlling the amount of these defects is of paramount importance to obtain high mobility, highly reliable AOS TFTs.

3. Recent Progress in Oxide Semiconductor Transistors

Before starting this section, it should be noted that an apple-to-apple comparison between the reported figures of merit, especially the mobility, in different literature reports may be inappropriate. Given that there are different mobility extraction methods[45–49] and each literature adopts specific one, the mobility value could be the parameter that has the largest disparity between the reports. Furthermore, a number of extrinsic factors to have a significant effect on the mobility extraction, such as device structure, electrical contact property, fabrication method, etc., are adopted differently, which makes it more difficult to directly compare the value of mobility between literature. Its exact and rigorous evaluation remains to be addressed, which should be compatible to the Si-based CMOS semiconductor industry.

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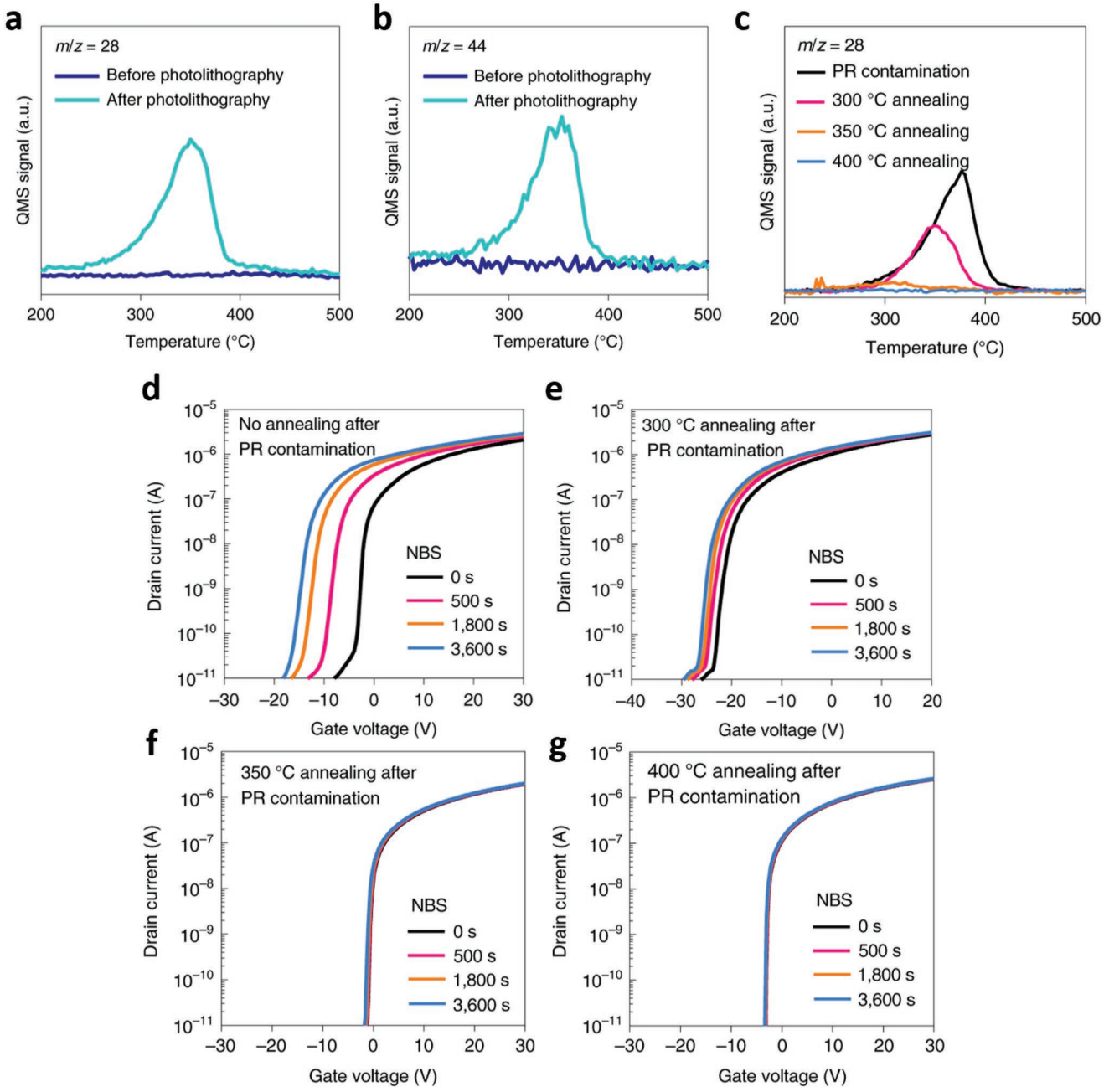


Figure 8. a–c) Thermal desorption spectroscopy (TDS) of different carbon-related defects: a) CO, b) CO_2, c) CO for photoresistance (PR)-contaminated amorphous IZTO thin films with different PDA temperatures. d–g) NBS results of IZTO TFTs with different PDA conditions: d) without PDA, e) 300 °C, f) 350 °C, g) 400 °C. a–g) Reproduced with permission.[44] Copyright 2021, The Authors, published by Springer Nature.

3.1. Composition and Crystalline Structure Engineering

3.1.1. Composition Control Approach

One of the most important objectives in the AOS field is to secure high μ_{FE} values comparable with those of Si-based semiconductor devices. Here, designing the multicomponent cation combination and relative cation fraction in AOS can be the simplest but imperative approach to improve the electrical characteristics of AOS devices. Thus, many studies have been conducted to find the optimal combination ratio and thus improve the μ_{FE} and reliability against external bias stress. For this purpose, the cosputtering method is a very effective approach to investigate the composition combinatorial dependence of IGZO on the performance of the resulting TFTs. Iwasaki et al. reported that device parameters such as μ_{FE}, V_{TH}, and current modulation ratio ($I_{ON/OFF}$) for the IGZO TFTs can be modified by varying the cation composition ratios. These TFTs were fabricated by cosputtering the separated three targets of In_2O_3, Ga_2O_3, and ZnO.[50] Very recently, Cho et al.

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reported a study on the control of In, Ga, Zn cation ratios of IGZO using ALD.[51] By controlling the cation composition, the bulk characteristics such as N_e, V_O, and density of states (DOS) can be adjusted, leading to promising device performances with a high μ_{FE} of 48.3 $cm^2\ V^{-1}\ s^{-1}$, a V_{TH} of −4.06 V, a subthreshold swing (SS) of 0.45 V dec^{-1}, and a $I_{ON/OFF}$ of $>10^7$ in the $In_{0.45}Ga_{0.15}Zn_{0.40}O$ TFT. Since then, studies to investigate the effect of composition ratio have been actively carried out on both IGZO and other promising AOS candidates such as In–Ga–O (IGO), In–Ga–Sn–O (IGTO), and IZTO using various deposition methods.[52–72]

Crystallization of AOS is also greatly affected by fabrication conditions such as the combination of multiple binary oxides, the fraction of each binary oxide, and growth/annealing temperature (refer to Section 2.1). Since Zn cations frustrate facile crystal growth in a-IGZO, however, a very high temperature (>700 °C) is required for its crystallization during the deposition and/or annealing process.[56,57] For this reason, Zn-free OSs, such as IGO and IGTO, have been actively studied by varying the fabrication conditions.[58–64] In case of In_2O_3, a cubic bixbyite crystalline phase easily grows at a low temperature (≈250 °C),[24,65] granting a remarkably high μ_{FE} through the high N_e and synergic effect of great In 5s orbital intercalation by crystallization. However, this promising electrical characteristic is somewhat compromised by uncontrollability of V_O creation, which leads to a highly negative V_{TH} and instability against the external bias stress. Furthermore, the random polycrystalline structures inevitably contain a number of large angle grain boundaries, degrading the uniformity on the large area. To overcome these disadvantages of the random polycrystalline structure, binary Ga_2O_3 has been incorporated into the In_2O_3 foundation as a crystallization retarder as well as electron suppressor. Yang et al. reported an in-depth, comprehensive study on the crystallization of ALD-derived a-IGO systems.[59] They investigated the effect of Ga cation fraction and annealing temperature in the IGO on the crystalline structure and electrical characteristics (**Figure 9**). As the Ga fraction increased from 25 to 34 at%, the crystalline structure of IGO thin films annealed at 400 °C is changed from a cubic bixbyite In_2O_3 (c-In_2O_3) to an amorphous structure. However, an $In_{0.66}Ga_{0.34}O$ thin film annealed at 700 °C surprisingly showed c-In_2O_3 structure with a preferential orientation of (222). Meanwhile, the crystalline structure of IGO thin films with the higher In fraction, i.e., the lower Ga fraction, had c-In_2O_3 structure with different faces such as (400) and (622) after PDA at 700 °C. It is also noteworthy that an IGO thin film with the higher Ga fraction had nearly amorphous structure, which might be attributed to the higher temperature for crystallization of Ga_2O_3 (≈1000 °C). Consequently, a preferentially oriented IGO TFT exhibited outstanding device performances with a μ_{FE} of 60.7 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.4 V dec^{-1}, a V_{TH} of −0.8 V, and ΔV_{TH} of +0.12 (−0.16) V against PBS (NBS) conditions.

In addition to the cation-dependent crystallization, other promising routes to crystallize AOS have been reported. Magari et al. crystallized binary In_2O_3 using H_2 gas ($R[H_2]$) during sputtering.[65] The as-deposited In_2O_3 thin films that exhibited the c-In_2O_3 phase with a preferential (222) orientation changed to amorphous structures by introduction of H_2 gas, which indicates that the incorporated H effectively suppresses the crystallization during deposition (**Figure 10**a). This amorphous hydrogenated In_2O_3 (In_2O_3:H) thin film was crystallized through PDA at a low temperature of 250 °C (Figure 10b). Furthermore, huge crystal grains with a size of 140 nm grew with a maximum area fraction of 23% in In_2O_3:H thin film at $R[H_2]$ of 5%, implying that the grain size was significantly enlarged through this method (Figure 10c–e). This high-quality crystallization could be attributed to a decrease in nuclei density in the as-deposited thin film through the incorporation of H. Note that similar results were confirmed irrespective of the PDA ambience. It is interesting that N_e rapidly decreased at a PDA temperature of >200 °C in the In_2O_3:H thin film with $R[H_2]$ of 5%, while the In_2O_3 thin film without the incorporation of H showed a PDA-independent N_e. Consequently, N_e of $2.0 \times 10^{17}\ cm^{-3}$ was obtained at a PDA of 300 °C for the 50 nm thick In_2O_3:H thin film with $R[H_2]$ of 5%. The resulting TFTs exhibited remarkable device performances with a μ_{FE} of 139.2 ± 3.0 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.19 ± 0.02 V dec^{-1}, and a V_{TH} of 0.2 ± 0.2 V, which could be attributed to the high crystallinity of thin film (Figure 10f–h). Park et al. reported another route for crystallization and its effect in the IGO thin film, where the crystalline structure was changed by controlling a partial oxygen plasma density (PO_2) from 0% to 30% during the thin-film deposition.[62] The c-In_2O_3 structure significantly grew in the IGO thin films with PO_2 = 10% even with the PDA at a low temperature of 200 °C. Note that the other IGO thin films with the different PO_2 showed amorphous structures. The resulting IGO TFT (PO_2 = 10%) exhibited outstanding device performances with a μ_{FE} of 56.0 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.1 V dec^{-1}, a V_{TH} of 0.1 V, and ΔV_{TH} of +0.8 (−1.0) V against the PBS (NBIS) condition. The transition in the crystalline structure by modulating the PO_2 was also reported in IGTO thin films.[64] The IGTO thin film with PO_2 = 0% had a c-In_2O_3 structure, while the amorphous structure was maintained in the others with different PO_2 contents (**Figure 11**). This difference resulted in a prominent gap in the electrical characteristics, where the polycrystalline IGTO (PO_2 = 0%) TFT revealed remarkable device performances with a μ_{FE} of 116.5 $cm^2\ V^{-1}\ s^{-1}$, a SS of 134 mV dec^{-1}, a V_{TH} of 0.47 V, and ΔV_{TH} of +0.01 (0.17) V against the PBS (PBTS) condition. It is worth mentioning that this facile crystallization occurring in the IGO and IGTO system is enabled because In, Ga, and Sn cations have the same octahedral motif (CN = 6). However, incorporation of Zn cations frustrates the facile crystallization (refer to Section 2.1), generally leading to an increase in the onset temperature of crystallization (T_{ON}) in the Zn-based multicomponent AOS. For this reason, a high temperature (>700 °C) during the deposition and/or annealing process is strongly required to fabricate a *c*-axis-aligned crystalline IGZO (CAAC-IGZO) or an IZTO.[57,66] Moreover, the CAAC-IGZO is not greatly affected by the composition ratio to the best of our knowledge. However, these OS thin films with the Zn cations also can be crystallized at low temperature (≈400 °C) using a metal capping layer on top of them, which will be discussed in the next section (**Table 1**).

3.1.2. Metal-Capping-Induced Crystallization

A metal capping approach is another simple but effective route to crystallize AOS (including Zn-based ones). Furthermore, T_{ON} can be significantly reduced using this approach, which

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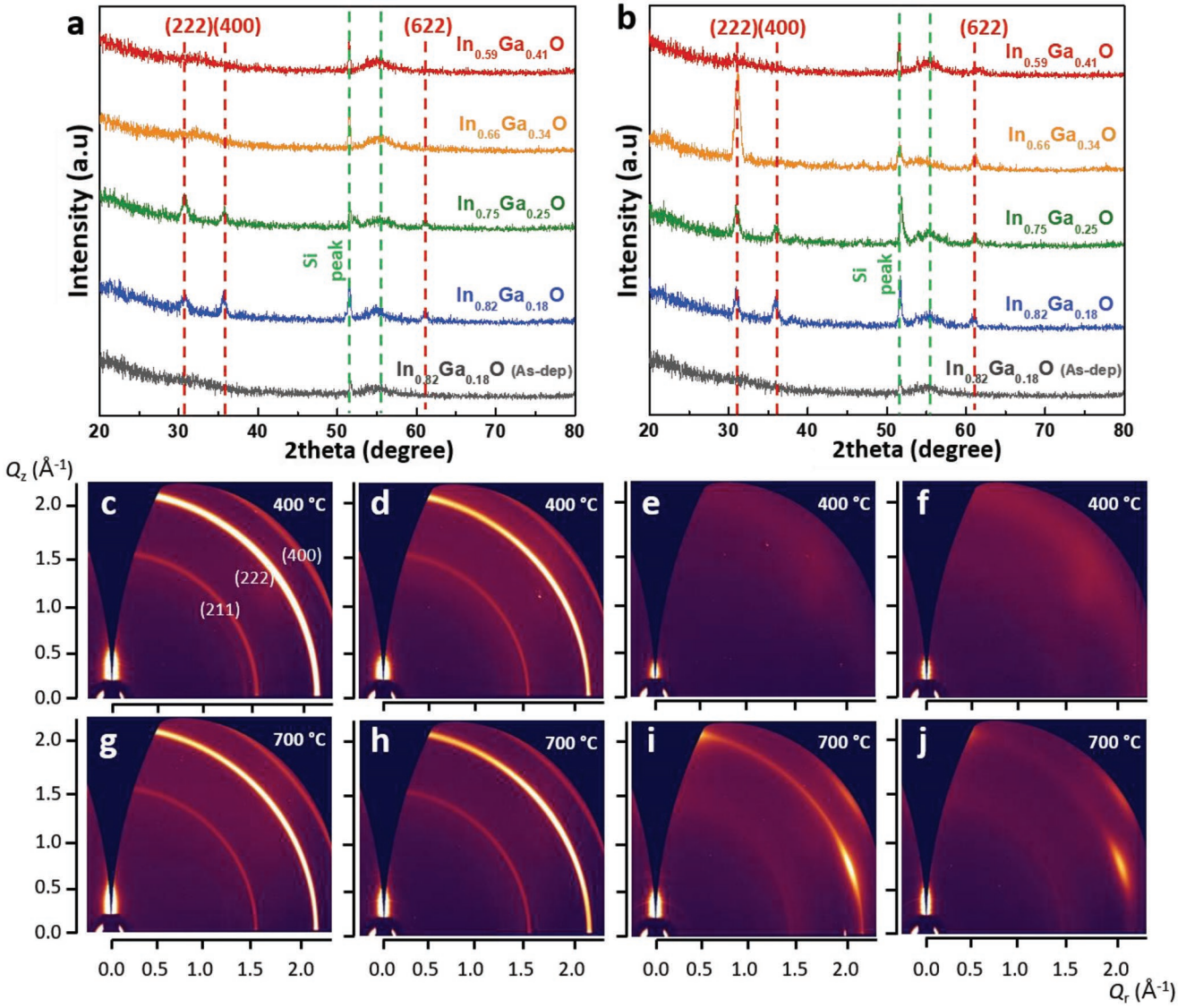


Figure 9. a,b) X-ray diffraction (XRD) patterns of IGO thin films with different In/Ga ratios depending on PDA conditions: a) 400 °C, b) 700 °C. c–j) 2D grazing-incidence XRD patterns of IGO thin films with different In/Ga ratios depending on PDA conditions: c–f) 400 °C, g–j) 700 °C. a–j) Reproduced with permission.[59] Copyright 2020, American Chemical Society.

is especially useful for crystallization of Zn-based AOS such as Zn–Sn–O (ZTO), IGZO, and IZTO. Main rationale to possibly occur during this crystallization route is as follows (**Figure 12**): 1) oxygen-related defect states in the underlying AOS thin film, such as weakly bonded oxygen and O_i, are removed because of the reduction effect during the formation of a metal–oxide layer, which is called the scavenge effect, 2) the catalytic metal layer on top of the AOS thin film can help break and rearrange the weak bonding between the cation and oxygen underneath during the PDA, which is called metal-induced crystallization (MIC). Because of these advantages, several studies have been carried out to improve the electrical characteristics of AOS devices using this metal capping approach.[73–76]

Hwang et al. reported tantalum (Ta)-induced crystallization in the ZTO thin films even through PDA at a low temperature of 300 °C.[73] Considering that the conventional T_{ON} (≈650 °C) for ilmenite ZTO is much higher than 300 °C, the effect of Ta capping is quite effective. The resulting ZTO TFT exhibited promising device performances with a μ_{FE} of 33.5 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.4 V dec^{-1}, and a V_{TH} of 1.8 V, which is much greater than a-ZTO TFT. Furthermore, Shin et al. successfully crystallized an IGZO thin film using a Ta capping layer even at 300 °C (**Figure 13**a,b), which led to a significant increase in μ_{FE} from 18.1 to 54.0 $cm^2\ V^{-1}\ s^{-1}$ in the Ta-capped IGZO TFT.[74] This metal capping approach has been extended to a Zn-oxynitride (ZnON) system.[75] Ta-capped ZnON TFTs exhibited improved device performances compared to those of control ZnON TFTs (μ_{FE} value of 89.4 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.33 V dec^{-1}, and a V_{TH} of −0.45 V). This improvement in the electrical characteristics is mainly derived from the scavenge effect, which results in uniform crystallization and an enhanced stoichiometric Zn_3N_2 portion, leading to fewer defect centers as well as higher N_e in the

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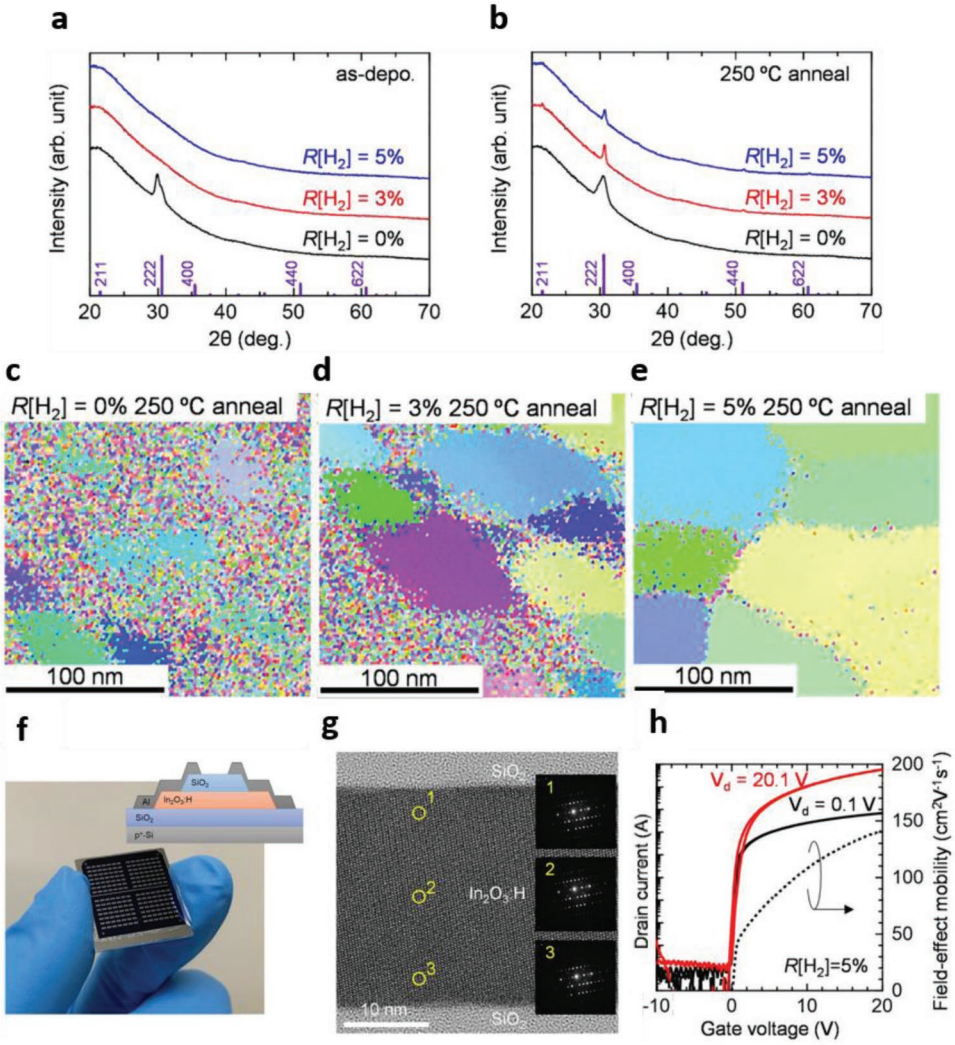


Figure 10. a,b) XRD patterns of In_2O_3 thin films deposited at different $R[H_2]$ values depending on PDA conditions: a) without PDA, b) 250 °C. c–e) Electron back-scattering diffraction images of 250 °C-annealed In_2O_3 thin films with different $R[H_2]$ values: c) 0%, d) 3%, e) 5%. f) Photograph and schematic of fabricated TFTs. g) Cross-sectional high-resolution transmission electron microscopy (HRTEM) image and selected-area electron diffraction (SAED) patterns of a 300 °C-annealed In_2O_3 thin film with $R[H_2]$ of 5%. h) Transfer characteristics of a TFT with a 300 °C-annealed In_2O_3 thin film with $R[H_2]$ of 5%. a–h) Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[65] Copyright 2022, The Authors, published by Springer Nature.

underlying ZnON thin film. This was supported by technology computer aided design (TCAD) simulation (Figure 13c–f), where the lower acceptor-like traps and large N_e allow the quasi-E_F to be easily elevated, and thus bulk accumulation-like transport is realized in the ZnON under the Ta capping layer.

Park et al. conducted a comparative study on the capping layer on top of Zn–Ba–Sn–O (ZBTO) thin films.[76] It is notable that this study did not address the crystallization approach. Nevertheless, it needs to be discussed because there is a similarity in that the device performance is greatly improved by utilizing a metal capping approach. Here, oxygen in the underlying ZBTO thin film is reduced through an Al capping layer during the formation of a metal–oxide interfacial layer (AlO_x), which results in a highly oxygen-deficient backchannel layer, i.e., degenerated (n^+) layer. A number of electrons introduced from the n^+ layer traveled in the front channel without noticeable scattering because a-ZBTO is a well-known material with a low defect density.[77] For this reason, the Al-capped ZBTO TFT showed a huge enhancement in the μ_{FE} from 20.8 (control) to 153.4 $cm^2\ V^{-1}\ s^{-1}$. Although Ti-capped ZBTO TFT showed an improvement in the μ_{FE} of ≈53.7 $cm^2\ V^{-1}\ s^{-1}$ (**Figure 14**a), the mobility boost was smaller due to the lower oxidation power compared to the Al. Note that the effect of the metal capping approach on the ZBTO is much greater than other OS such as ZTO and In–Zn–O (IZO), which shows the importance of the adoption of a channel layer with the low defect density. It is also noteworthy that there is no apparent degradation in the other electrical figures of merit (**Table 2**). The resulting Al-capped ZBTO TFT revealed outstanding device performances with a μ_{FE} of 153.4 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.37 V dec^{-1}, a V_{TH} of –0.77 V, and a ΔV_{TH} of +0.36 (–0.46) V against the PBS (NBS) condition (Figure 14b–d).

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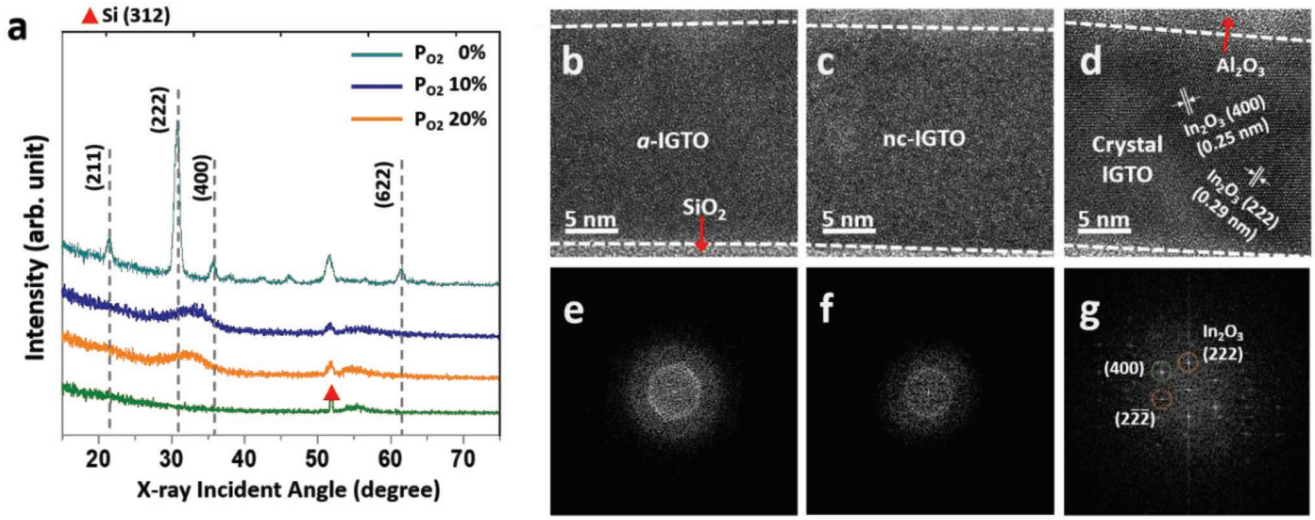


Figure 11. a) XRD patterns of 400 °C-annealed IGTO thin films with different PO_2 ratios. b–g) Cross-sectional HRTEM images and the corresponding SAED patterns of IGTO thin films with different PO_2 ratios: b,e) 20%, c,f) 10%, d,g) 0%. a–g) Reproduced with permission.[64] Copyright 2021, IEEE.

To sum up Section 3.1, controlling the composition and chemical ratio of OS is an effective approach to improve the electrical characteristics of resulting OS TFTs because each binary oxide has its own role in this multicomponent materials. Here, it is important to note that this also has a close relationship with crystallization of OS. For example, Zn-free OS can be easily crystallized at a low temperature (≈400 °C) using this approach because the representative In, Ga, and Sn cations have the same octahedral motif. However, it simultaneously hints that it is difficult to crystallize Zn-based OS at a low temperature by controlling the composition. For this reason, other routes to crystallize the Zn-based OS have been studied, and the metal capping on top of them can be a promising approach.

Table 1. Summary of device performances of oxide semiconductor TFTs addressed in Section 3.1.1.

Material	Process	Thickness [nm]	Channel dimension [W/L] [μm]	Mobility [$cm^2 V^{-1} s^{-1}$]	Ref.	Year
IZTO	Sputter	20	60/30	70	[44#44]	2021
a-IGZO	Sputter	50	100–300/10–50	12	[50#50]	2007
a-IGZO	ALD	13	10/50	48.3 ± 1.7	[51#51]	2019
a-IGZO	ALD	20	40/20	74	[52#52]	2019
a-IGZO	ALD	20	40/20	15.1 ± 0.31	[53#53]	2019
a-IGZO	ALD	20	40/20	27.6	[54#54]	2021
a-IGZO	Sputter	40	1000/150	12.0 ± 0.7	[55#55]	2021
a-IGO	ALD	20	40/20	9.45	[58#58]	2017
pc-IGO	ALD	13	40/20	60.7 ± 1.0	[59#59]	2020
c-IGO	Pulsed laser deposition (PLD)	–	–	–	[60#60]	2021
IGO	PEALD	20	40/20	36.7	[61#61]	2021
pc-IGO	Sputter	12	1000/300	56.0	[62#62]	2022
a-IGTO	Sputter	20	1000/300	35.0 ± 1.7	[63#63]	2018
pc-IGTO	Sputter	20	1000/300	116.5 ± 13.3	[64#64]	2021
c-In_2O	Sputter	50	300/300	139.2 ± 3.0	[65#65]	2022
pc-IZTO	Sputter	19	1000/300	39.2	[66#66]	2020
IGZO, IZO	Sputter	13	–	40	[67#67]	2022
a-IGTO	Sputter	40	20/15	39.1	[68#68]	2019
IGTO	Solution	–	1000/50	2.13	[69#69]	2021
IZTO	Sputter	50	270/15	52.4	[70#70]	2014
IZTO	ALD	20	40/20	27.8	[71#71]	2019
IZTO	PEALD	8	300/50	22	[72#72]	2019

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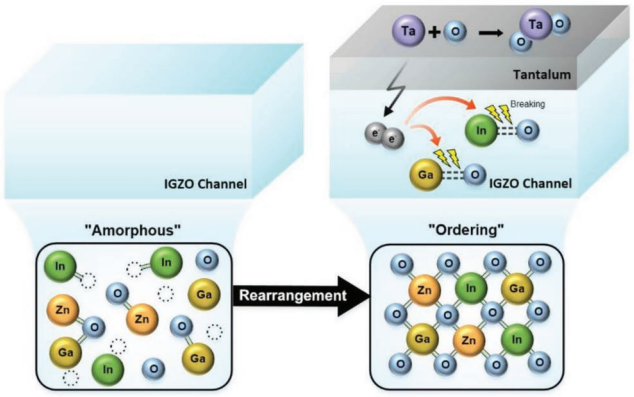


Figure 12. Illustration of crystallization mechanism induced by a metal catalytic layer (herein, a Ta metal is used). Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[74] Copyright 2017, The Authors, published by Springer Nature.

3.2. Multichannel and Gate Dielectric Layer Engineering

3.2.1. Multilayer Structure Oxide Channel

Another strategy to improve device performances of OS TFTs is to employ a heterostructure channel.[78–86] The exact mechanism providing an enhancement in device performance using heterostructure channels, even multilayer channels, is still elusive in this field. Nonetheless, formation of a 2D electron gas (2DEG) layer at the heterointerface can be the strongest rationale of the numerous explanations to elucidate this phenomenon. The 2DEG is created by electron transfer and confinement at the heterointerface originating from the CB edge offset, which is frequently observed in high electron mobility transistors (HEMTs) using III–V compound semiconductors.[87–89] However, high-quality crystalline structure is generally required to create the 2DEG in the heterojunction and superlattice devices. For this reason, the explanation using the 2DEG creation in the multilayer OS is still under debate because the high quality of the heterointerface cannot be guaranteed by the countless disordered states. This point must be clearly acquainted.

An impressive study of a heterostructure OS TFT was reported by Faber et al., where μ_{FE} was significantly improved from ≈16 (In_2O_3 only) to ≈45 $cm^2\ V^{-1}\ s^{-1}$ using an In_2O_3/ZnO heterostructure channel layer (**Figure 15**a,b).[78] It is interesting to note that the dominance of electron transport is changed from TLC (In_2O_3 only) to PC (In_2O_3/ZnO) through the heterostructure, which was demonstrated by investigating the temperature dependence of μ_{FE} (Figure 15c). This can be attributed to the synergic effects of: 1) high-quality crystallinity of the In_2O_3 front layer, 2) formation of an atomically sharp, dislocation-free In_2O_3/ZnO heterointerface, and 3) creation of a 2DEG confined in the vicinity of the heterointerface due to a discontinuity between the CB edges (Figure 15d,e). Each factor should be considered as a prerequisite to fully utilize the effects of 2DEG creation in this OS field. Seul et al. also conducted an in-depth study that demonstrated a significant improvement in device performance through the HEMT structure and showed the importance of selecting a front channel layer.[85] In general, μ_{FE} increases as the fraction of In cation increase, but V_{TH} is negatively shifted in a single In-based OS channel layer. To overcome this problem, a 3 nm thick ZnO thin film that has a wider bandgap (≈3.98 eV) than the IGO film layer (≈3.68 eV) was deposited on top of the IGO thin film as a barrier layer (**Figure 16**a,b). As a result, device performances were improved irrespective of composition of the front IGO channel layer. Among them, the one with In cation of 83 at% showed the largest enhancement with a μ_{FE} of 63.2 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.26 V dec^{-1}, and a V_{TH} of –0.84 V. Note that the 10 nm thick, single IGO TFTs with In cations of 83 at% exhibited a μ_{FE} of 41.7 $cm^2\ V^{-1}\ s^{-1}$, a SS of 0.44 V dec^{-1}, and a V_{TH} of –1.55 V (Figure 16c). This could be attributed to the effective formation of a quantum well responsible for the 2DEG-like electron path at the heterointerface because of the difference in the E_F, which can be elucidated through an energy band diagram (Figure 16d–g). Furthermore, it might be partially related to the low trap density originating from the polycrystalline structure of the corresponding IGO thin film unlike the other IGO thin films with different amounts of In cation. The ZnO/IGO TFT with In cation of 83 at% revealed improved reliability with a ΔV_{TH} of +0.58 (–0.39) V against the PBS (NBS) condition. Note that the single IGO TFT showed behaviors with ΔV_{TH} of +1.96 (–1.99) V against the identical PBS (NBS) condition. This superior reliability of the heterostructure TFT can be mainly attributed to the reduction of adverse gap states. This stems from the elevated E_F in the underlying IGO layer, which could increase the formation energy of V_O.

A superlattice structure with multiple heterointerfaces has been applied for high-performance OS TFTs,[90–93] wherein the term superlattice can be somewhat controversial due to the lack of high-quality crystallinity, but it is used for simplicity throughout this paper. Khim et al. reported a comprehensive, interesting study that showed the strong potential of a superlattice OS TFT.[94] They confirmed that the TFTs using a In_2O_3/ZnO stack exhibited device performances superior to those of IZO TFTs (**Figure 17**a), which could be attributed to incomplete decomposition of the precursors to In–O–Zn because of insufficient thermal energy (200 °C). It also showed greater device performances than a single In_2O_3 TFT, which may originate from the 2DEG formation at the heterointerface (Figure 17b,c). Furthermore, this study investigated the importance of stack sequence and the effect of the number of critical heterointerfaces, which compared two types of series (Figure 17d–g). Type-I includes In_2O_3, In_2O_3/ZnO (IZ), In_2O_3/ZnO/In_2O_3 (IZI), and In_2O_3/ZnO/In_2O_3/ZnO (IZIZ). Type-II includes ZnO, ZnO/In_2O_3 (ZI), ZnO/In_2O_3/ZnO (ZIZ), and ZnO/In_2O_3/ZnO/In_2O_3 (ZIZI). Interestingly, N_e was modulated by controlling the number of alternating In_2O_3 and ZnO layers. More importantly, the change in electrical characteristics depended on the number of heterointerface and was quite different between types I and II. As a result, the electrical characteristics are improved only when the ZnO layer is deposited on the In_2O_3 layer in both cases, whose effect stands out in the type-I devices. Consequently, the IZIZ TFT exhibited device performance with a μ_{FE} of 11.4 $cm^2\ V^{-1}\ s^{-1}$, and a V_{TH} of 8.3 V, which is superior to those (a μ_{FE} of 1.6 $cm^2\ V^{-1}\ s^{-1}$, and a V_{TH} of 13.5 V) of the single In_2O_3 TFT. To elucidate the different

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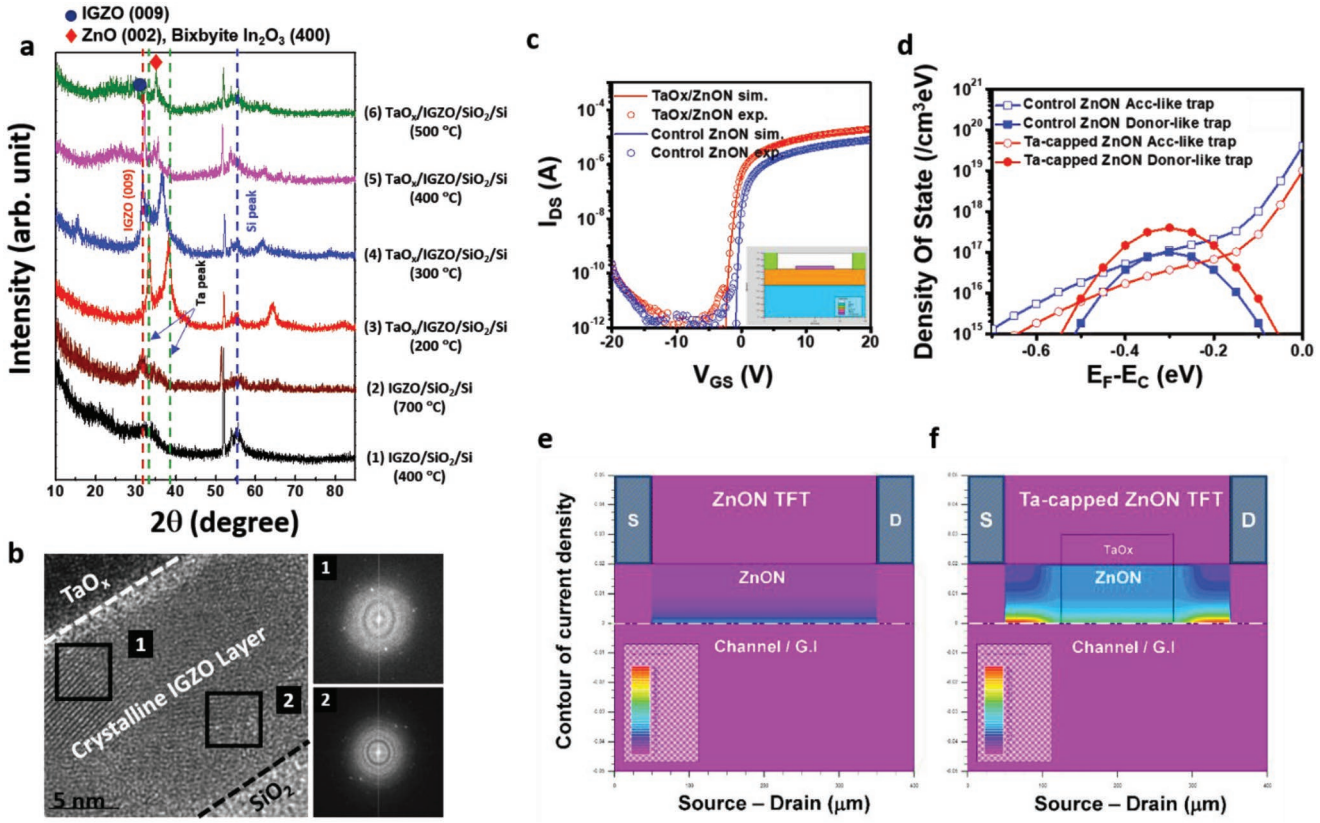


Figure 13. a) XRD patterns of IGZO thin films with different PDA conditions depending on a Ta catalytic layer. b) Cross-sectional HRTEM images and the corresponding SAED patterns of a 400 °C-annealed IGZO thin film with a Ta catalytic layer. a,b) Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[74] Copyright 2017, The Authors, published by Springer Nature. c) Simulated/measured transfer characteristics of ZnON TFTs depending on a Ta catalytic layer at V_{DS} = 0.1 V. d) Extracted DOS parameters of ZnON thin films depending on a Ta catalytic layer. e,f) Current density profiles of ZnON TFTs at V_{DS} = 0.1 V and V_{GS} = 4 V according to the position of a device with different catalytic layer conditions: e) without a Ta catalytic layer, and f) with a Ta catalytic layer. c–f) Reproduced with permission.[75] Copyright 2019, American Chemical Society.

operating characteristics, a thin-film surface was examined depending on the layer configuration, where it was confirmed that the surface roughness was significantly degraded from 0.15 (In_2O_3), 0.59 (IZ), and 0.67 (ZnO) to 1.55 nm (ZI) in the case of depositing the In_2O_3 layer on top of the ZnO layer. This poor surface quality can lead to a critical obstacle to form a 2DEG at the heterointerface, which results in deterioration of device performance. This result emphasizes the importance of stack sequence and quality of the heterointerface to implement a quasi-superlattice OS device. Another interesting study was reported, where a hybrid organic–inorganic superlattice film composed of 4-mercaptophenol (4MP) organic monolayers and ZnO nanolayers was used as a channel layer (**Figure 18**a,b).[95] Free electrons can be confined in each potential well at the ZnO nanolayer in the vicinity of heterointerfaces with the 4MP monolayer, which creates the quasi-2DEG layer (Figure 18c). Indeed, it was confirmed that N_e shows two additional peaks in the hybrid superlattice. This results in band-like transport in the superlattice channel, which leads to a considerable improvement in device performances. Consequently, μ_{FE} and SS are improved from 30 and 165 to 73 $cm^2\ V^{-1}\ s^{-1}$ and 110 mV dec^{-1}, respectively, in the hybrid superlattice TFT with triple quantum wells (Figure 18d). This result demonstrates the potential of the hybrid materials. It is worth mentioning that these hybrid TFTs show much greater device performances than pure ZnO TFTs.

In this section, several studies on basis of multilayer structure OS channels were addressed. To summarize the key points of this approach, OS TFTs with heterostructure channels generally exhibit improved device performances compared to those with single layer channels, which could be attributed to quasi-2DEG formation at the heterointerface. Here, it is quite important to select the suitable material for front and back channel layers because electron transfer and confinement should occur through the CB edge offset for the 2DEG formation. Moreover, the high-quality crystallinity of front channel and smooth heterointerface are needed to make the best use of this approach. It should also be noted that the stack sequence can be an important variable (**Table 3**).

3.2.2. High-κ Gate Dielectric for Oxide Transistors

Power efficiency in electronic devices is an increasingly important subject (refer to Section 1.2). To improve efficiency, reducing a driving voltage by scaling down the GI thickness can be the simplest step. However, the widely used silicon dioxide

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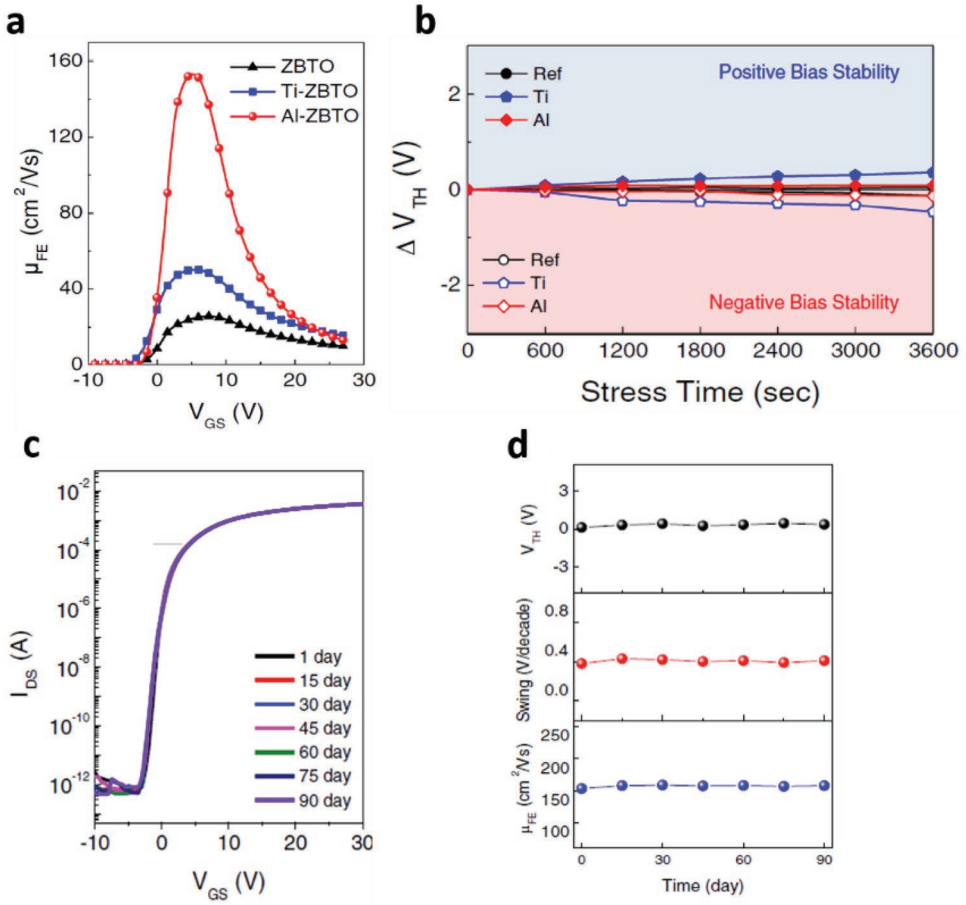


Figure 14. a) Saturation mobility plots of ZBTO TFTs with different catalytic layer conditions. b) ΔV_{TH} values of ZBTO TFTs with different catalytic layer conditions under NBS/PBS conditions. c,d) Time-dependent electrical characteristics of a ZBTO TFT with an Al catalytic layer: c) transfer characteristics, and d) electrical parameters extracted from transfer characteristics. a–c) Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[76] Copyright 2020, The Authors, published by Springer Nature.

(SiO_2) has reached its thickness limit, making it vulnerable to carrier tunneling, i.e., leakage current, such that it cannot be employed in the semiconductor industry. To overcome this issue, the conventional SiO_2 was replaced with other dielectrics which have a higher value of κ such as hafnia (HfO_2),[96–98] zirconia (ZrO_2),[99,100] alumina (Al_2O_3),[101–103] and their mixtures.[104–108] This is because the high value of κ enables equivalent oxide thickness (EOT) to be reduced to more than the physical limit of SiO_2, even if the GI's thickness becomes thicker than the previously used SiO_2 thickness. Moreover, the thicker GI effectively suppresses the leakage current due to an exponential drop in tunneling probability.[12] For this reason, Si-based devices have been improved by the way the EOT is reduced using the high-κ dielectric materials as a GI.[12,109] Likewise, OS TFTs using the high-κ GI have been frequently reported to improve device performances.[110–113] Because of the short study period, however, there is still much room to be explored in the OS TFT adopting the high-κ GI, such as the interface matching between the OS and GI, and the effect of Coulomb scattering.

Choi et al. reported the effect of PO_2 during deposition of an Al_2O_3 GI using plasma-enhanced atomic layer deposition (PEALD) on top of an IGTO thin film at a low temperature (≤150 °C), where it was confirmed that it modulated the density of both oxygen-related defects and H-related defects in the Al_2O_3 thin film.[114] An IGTO TFT using the Al_2O_3 GI with PO_2 = 2.5% exhibited a conductor-like behavior with a poor switching capability due to the high N_e value (≈10^{19} cm^{-3}) in

Table 2. Summary of oxide semiconductor TFTs with the metal capping approach.

Material	Capping layer	Process	T_{ON} [°C]	Channel dimension [W/L] [μm]	Mobility [cm^2 V^{-1} s^{-1}]	Ref.	Year
ZTO	Ta	Sputter	300	1000/300	33.5	[73#73]	2016
IGZO	Ta	Sputter	300	1000/300	54.0	[74#74]	2017
ZnON	Ta	Sputter	200	1000/300	89.4	[75#75]	2019
ZBTO	Al	Sputter	NA	800/200	153.4	[76#76]	2020

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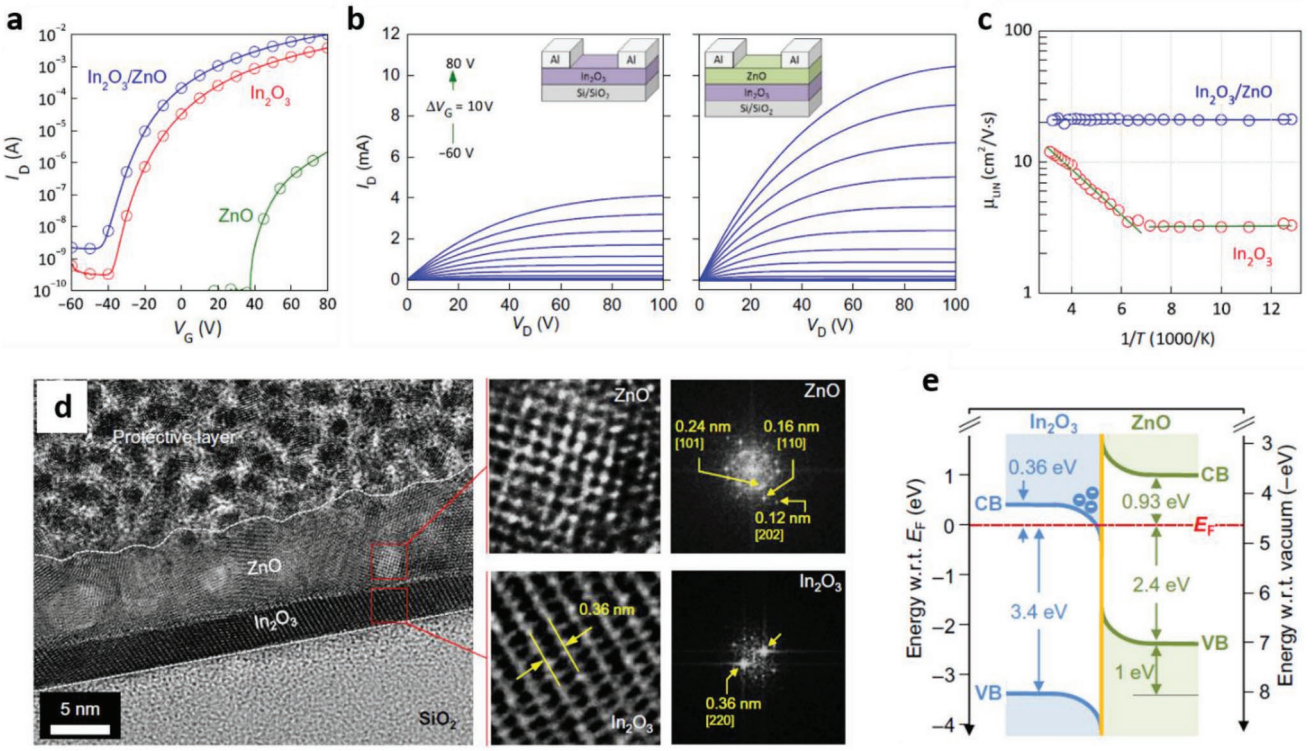


Figure 15. a) Transfer characteristics of single-layer ZnO and In_2O_3 and In_2O_3/ZnO heterojunction TFTs at V_{DS} = 100 V. b) Output characteristics of single-layer In_2O_3 and In_2O_3/ZnO heterojunction TFTs. c) Arrhenius plot of electron mobility of single-layer In_2O_3 and In_2O_3/ZnO heterojunction TFTs at $V_{GS} - V_{TH}$ = 20 V. d) Cross-sectional HRTEM images and the corresponding SAED patterns of an In_2O_3/ZnO heterojunction thin film. e) A schematic of energy band diagram of In_2O_3/ZnO heterojunction. a–e) Reproduced with permission.[78] Copyright 2017, The Authors, published by American Association for the Advancement of Science (AAAS). From ref. [78]. © The Authors, some rights reserved; exclusive licensee AAAS. Distributed under a CC BY-NC 4.0 license http://creativecommons.org/licenses/by-nc/4.0/. Reprinted with permission from AAAS.

the IGTO channel layer (**Figure 19**a). Meanwhile, an IGTO TFT using the Al_2O_3 GI with PO_2 = 5% showed outstanding device performance with a μ_{FE} of 58.5 cm^2 V^{-1} s^{-1}, a SS of 0.12 V dec^{-1}, a V_{TH} of 0.5 V, and ΔV_{TH} of +0.75 (–0.42) V against PBS (NBS) (Figure 19b). However, device performances were degraded as the PO_2 increased from 5% to 20%. In particular, the N_e in the IGTO thin film consistently decreased, which resulted in a decrease in the μ_{FE} from 58.8 to 37.8 cm^2 V^{-1} s^{-1} and a positive shift of V_{TH} from 0.5 to 6.2 V (Figure 19a–d). To understand the effect of PO_2, time-of-flight secondary ion mass spectroscopy and TDS were carried out, where the H- and oxygen-related defects consistently decreased and increased, respectively (Figure 19e–h). Consequently, the high N_e in the IGTO thin film using the Al_2O_3 with PO_2 = 2.5% originated from a number of H-related defects, and this H could be derived from either unreacted CH_3 ligands of the trimethyl-aluminum precursor or unreacted OH groups. On the other hand, the IGTO TFT with PO_2 = 20% showed a low content of H-related defects but high content of oxygen-related defects. The latter can act as acceptor-like traps in the GI thin film, degrading the device performance as PO_2 increases. This study shows the effect of point defects in the Al_2O_3 GI on the device performances of IGTO TFTs. More importantly, however, this phenomenon can be applied to other OS TFTs using a GI thin film fabricated through the (PE)ALD process at a low temperature.

Interface matching between OS channel and dielectric layer, which is another important factor to determine device performances of OS TFTs, has been researched using a high-κ GI.[115,116] Islam et al. reported the effect of Ar/O_2 plasma treatment on the interface between an IGZO channel and a Zr–Al–O (ZAO) GI.[117] After the plasma treatment, the ZAO thin films became smoother and denser than the pristine condition (**Figure 20**a–d). Furthermore, this step provided a high portion of metal–oxygen (M—O) bonds on their surface, reducing trap sites at the interface. This led to an improvement in device performance of IGZO TFTs, where μ_{FE}, SS, and V_{TH} were enhanced from 2.67, 135, and 2.25 to 15.04 cm^2 V^{-1} s^{-1}, 115 mV dec^{-1}, and 1.46 V, respectively. The importance of the interface was also demonstrated by Cho et al.[118] IGZO TFTs with a single HfO_2 GI were successfully operated in a low voltage range. However, it was also confirmed that there is a non-negligible hysteresis, which can be attributed to the mismatch of an interface between the GI and channel and/or electron trapping due to the high polarity in the HfO_2 bulk layer. To solve this problem, a 4 nm thick Al_2O_3 interfacial layer was inserted between the HfO_2 and IGZO thin films (Figure 20e,f), which improved device performances with the negligible hysteresis. Specifically, μ_{FE}, SS, and V_{TH} were enhanced from 70.5, 0.17, and –0.25 to 74.0 cm^2 V^{-1} s^{-1}, 0.13 V dec^{-1}, and 0.2 V, respectively, in the low driving range (≤2 V). Choi et al. investigated the effect of thickness of a SiO_2 interfacial layer inserted

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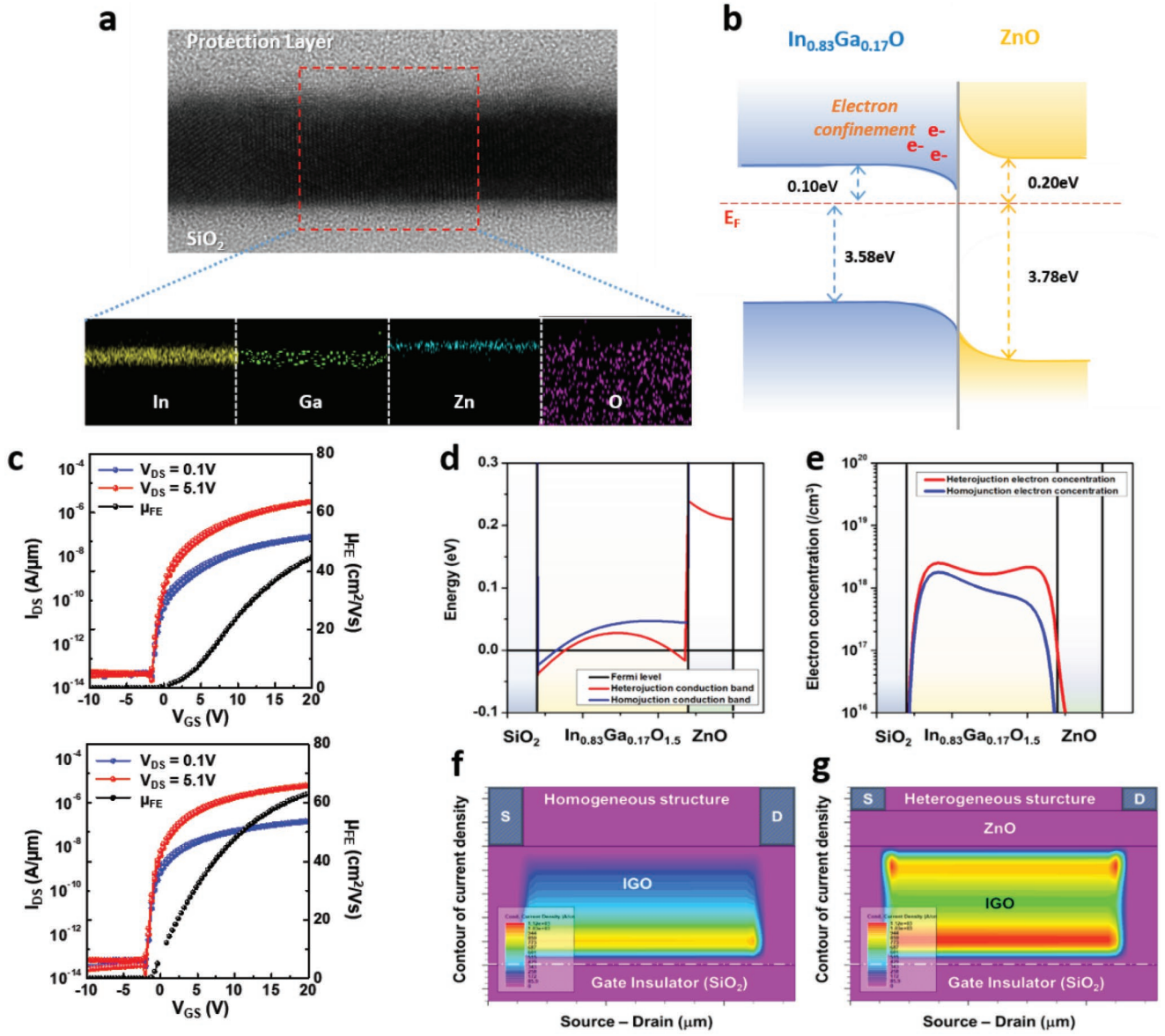


Figure 16. a) Cross-sectional scanning TEM (STEM) image and the corresponding electron-dispersive microscopy (EDS) compositional mappings of a ZnO/IGO heterojunction thin film. b) A schematic of energy band diagram of ZnO/IGO heterojunction. c) Transfer characteristics of single-layer IGO (top) and ZnO/IGO (bottom) TFTs. d,e) Simulation results along vertical direction of single-layer IGO and ZnO/IGO heterojunction TFTs: d) energy band diagrams, e) electron concentration. f,g) Current density profiles of single-layer IGO and ZnO/IGO heterojunction TFTs at $V_{GS} - V_{TH} = 10$ V and $V_{DS} = 0.1$ V: f) single-layer IGO, g) ZnO/IGO heterojunction. a–g) Reproduced with permission.[85] Copyright 2020, American Chemical Society.

between a ZrO_2 GI and IZTO on device performance.[119] Here, the stack structures of ZrO_2 and SiO_2 with various SiO_2 thicknesses were examined to achieve a highly stable, high-κ GI. As a result, a IZTO TFT using a SiO_2 interfacial layer with a critical thickness of 8 nm exhibited both a high value of κ (≈17.4) and a μ_{FE} of 27.7 cm^2 V^{-1} s^{-1}, a SS of 0.17 V dec^{-1}, and a V_{TH} of –1.1 V. Note that the critical thickness of the interfacial layer can be different depending on the GI, interfacial layer, and channel layer.

As such, OS TFTs with high-κ dielectrics have been intensively studied recently. Given that OSs are significantly vulnerable to H- or oxygen-related defects which can be easily introduced during the fabrication of the high-κ dielectrics, it should be noted that optimizing the deposition process of dielectrics, such as controlling the deposition ambience, is a primary step. Besides, this optimization procedure affects the interface quality between the dielectric and OS, which further emphasizes its importance. The interface can be also improved by using a stable, defectless interfacial layer. Here, it is noteworthy that finding its critical thickness should be simultaneously conducted in case of using the interfacial layer (**Table 4**).

3.3. Thickness and Gate Length Scaling

Electrostatic potential dictated by a gate electrode through an interjacent GI layer modulates the semiconducting channel's

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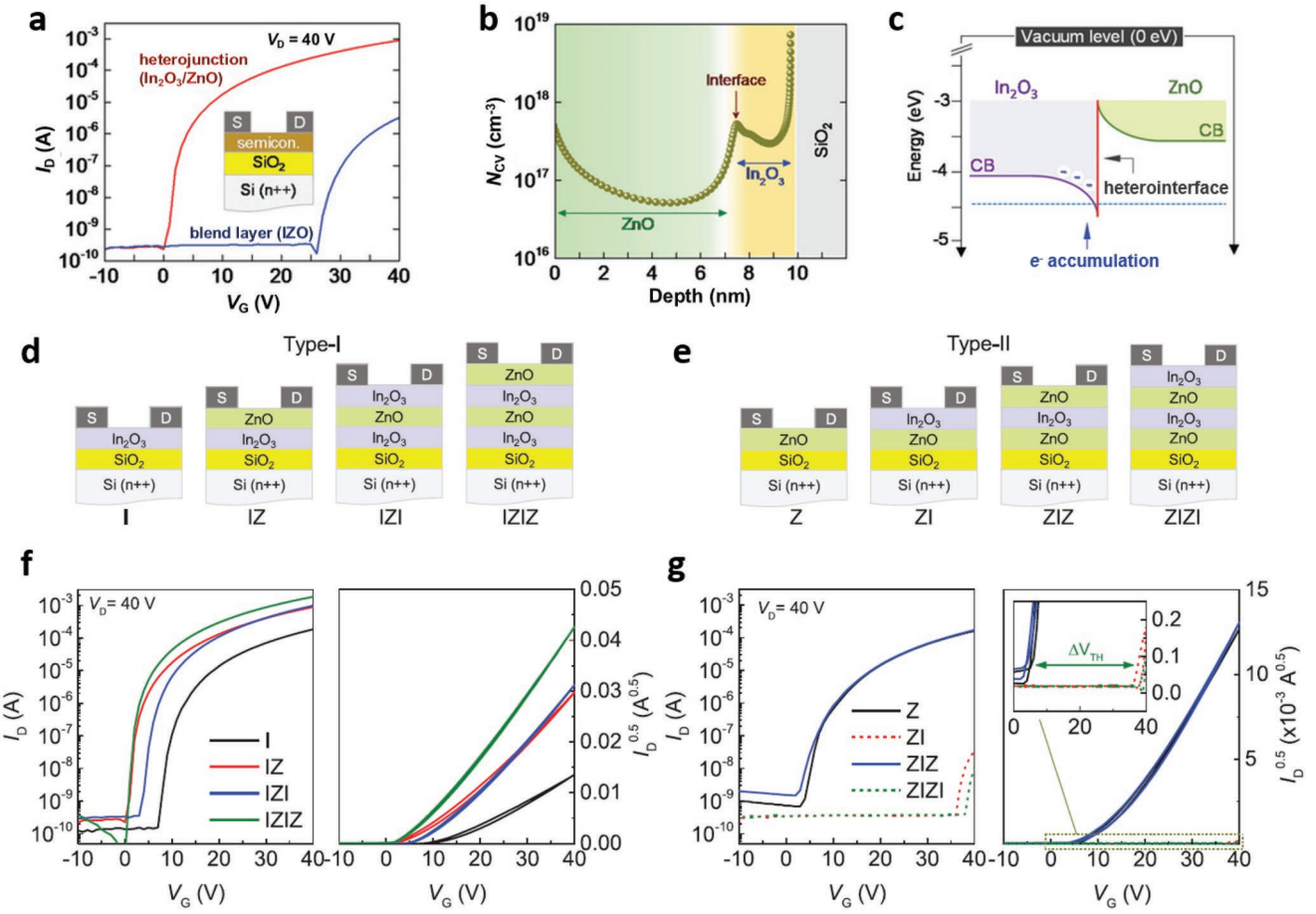


Figure 17. a) Transfer characteristics of single-layer IZO and In_2O_3/ZnO heterojunction TFTs. b) Apparent carrier density distribution as a function of depth of a In_2O_3/ZnO heterojunction thin film, which was extracted from capacitance–voltage results. c) A schematic of energy band diagram of In_2O_3/ZnO heterojunction. d,e) Schematics of devices with different superlattice types: d) type-I TFTs, and e) type-II TFTs. f,g) Transfer characteristics and the corresponding $I_D^{0.5}$–V_G plots for TFTs with different superlattice types: f) type-I TFTs, g) type-II TFTs. The inset shows zoom-in region of the plot highlighting the huge difference in threshold voltage between type-II TFTs based on odd and even number of alternating oxide layers. a–g) Reproduced with permission.[94] Copyright 2019, Wiley-VCH.

conductance in FET. Here, the potential is governed by the 1D Poisson equation in a single gate FET, as follows[12,120]

$$\frac{d^2\varphi(x)}{dx^2} - \frac{\varphi(x)}{\lambda^2} = 0 \text{ with } \lambda = \sqrt{\frac{t_b t_{OX} \varepsilon_b}{\varepsilon_{OX}}} \quad (1)$$

where $\varphi(x)$ is the potential distribution in the horizontal direction from source to drain electrodes, λ is the natural length that determines the transistor size, and t_b, t_{OX}, ε_b, and ε_{OX} are the thicknesses and dielectric constants of the body channel and GI layer, respectively. Note that the λ can also be described as follows

$$\lambda = \sqrt{\frac{\varepsilon_b}{\varepsilon_{Si}} \text{EOT} t_b} \quad (2)$$

where ε_{Si} and EOT are the dielectric constants of Si and equivalent oxide thickness, respectively. Following this rule, academic/practical endeavors have been made to decrease the EOT, which enables device scaling with a shorter L_g, whereas the gate controllability was efficiently retained. In parallel, approaches to reduce the t_b have been actively investigated to further decrease the λ. An ultrathin body (fully depleted) silicon-on-insulator (FDSOI) and FinFET are representative examples that successfully extend the Moore's law down to sub-10 nm technology nodes. Nevertheless, the endless scaling for the shorter L_g requires a further decrease in the t_b or fin width, which triggers severe hindrances including thickness variation and surface roughness.[12] These nonidealities significantly induce adverse carrier scattering, which drastically worsens with decreasing t_b, substantially degrading μ_{FE} due to carrier scattering induced by thickness fluctuation.[121–123] Moreover, a degradation of off-current (I_{OFF}) originating from direct/thermionic tunneling in the short channel Si-based devices becomes inevitable because of the small E_g (≈1.1 eV). Here, ALD-derived ultrathin OS is expected to be immune to the abovementioned challenges due to their wide E_g, atomically smooth surface, low ε_b, and high μ_{FE}. Furthermore, the ALD process (a state-of-the-art thin-film technology) has a huge advantage in terms of depositing conformal, nanometer-thick thin films through self-limiting surface reactions by sequentially introduced gaseous precursors. Simultaneously, it enables a precise composition

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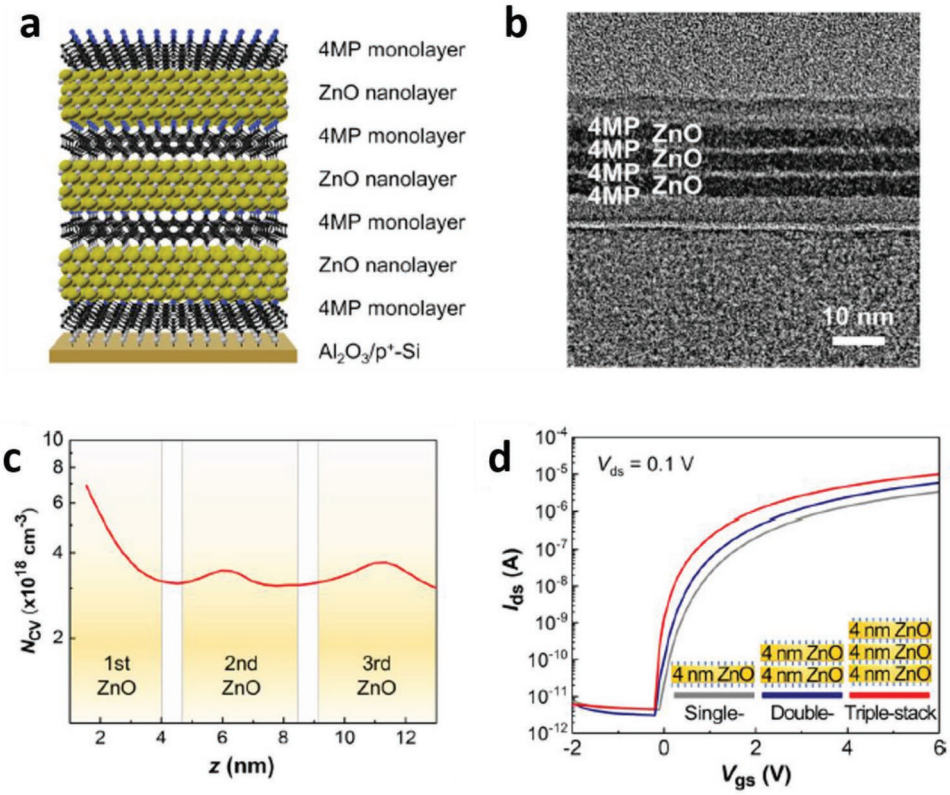


Figure 18. a) A schematic of complementary 4MP/ZnO hybrid superlattice. b) Cross-sectional HRTEM of a superlattice thin film. c) Charge-carrier distribution as a function of depth of superlattice. d) Transfer characteristics of superlattice TFTs with different number of stacks. The insets show schematics of the active superlattice channels. a–d) Reproduced with permission.[95] Copyright 2020, American Chemical Society.

control of multicomponent materials even in complex geometries including 3D-shaped devices. For this reason, numerous studies regarding the ultrathin ALD-derived OS TFTs, especially In_2O_3 or In–Sn–O (ITO) with scaled L_g, have been intensively reported recently.[124–132] In this section, studies that demonstrate a strong potential of the scaled OS for BEOL-compatible transistors fabricated by both the conventional physical vapor deposition (PVD) method and the ALD method are introduced. It should be noted that there are huge uncertainties in the commonly used figures of merit, especially for μ_{FE}, in short-channel devices.[12] Instead, an on-current density (I_{ON}) could be a more accurate, direct, and reliable parameter. Thus, it is characterized rather than the μ_{FE} in this section.

3.3.1. Scaling Effect on Binary In_2O_3 or Doped In_2O_3

An impressive study on ITO TFTs was reported by Li et al.,[124] where the effect of L_g ranging from 30 µm to 40 nm was systematically investigated by varying the t_b, t_{OX}, and ε_{OX}

Table 3. Summary of device performances of multilayer structure oxide semiconductor TFTs.

Material	Mobility [$cm^2 V^{-1} s^{-1}$]	SS [$mV dec^{-1}$]	V_{TH} [V]	Deposition method	Ref.	Year
In_2O_3–ZnO	45	–	–7.3	Solution	[78#78]	2017
a-HIZO	23	–	–	Sputter	[79#79]	2010
AlInZnO (AIZO–IZO)	1.57	590	1.28	Solution	[80#80]	2011
ZTO–ITO	43.2	180	–1.03	Sputter	[81#81]	2011
ZTO–IZO	32.3	120	0.5	Sputter	[82#82]	2014
IGZO–ITZO	22.16	300	≈0	Solution	[83#83]	2014
IGO–IGO:N	25.0	390	0.43	Sputter	[84#84]	2019
ZnO–IGO	63.2	260	–0.84	ALD	[85#85]	2020
IZIZ	11.4	430	0.83	Solution	[94#94]	2019
ZnO–4MP	73	110	–	ALD	[95#95]	2020

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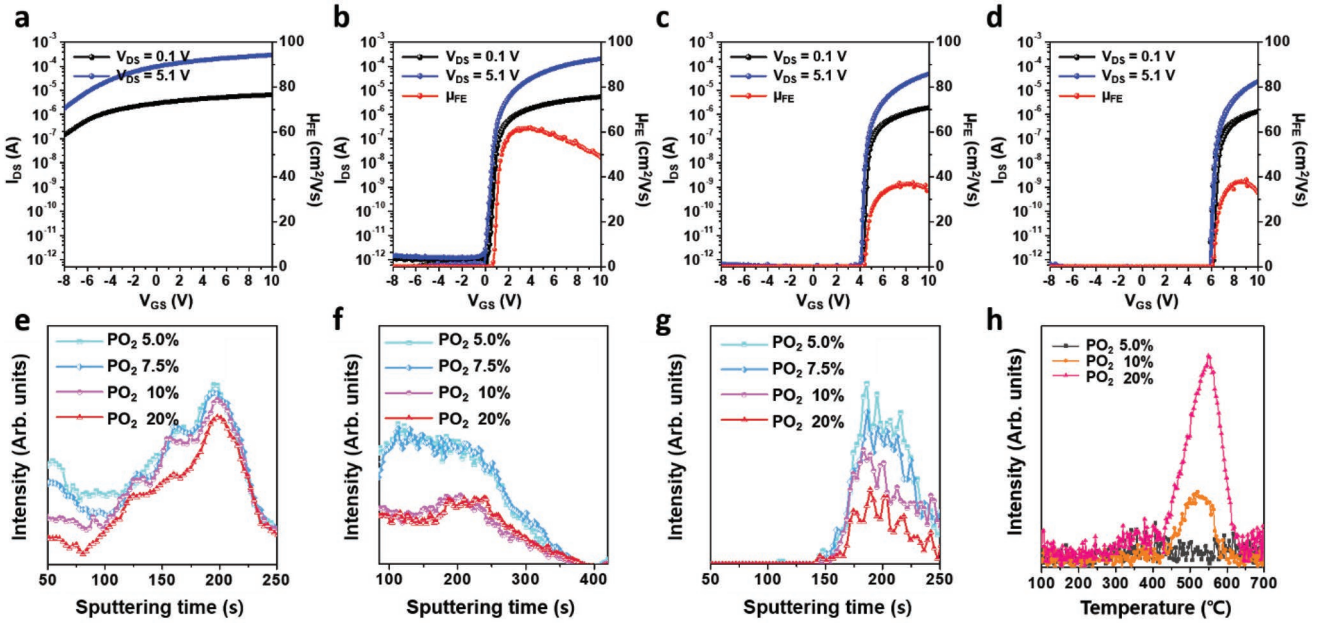


Figure 19. a–d) Transfer characteristics of IGTO TFTs with different PO_2 ratios of Al_2O_3 GI layers: a) 2.5%, b) 5.0%, c) 10%, d) 20%. e–g) SIMS depth profiles of Al_2O_3/IGTO stacks with different PO_2 ratios: e) H, f) AlOH, g) InH. h) TDS results of Al_2O_3 thin films with different PO_2 ratios. a–h) Reproduced with permission.[114] Copyright 2021, American Chemical Society.

(**Figure 21**a–f). First, the effect of decreasing (increasing) t_{OX} (ε_{OX}) in 10 nm thick ITO TFTs with L_g of 30 µm was confirmed by changing a GI layer from the conventional 90 nm thick SiO_2 to 20 nm thick La-doped HfO_2 (HfLaO). This showed improved device performances with enhancement-mode operation and better SS with a drain–source voltage (V_{DS}) of 2 V

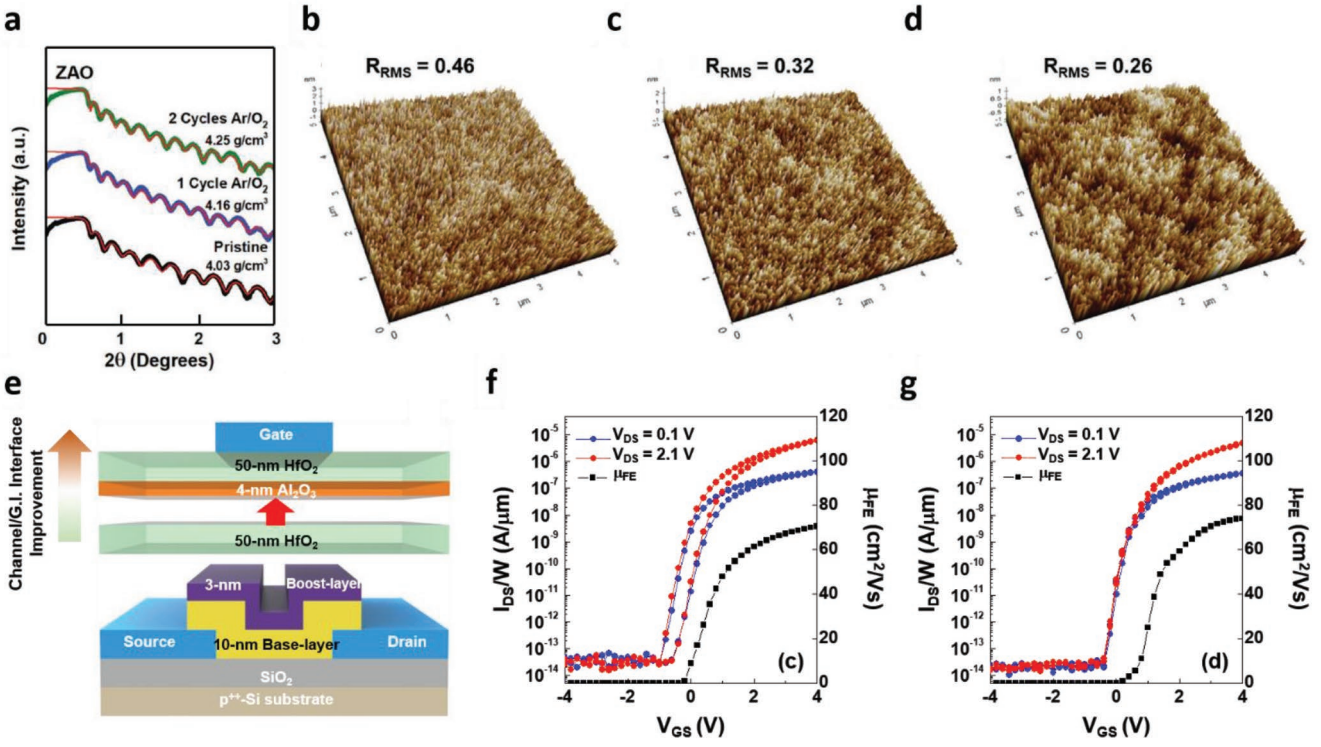


Figure 20. a) X-ray reflectivity analyses of ZAO thin films with different Ar/O_2 plasma cycles. b–d) Atomic force microscopy (AFM) images of ZAO thin films with different Ar/O_2 plasma cycles: b) pristine, c) 1 cycle, d) 2 cycles. a–d) Reproduced with permission.[117] Copyright 2021, Wiley. e) A schematic of a bilayer IGZO TFT with an Al_2O_3 (4 nm)/HfO_2 (50 nm) GI stack. f,g) Transfer characteristics of bilayer IGZO TFTs with different GI layers: f) HfO_2 (50 nm), and g) Al_2O_3 (4 nm)/HfO_2 (50 nm) stack. e–g) Reproduced with permission.[118] Copyright 2021, American Chemical Society.

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Table 4. Summary of device performances of oxide semiconductor TFTs using a high-κ GI layer.

Material [Channel/GI]	T_{OX} [nm]	Mobility [$cm^2\ V^{-1}\ s^{-1}$]	SS [mV dec^{-1}]	Driving voltage [V]	Deposition method [channel/GI]	Ref.	Year
IGZO/HfO_2	5	4.6	75	1	Sputter/ALD	[96#96]	2018
IGZO/HfO_2	20	9.7	93	2	Sputter/Sputter	[97#97]	2015
IGZO/HfO_2	20	19.7	87	4	Sputter/ALD	[98#98]	2020
In_2O_3/ZrO_2	≈18	11.3	130	2	Solution	[99#99]	2018
In_2O_3/ZrO_2	13	11.0	100	2	Solution	[100#100]	2016
IGZO/Al_2O_3	50	11.3	190	–	Sputter/ALD	[101#101]	2017
In_2O_3/Al_2O_3	15	12	150	2	Solution	[102#102]	2018
IGZO/Al_2O_3	5	6.3	100	1	Sputter/ALD	[103#103]	2018
IGZO/HfAlO	10	4.5	176	2	Sputter/ALD	[106#106]	2019
IGZO/HfAlO	30	18.4	105	4	PLD/ALD	[107#107]	2022
IGZO/$SrTa_2O_5$	100	11.1	163	5	Sputter	[112#112]	2019
IGTO/Al_2O_3	50	58.8	120	4	Sputter/PEALD	[114#114]	2021
IGZO/ZrAlO	25	15.0	115	4	Sputter/Solution	[118#117]	2021
IGZO/Al_2O_3–HfO_2	54	74	130	4	ALD	[119#118]	2021
ITZO/SiO_2–ZrO_2	98	27.7	170	10	Sputter/PEALD	[120#119]	2020

(Figure 21a,d). This improvement was attributed to enhanced electrostatic control by the reduced EOT. Then, L_g was reduced from 30 to 3 μm in the 10 nm thick ITO TFT, where it was confirmed that the V_{TH} negatively shifted by 5 V. This V_{TH} degradation was prevented by fully depleting the channel through reducing t_b from 10 to 4 nm, where the 4 nm thick ITO exhibited a positive shift by >3 V (Figure 21b,e). Finally, L_g was further reduced to 40 nm in 4 nm thick ITO TFTs with the different t_{OX} values of 20 and 5 nm, comparing the effect of SCE (Figure 21c,f). As a consequence, the resulting ITO TFT with 5 nm thick HfLaO GI (EOT = 0.8 nm) revealed outstanding device performances with a SS of 66 mV dec^{-1}, an I_{OFF} of <100 fA μm^{-1}, and a $I_{ON/OFF}$ of >10^9 at V_{DS} = 2 V. More importantly, a negligible SCE was observed, which can be ascribed to the low λ value [≈1.8 nm, refer to Equation (2)] of the ITO thin film based on a combination of the small values of EOT (0.8 nm), t_b (4 nm), and ε_b (≈4). Note that these device performances exceed those of traditional SOI. This study demonstrated the strong feasibility of the degenerated, conducting oxides such as ITO and In_2O_3 for use as ultrathin channel layers. Afterward, Si et al. implemented 1 and 2 nm thick ITO TFTs with L_g values of 800 and 600 μm, respectively, on ferroelectric Hf–Zr–O (HZO)/Al_2O_3 GI stacks.[125] The former and the latter exhibited maximum I_{ON} and μ_{FE} of 0.243 and 1.06 A mm^{-1} and 6.5 and 27.0 $cm^2\ V^{-1}\ s^{-1}$, respectively. Such outstanding electrical characteristics may come from a high 2D electron density (n_{2D}) of ≈8×10^{13} cm^{-2} (calculated). Note that this value is much higher than that (≈0.7×10^{12} cm^{-2}) from the TFTs on the conventional 90 nm thick SiO_2 GI, which might be attributed to the ferroelectric gating by the HZO/Al_2O_3 (20/3 nm) GI layer. Li et al. realized 3.5 nm thick, 10 nm long ITO TFTs having a 5 nm thick HfLaO GI layer, where remarkable device performances were obtained with an I_{ON} of 1860 μA μm^{-1} and a transconductance (G_m) of >1000 μS μm^{-1} at V_{DS} = 1 V as well as a $I_{ON/OFF}$ of >10^{10} and an I_{OFF} of ≈40 fA μm^{-1} at V_{DS} = 0.5 V (**Figure 22**a–c).[126] It should be emphasized that these electrical figures of merit obtained from the resulting ITO TFTs are much greater than other promising candidates such as black phosphorus (BP) and MoS_2, as shown in Figure 22e,f. Besides, Chakraborty et al. demonstrated 7 nm thick W-doped In_2O_3 (IWO) TFTs.[127] 100 nm long dual gate IWO TFTs exhibited outstanding device performances with a SS of 73 mV dec^{-1}, an I_{ON} of 370 μA μm^{-1}, and a $I_{ON/OFF}$ of >4×10^9 at V_{DS} = 1 V.

In addition to doped In_2O_3, ultrathin binary In_2O_3 has been intensively investigated and showed potential for advanced low power electronics.[128–132] Si et al. implemented enhancement-mode 1 nm thick, 40 nm-long In_2O_3 TFTs on a 5 nm-thick HfO_2 GI, where it was confirmed that the resulting devices exhibited a high I_{ON} of 1 A mm^{-1} at V_{DS} of 1 V.[128] These results were attributed to a high n_{2D} of 4.5×10^{13} cm^{-2} at the HfO_2/In_2O_3 interface. It should be noted that the V_{TH} and n_{2D} can be tuned by t_b as described by the trap neutral level (TNL) model.[129] To be specific, E_F tends to align to the TNL, and E_g enlargement by the quantum confinement effect shifts the location of TNL from inside to deeply below the CB edge in the decreased t_b (≤1 nm) In_2O_3. For this reason, the operation mode of In_2O_3 TFTs can be changed from depletion to enhancement mode or vice versa depending on the t_b. This physical rationale was corroborated by DFT calculations (**Figure 23**a,b). Further thickness scaling was demonstrated down to 0.7 nm thick In_2O_3 thin films, whose device performances exhibited a maximum I_{ON} of 0.5 μA μm^{-1} at V_{DS} = 4 V, a $I_{ON/OFF}$ of >10^7, and a V_{TH} of 4.9 V with an excellent immunity against the SCEs (Figure 23c,d). Si et al. also observed the impact of low-temperature PDA from 250 to 350 °C on highly scaled (≈1.2 nm) In_2O_3 TFTs with L_g down to ≈40 nm.[130] The PDA was conducted under various atmospheres including N_2, O_2, and forming gas (FG, 4% H_2), which showed a positive shift of V_{TH} and led to enhancement-mode operation regardless of the PDA environment. These results can be attributed to the reduction of V_O by excess oxygen during the PDA. Consequently, 1.5 nm thick, 40 nm long In_2O_3 TFTs annealed at 350 °C under the O_2 ambience

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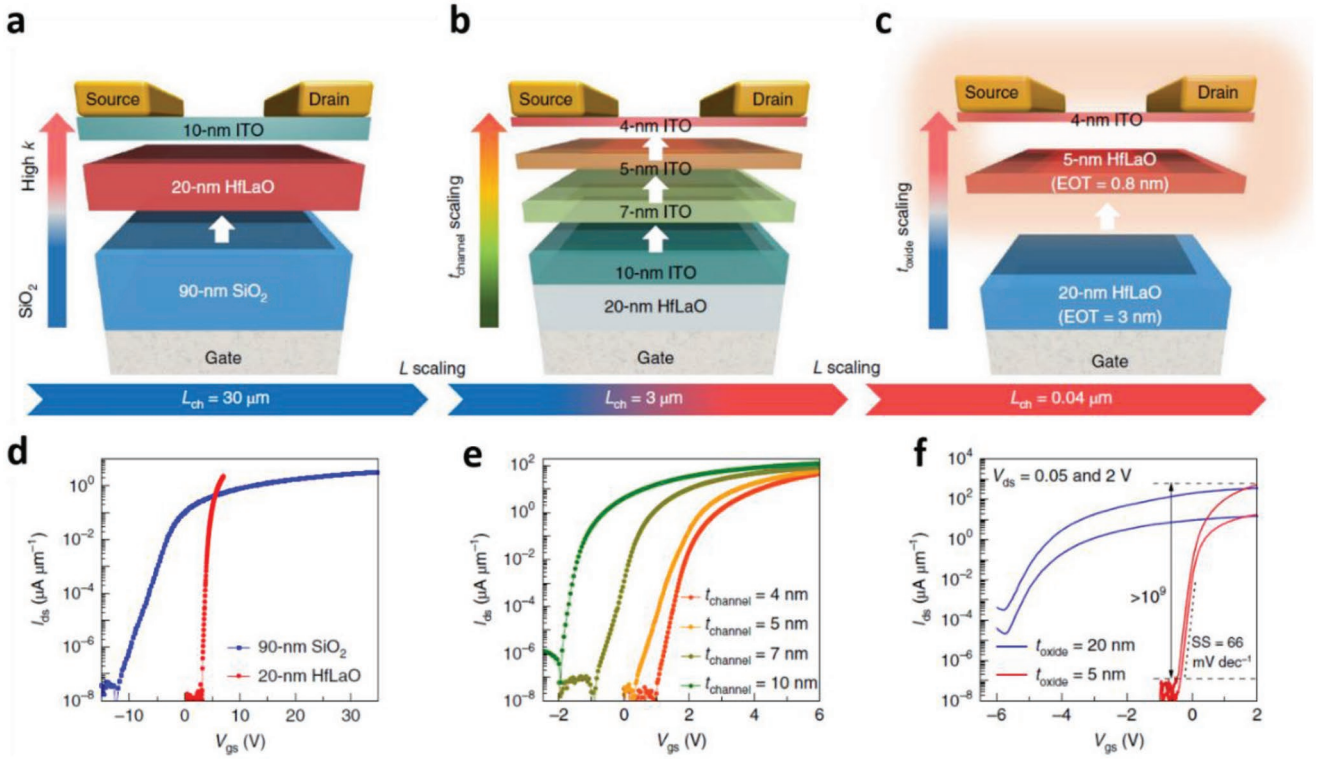


Figure 21. a–c) Schematics of ITO TFTs with different constituent conditions: a) 30 μm long, 10 nm thick ITO channel, and 20 nm thick HfLaO or 90 nm thick SiO_2 GI layers, b) 3 μm long, 4, 5, 7, or 10 nm thick ITO channel, and 20 nm thick HfLaO GI layers, c) 40 nm long, 4 nm thick ITO channel, and 5 or 20 nm thick HfLaO GI layers. d–f) Transfer characteristics corresponding to (a)/(d), (b)/(e), and (c)/(f). a–f) Reproduced with permission.[124] Copyright 2019, The Authors, published by Springer Nature.

revealed a high I_{ON} of 2.2 A mm^{-1} and a V_{TH} of 0.02 V at V_{DS} of 0.7 V (**Figure 24**a,b). More importantly, scaling metrics of the resulting TFTs with L_g from 0.8 μm to 40 nm were observed depending on the PDA temperature and ambience. Remarkably, the V_{TH} was independent of L_g in 1.2 nm thick In_2O_3 TFTs annealed at 350 °C, irrespective of the ambience. Note that it was significantly affected by L_g in the as-deposited In_2O_3 TFTs. It is also noteworthy that the In_2O_3 TFTs annealed under the FG ambience exhibited stable behaviors (Figure 24c–f), implying that the In_2O_3 could be less affected by H defects than the conventional IGZO. The effect of oxygen plasma treatment at room temperature was also examined on 1.5 nm thick In_2O_3 TFTs with L_g ranging down to 40 nm.[131] The 40 nm long devices treated for 10 min exhibited enhancement-mode operation (V_{TH} of 0.56 V) and an I_{ON} of 2.2 A mm^{-1} at V_{DS} of 0.7 V. Importantly, the oxygen plasma treatment resulted in insensitivity of the V_{TH} on L_g to the resulting TFTs. Recently, Si et al. demonstrated ultrascaled In_2O_3 TFTs with t_b and L_g values down to 0.5 and 8.0 nm, respectively, on the 3.0 nm thick HfO_2 GI with EOT of 0.84 nm.[132] It is noteworthy that the fabricated devices have W of 2.0 μm for $L_g \geq 40$ nm, whereas W of 0.6 μm was prepared for devices with $L_g < 40$ nm to minimize the self-heating effect. In this study, scaling down the t_b of In_2O_3 TFT on the 5 nm thick HfO_2 was achieved to 0.5 nm, and these device performances are shown in **Figure 25**a–c. As a result, the devices with the t_b of 0.8, 0.65, and 0.5 nm exhibited outstanding SS values of 109.0, 114.3, and 108.2 mV dec^{-1}, respectively, with $I_{ON/OFF} > 10^7$. With the t_b linearly decreasing from 0.8 to 0.5 nm, drain current was exponentially reduced by the quantum confinement on the band structure of the In_2O_3 thin film and its metrics depending on the scaling deviated from $1/L$ (Figure 25a). This indicates that the electron accumulation becomes more difficult in the long channel devices. It should be emphasized that scaling down the t_b as low as 0.5 nm for transistor applications is highly challenging because of difficulties in forming an atomically conformal, uniform semiconducting thin film, which illustrates the superb advantage of ALD methods. Note that there is no detectable drain current in In_2O_3 TFTs with t_b of 0.4 nm. Electrical characteristics of 3.5 nm thick In_2O_3 TFTs with L_g ranging from 1 μm to 50 nm were investigated depending on EOT scaling (Figure 25d). The device with L_g of 50 nm and EOT of 0.84 nm showed promising performances including a SS of 387.3 mV dec^{-1} and an I_{ON} of 2.4 A mm^{-1} at V_{DS} of 0.7 V. Nevertheless, drain-induced barrier lowering (DIBL) was observed in the TFTs, which might come from the leakage current of the highly scaled EOT. Finally, 2.5 nm thick In_2O_3 TFTs with ultrascaled L_g of 8 nm were demonstrated on top of a GI with an EOT of 0.84 nm, which exhibited the highest I_{ON} of 3.1 A mm^{-1} (Figure 25e,f). The implemented devices in this study showed highly promising electrical characteristics of ultrascaled OS TFTs compared to state-of-the-art performances of transistors using ultrathin semiconducting thin films including 2D materials (Figure 25g,h), which demonstrates the huge potential of the ALD-derived ultrathin In_2O_3 thin film for BEOL-compatible transistors for M3D integration and semiconducting channels for 3D VNAND.

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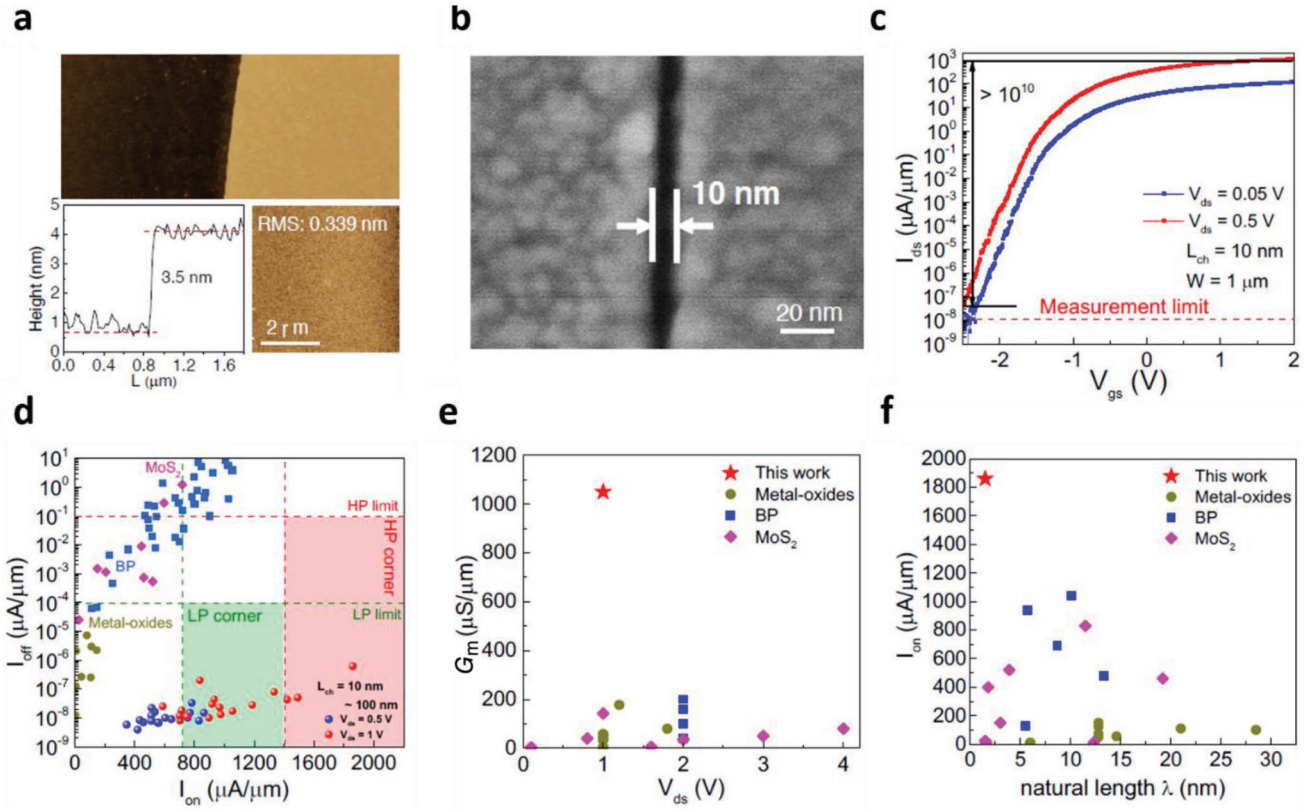


Figure 22. a) AFM image of a 3.5 nm thick ITO thin film. b) Scanning electron microscopy (SEM) image of a 10 nm-long ITO thin film. c) Transfer characteristics of a 10 nm long ITO TFT. d–f) Benchmarks of electrical parameters for metal–oxides, BP, and MoS_2 TFTs with ITO TFTs: d) I_{ON} and I_{OFF}, e) G_m, f) I_{ON} and natural length. a–f) Reproduced with permission.[126] Copyright 2020, IEEE.

3.3.2. Scaling Effect on IGZO

Many efforts to investigate the impact of thickness/length scaling in IGZO TFTs have also been made.[133–135] On et al. explored the effect of SCE and hot carrier stress (HCS) in 40 nm thick IGZO TFTs with different L_g ranging from 10 to 2 μm.[133] It was confirmed that reducing the L_g is accompanied by an increase in a negative shift of V_{TH}. Such displacement became severe when V_{DS} increased from 1 to 13 V. With PO_2 increasing from 40% to 80%, however, it decreased from 1.7 to 1.0 V (**Figure 26**a–c). This result can be attributed to a decrease in lateral diffusion (ΔL) of V_O defects from the n^+ contact region under the source/drain (S/D) electrodes, leading to an increase in an effective channel length (L_{eff}), as shown in Figure 26d–g. By contrast, HCS-induced degradation was worst in the IGZO TFT with PO_2 = 80%, leading to significant stretch-out, a positive shift of V_{TH}, and a drop in I_{ON}. Note that the devices with PO_2 = 40% or 64% showed strong immunity with the negligible V_{TH} shift. The HCS-induced degradation is strongly related to the distribution of the local electric field. It is important to note that a maximum electric field (E_{max}) is the highest in the IGZO TFT with PO_2 = 80% because of the shortest ΔL, which can create acceptor-like trap centers like O_i^{2-} defects through impact ionization on the region near n^+-drain with a relatively weak bonding nature. Thus, this prominent difference can be mainly attributed to the E_{max}. This study showed that the composition ratio should be carefully controlled to obtain excellent immunity against both the SCE and HCS in the IGZO. Han et al. demonstrated 15 nm thick, 160 nm long IGZO TFTs having a 5 nm thick HfO_2 GI layer (EOT ≈ 1.2 nm), which exhibited promising device performances with a SS of 80 mV dec^{-1}, a DIBL of 17.6 mV V^{-1}, and a G_m of 62 μS $μm^{-1}$ at V_{DS} = 2 V.[134] Given that the L_g should be larger than λ by fivefold to tenfold to be insensitive to the SCEs,[134] these excellent electrical characteristics, especially the strong immunity against the DIBL, were mainly attributed to the highly scaled EOT that allowed λ to decrease to 8 nm. Samanta et al. examined the effect of t_b in the sub-10 nm regime on the device performances of IGZO TFTs having a 10 nm thick HfO_2 GI layer.[135] There was no obvious degradation of μ_{FE} depending on t_b when scaling from 6 down to 3.6 nm in 5 μm long IGZO TFTs (**Figure 27**a), which was maintained around 32.5 cm^2 V^{-1} s^{-1}. Furthermore, the resulting 3.6 nm thick IGZO TFTs exhibited a SS of 74.4 mV dec^{-1} and a $I_{ON/OFF}$ of ≈10^7 at V_{DS} = 0.1 V. This might be attributed to the high quality IGZO thin films as well as the HfO_2/IGZO interface. It is also noteworthy that similar μ_{FE} values were obtained even at different temperatures ranging from 77 to 300 K, indicating that phonon scattering has a negligible effect on the μ_{FE} in this ultrathin IGZO thin film. Finally, IGZO TFTs with L_g of 38 nm were also demonstrated (Figure 27b,c), which showed great device performances with a $I_{ON/OFF}$ of >10^7, a G_m of ≈125 μS $μm^{-1}$ at V_{DS} of 1 V, a DIBL of 187 mV V^{-1}, and an I_{ON} of ≈350 μA $μm^{-1}$ at V_{DS} = 2.5 V (Figure 27d–f). Importantly, the SCEs were decently controlled in the ultrathin IGZO TFTs with short L_g of 38 nm, which paved the way for scaled IGZO TFTs to be realized in BEOL-compatible devices.

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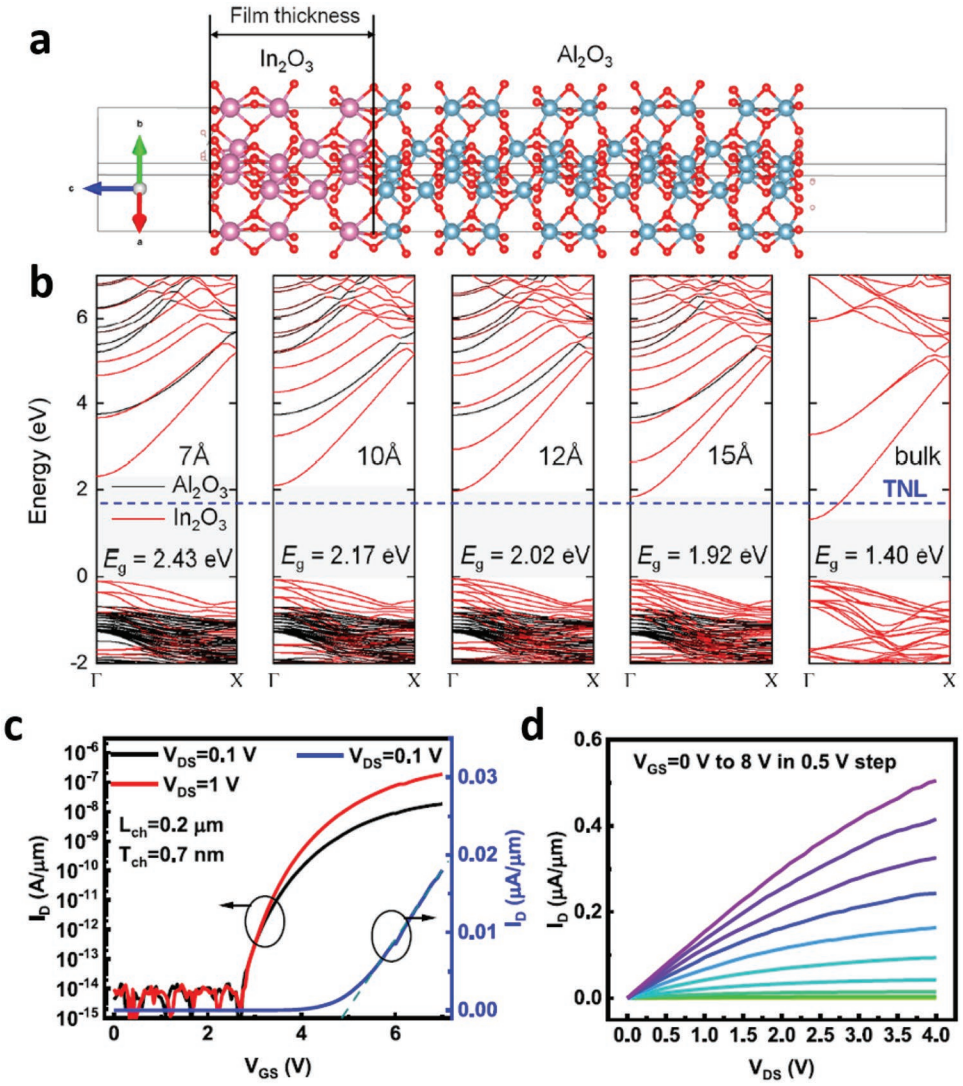


Figure 23. a) Atomic structure of In_2O_3 thin film used in DFT calculation, wherein the thickness of In_2O_3 layer is varied to simulate the quantum confinement effect. b) Calculated band structure of In_2O_3 thin films with different thicknesses of 0.7, 1.0, 1.2, 1.5 nm, and bulk along the Γ- to *X*-direction. The red and black lines indicate the band structure from In_2O_3 and Al_2O_3, respectively. The shadowed areas represent the bandgaps of In_2O_3. c,d) Electrical characteristics of an In_2O_3 TFT with a L_g of 0.2 μm and a t_b of 0.7 nm: c) transfer characteristics, d) output characteristics. a–d) Reproduced with permission.[129] Copyright 2021, American Chemical Society.

As described above, scaling L_g is closely related to EOT and t_b, which indicates that these three metrics must be simultaneously scaled to effectively endure the notorious SCEs. Here, the thickness variation, which induces considerable carrier scattering, in ultrascaled thin films can be a critical obstacle to device miniaturization. In this regard, ALD process is the most effective fabrication routes to deposit ultrathin semiconducting thin films, and OSs can be the best candidate to use this advanced technology. Owing to this significant importance, Section 3.3 focused on studies investigating the effects of miniaturization of OS TFTs. These studies have been mainly adopted highly conductive materials such as In_2O_3 and ITO, because they enable ultrathin body with a nanometer or even subnanometer thickness for transistor applications. Consequently, ultrascaled OS TFTs have greatly outstanding device performances including the promising immunity against the SCEs. It is noteworthy that the electrical characteristics of ultrathin OSs correspond or even surpass those of 2D materials. In addition to the highly conductive OSs, scaling effects of the representative IGZO TFT have also been investigated, where outstanding device performances including the immunity against the SCEs have been exhibited. Before ending this section, it needs to be emphasized that one of main factors granting the advantages in the miniaturization of OS TFTs is their intrinsic wide E_g nature (**Table 5**).

3.4. Device Structure Engineering

To obtain more efficient gate controllability in scaled devices, nonplanar structures such as FinFETs and GAAFETs have been employed in conventional Si-based devices (refer to Sections 1.2

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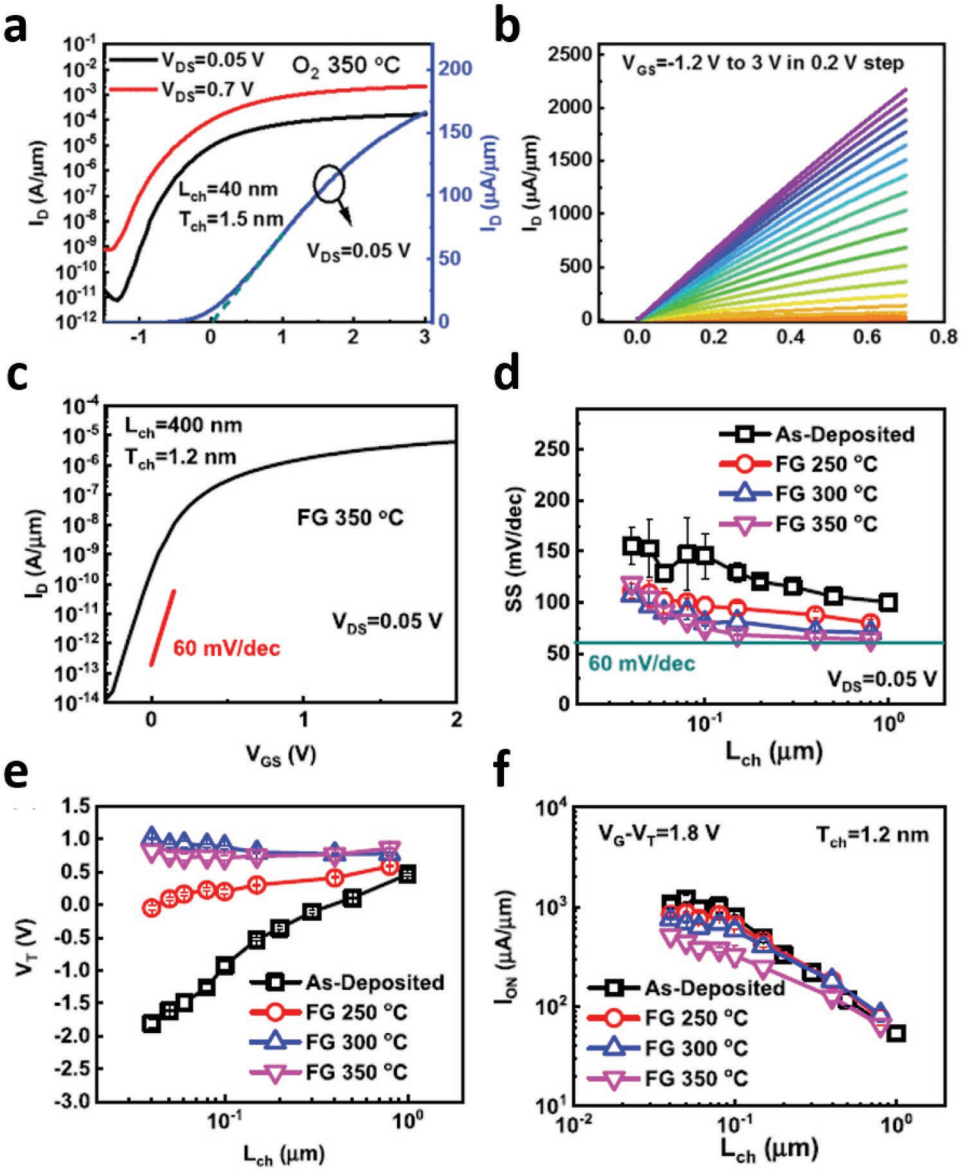


Figure 24. a,b) Electrical characteristics of a 40 nm-long, 1.5 nm-thick In_2O_3 TFT annealed at 350 °C under an O_2 ambience: a) transfer characteristics, b) output characteristics. c) Transfer characteristics of a 400 nm long, 1.2 nm-thick In_2O_3 TFT annealed at 350 °C under a FG ambience. d–f) Scaling metrics of electrical parameters of In_2O_3 TFTs: d) SS, e) V_{TH}, f) I_{ON}. a–f) Reproduced with permission.[130] Copyright 2021, IEEE.

and 3.3). Here, it is important to note that the λ in a multi-gate FET can be defined differently than a single gate FET, as follows

$$\lambda = \sqrt{\frac{t_b t_{OX} \varepsilon_b}{N \varepsilon_{OX}}} \tag{3}$$

Here, N is the effective gate number.[120] For this reason, nonplanar devices with multigate electrodes should have significant advantages in both integration density and device performance including enhanced immunity against SCEs. As such, the need to demonstrate nonplanar structure OS devices is increasing. This section reviews several studies where various kinds of vertical devices, such as step structure, fin structure, and GAA structure TFTs using OS channel layers have been implemented, and we introduced the state of the art of nonplanar structure OS devices.

3.4.1. Step Structure Oxide TFT

Baek et al. demonstrated 10 nm thick IZTO TFTs having a step structure with depth of 900 nm, where promising device performances were exhibited with a μ_{FE} of 10 cm^2 V^{-1} s^{-1}, a SS of 0.24 V dec^{-1}, a $I_{ON/OFF}$ of ≈10^8 (**Figure 28**).[72] Note that effective L and W were 4.9 and 26 μm, respectively. Given that the planar TFT with an 8 nm thick ZTO channel layer revealed device performances with a μ_{FE} of 22 cm^2 V^{-1} s^{-1} and a SS of 0.15 V dec^{-1}, the performance degradation originating from adopting a complex device structure is not severe.

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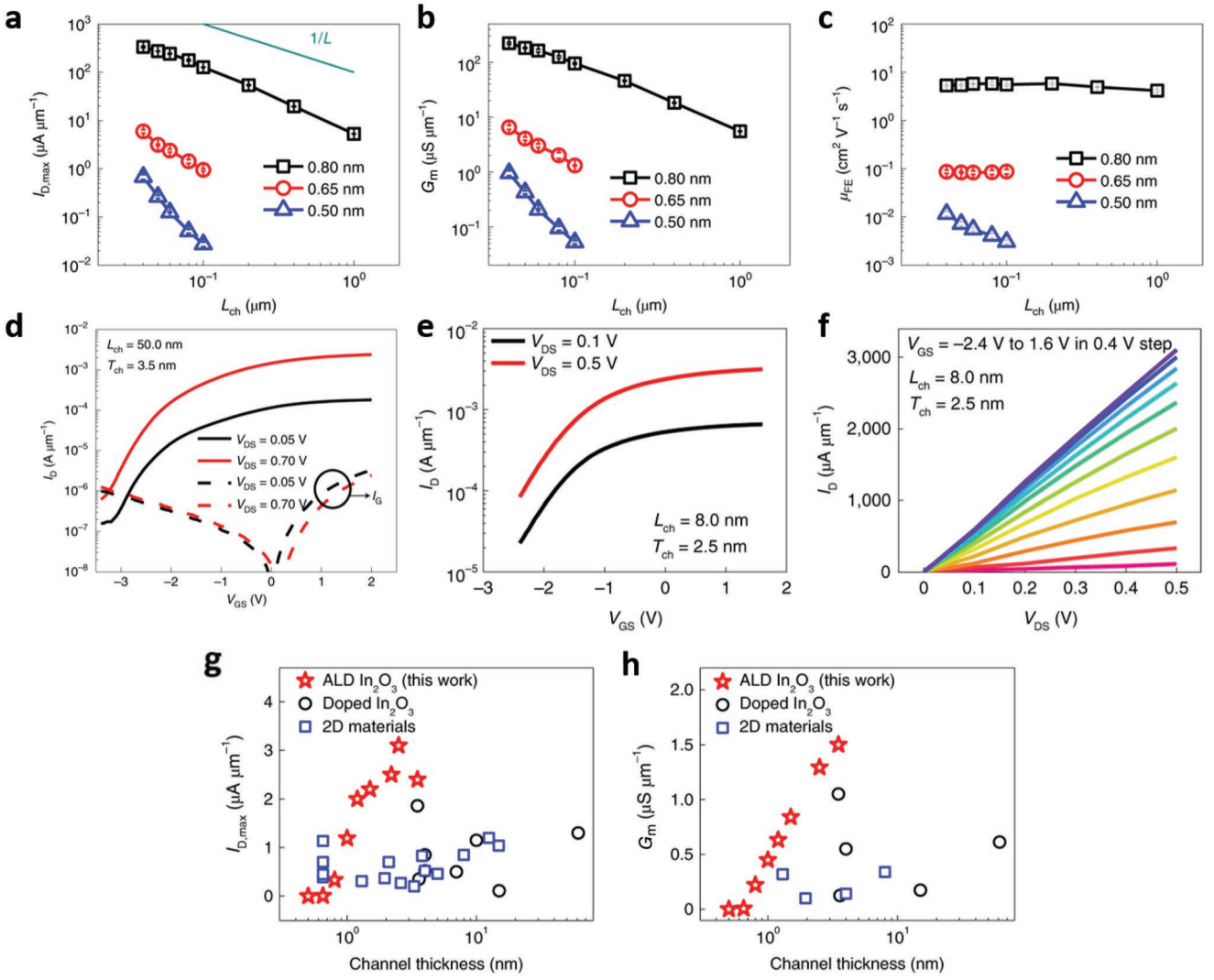


Figure 25. a–c) Scaling metrics of electrical parameters of 0.8 nm-, 0.65 nm-, and 0.5 nm-thick In_2O_3 TFTs with 5 nm-thick HfO_2 GI layers: a) $I_{D,max}$, b) G_m, c) μ_{FE}. d) Transfer characteristics of a 50 nm-long, 3.5 nm-thick In_2O_3 TFT with a 3.0 nm-thick HfO_2 GI layer (EOT of 0.84 nm). e,f) Electrical characteristics of a 8.0 nm-long, 2.5 nm-thick In_2O_3 TFT (EOT of 0.84 nm): e) transfer characteristics, f) output characteristics. g,h) Benchmarks of electrical parameters of atomically thin In_2O_3 and other ultrathin semiconductors: (g) $I_{D,max}$, (h) G_m. a–h) Reproduced with permission.[132] Copyright 2022, The Authors, published by Springer Nature.

Nevertheless, the performance deterioration, especially for the μ_{FE}, is attributed to the increased scattering by the thinner t_b of the channel layer deposited at the rougher side wall. The step structure TFT with depth of 350 nm was also implemented using a 12 nm thick IGO channel layer.[59] The resulting TFT with the effective L and W of 30 and 50 μm, respectively, exhibited a μ_{FE} of 14.28 cm^2 V^{-1} s^{-1}, a SS of 0.34 V dec^{-1}, a V_{TH} of −0.39 V, and a $I_{ON/OFF}$ of ≈10^9. Yin et al. implemented an 80 nm thick IZTO TFT having a step structure with a depth of 300 nm and an overlap area (OA) of 0.5 μm between S/D electrodes,[136] which revealed a μ_{FE} of 5.96 cm^2 V^{-1} s^{-1}, a SS of 0.21 V dec^{-1}, and a $I_{ON/OFF}$ of 3.39×10^7 (**Figure 29**a). Note that the effective L and W were 0.3 and 8 μm, respectively. More importantly, this study investigated the effect of OA on device performances of the TFT, where the OA was changed from 0.5 to 3 μm (Figure 29b). Consequently, the SS was degraded with increasing OA. The surface potential of the channel layer is governed by the electric field between the S/D electrodes as well as the gate voltage because of a thin vertical t_b and short L. Furthermore, the impact of electric field between S/D electrodes increases with increased OA, which hampers the channel modulation by the gate voltage and thus leads to degradation in device performance. This finding was corroborated through TCAD simulations (Figure 29c,d), where it was confirmed that the gate controllability on the channel layer is weakened in the step structure TFT with a longer OA. This study provides guidelines to design high-performance TFTs having a step structure.

3.4.2. Fin Structure Oxide TFT

Wu et al. fabricated crystalline IGZO transistors having finger structures with L and W of 60 and 60 nm, respectively

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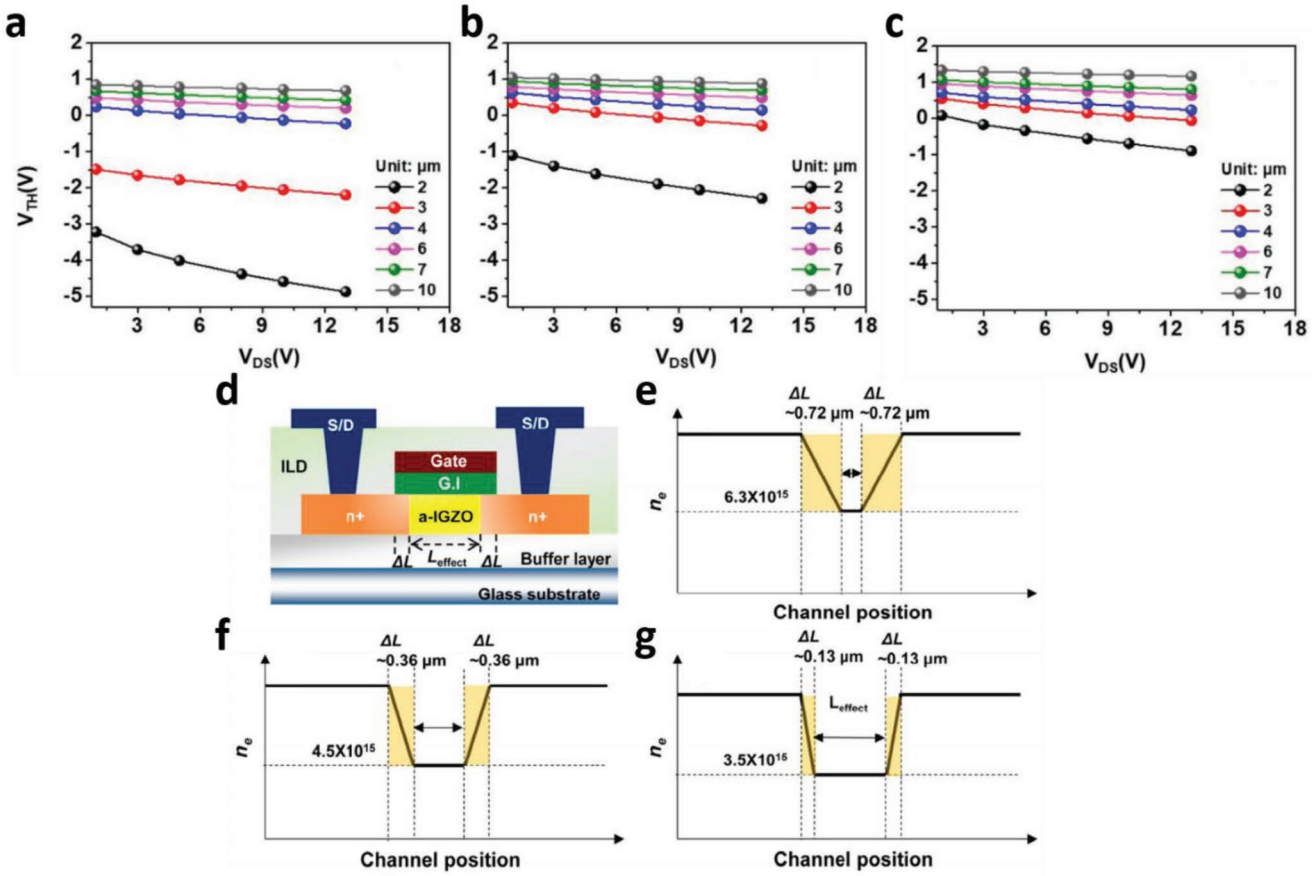


Figure 26. a–c) V_{DS}-dependent V_{TH} variations of IGZO TFTs with different PO_2: a) 40%, b) 64%, c) 80%. d) A schematic of a self-aligned IGZO TFT. e–g) Schematics of carrier distribution between source and drain electrodes in the self-aligned IGZO TFT with different PO_2: e) 40%, f) 64%, g) 80%. a–g) Reproduced with permission.[133] Copyright 2020, IEEE.

(**Figure 30**a).[137] The fabricated TFT with two fingers revealed almost twofold higher I_{ON} (19 μA) than that (9.0 μA) with a single finger (Figure 30b–d). It is also noteworthy that the TFTs had similar SS of ≈98 mV dec^{-1} and extremely low I_{OFF}, regardless of the number of fingers. Kunitake et al. implemented CAAC-IGZO fin transistors with L of 21 nm (**Figure 31**a–c),[138] where device performances were exhibited with a μ_{FE} of 10.5 cm^2 V^{-1} s^{-1}, and a SS of 87 mV dec^{-1} at V_{DS} = 1.2 V. Furthermore, the effect of back gate (V_{BG}) below the channel layer on its V_{TH} was investigated (Figure 31d,e), and it was confirmed that the V_{TH} is changed by –80 mV V^{-1} in accordance with the V_{BG}. Note that the SS and G_m are not significantly affected. Moreover, temperature-dependent electrical characteristics were observed (Figure 31f), where the V_{TH} was negatively shifted and SS was degraded with the increased temperature from –40 to 150 °C. Nevertheless, the μ_{FE} were maintained, and more importantly, the resulting TFTs showed an extremely low I_{OFF} below the measurement limit irrespective of temperature. Si et al. demonstrated 1.5 nm thick In_2O_3 3D fin transistors with L of 2 μm, and fin height/width of 180/130 nm (**Figure 32**a–d).[139] The resulting devices exhibited promising device performances with a maximum I_{ON} of 180 μA μm^{-1} (Figure 32e,f), which is about 2 times larger than that from its top-gate planar counterpart, indicating the efficient gate controllability of the fin structure. This shows the feasibility of 3D fin transistors using OS channel layers.

3.4.3. GAA Structure Oxide TFT

Fujiwara et al. demonstrated surrounding gate vertical-channel TFTs using an In–Al–Zn–O (IAZO) channel layer (**Figure 33**a),[140] where the diameter of the channel hole and L_g are ≈75 and 40 nm, respectively. The resulting TFT revealed device performances with a DIBL of 300 mV V^{-1}, a SS of 130 mV dec^{-1}, an I_{OFF} below the detection limit (≈10^{-12} A), and an I_{ON} of 5.8 μA at V_{DS} = 2 V and V_{GS} = 3 V (Figure 33b,c). Drain-bias stress instability was also investigated, which showed a negligible shift of V_{TH} for at least 10^3 s. This indicated that the floating body effect does not occur in the resulting TFT because of the junctionless structure and the lack of hole carriers and/or its very low mobility. Han et al. implemented 100 nm long IGZO nanowire (NW) TFTs with W scaled down to 20 nm (Figure 33d),[141] where a novel digital etch (DE) technique was used for realizing NW patterning. Note that the DE process etches the IGZO channel layer in a cycle-by-cycle manner with a rate of 1.5 nm per cycle. The resulting NW TFT having a 5 nm thick HfO_2 GI layer exhibited a $I_{ON/OFF}$ of

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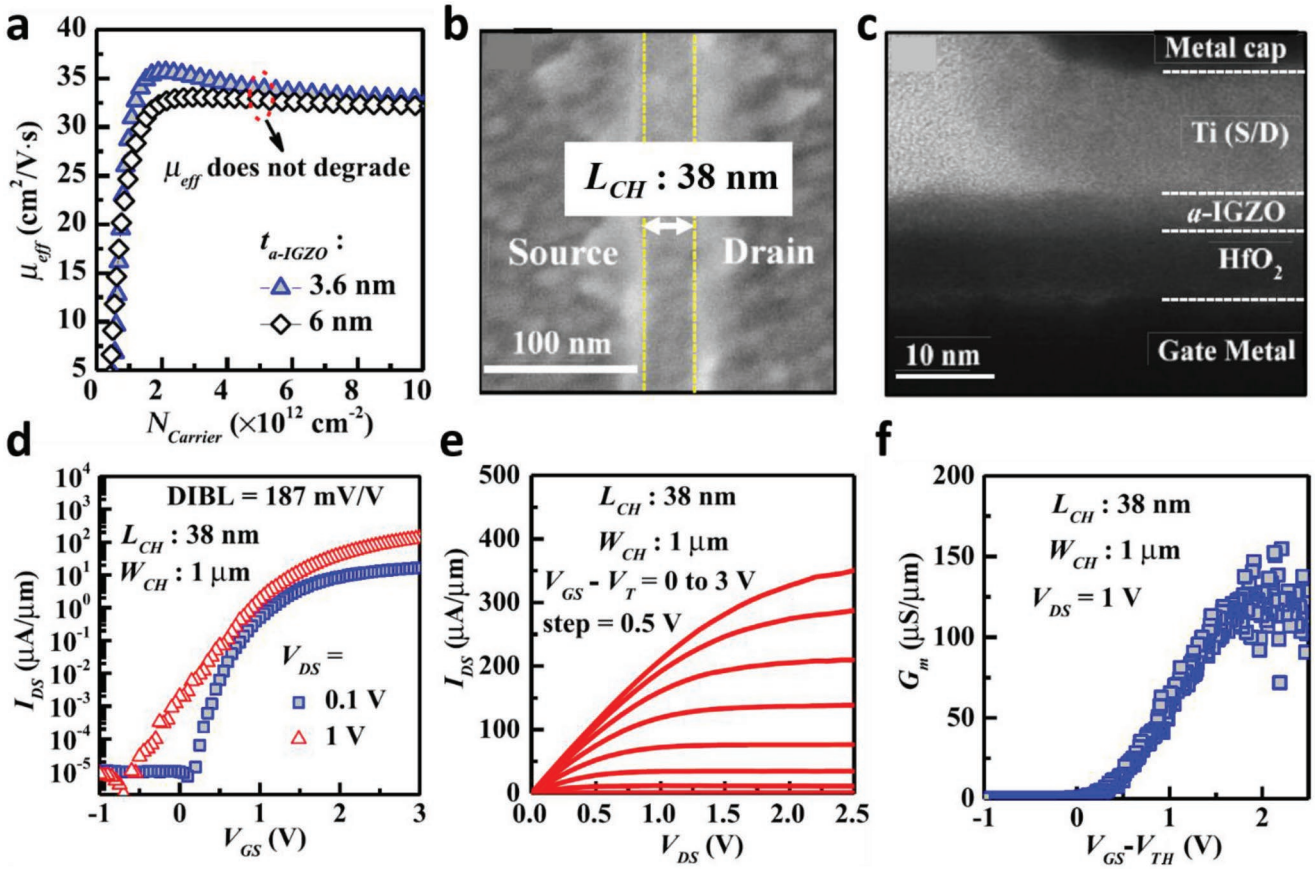


Figure 27. a) Carrier-density-dependent mobility of IGZO TFTs with different channel thicknesses. b) SEM image of a 38 nm-long IGZO thin film. c) Cross-sectional HRTEM image of a 3.6 nm-thick IGZO TFT. d,e) Electrical characteristics of a 38 nm long IGZO TFT: d) transfer characteristic, e) output characteristics. f) G_m as a function of $V_{GS} - V_{TH}$ of the corresponding IGZO TFT. a–f) Reproduced with permission.[135] Copyright 2021, IEEE.

10^8, a SS of 80 mV dec^{-1}, an I_{ON} of 620 μA μm^{-1}, and a G_m of 456 μS μm^{-1} at V_{DS} = 1 V (Figure 33e–g). Furthermore, the effect of scaling of W on device performances of IGZO NW TFTs was examined with different L values. Consequently, decreasing W resulted in a greater SS and did not significantly degrade the I_{ON} and G_m. This study successfully demonstrated the potential of ultrascaled NW TFTs using OS channel layers (**Table 6**).

Table 5. Summary of ultrascaled oxide semiconductor TFTs.

Material	Process	t_b [nm]	L_g [nm]	GI	t_{OX} [nm]	G_m [μS μm^{-1}]	I_{ON} [μA μm^{-1}]	Ref.	Year
ITO	Sputter	4	40	HfLaO	20	–	520	[125#124]	2019
ITO	Sputter	1	800	HZO/Al_2O_3	20/3	–	243	[126#125]	2020
ITO	Sputter	3.5	10	HfLaO	5	1050	1860	[127#126]	2020
IWO	Sputter	7	100	HfO_2	5	–	370	[128#127]	2020
In_2O_3	ALD	1	40	HfO_2	5	>100	200	[129#128]	2021
In_2O_3	ALD	0.7	200	HfO_2/Al_2O_3	10/1	–	0.5	[130#129]	2021
In_2O_3	ALD	1.2	40	HfO_2	5	–	220	[131#130]	2021
In_2O_3	ALD	1.5	40	HfO_2	5	940	220	[132#131]	2021
In_2O_3	ALD	2.5	8	HfO_2	3	>1000	3100	[133#132]	2022
a-IGZO	Sputter	40	2000	SiO_2	140	–	–	[134#133]	2020
a-IGZO	Sputter	15	160	HfO_2	5	62	–	[135#134]	2021
a-IGZO	Sputter	3.6	38	HfO_2	10	125	350	[136#135]	2021

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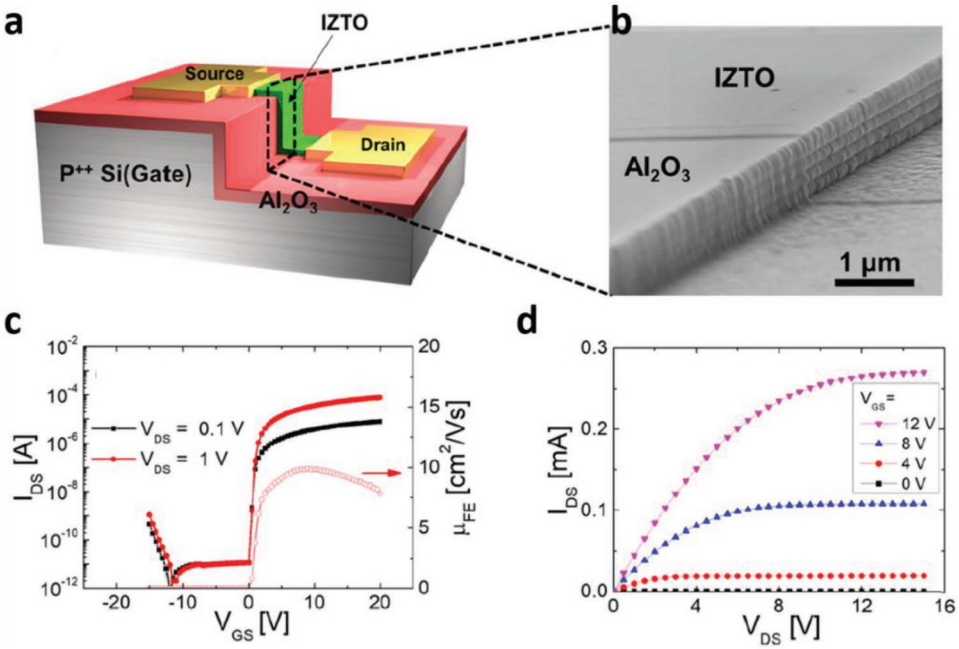


Figure 28. a) A schematic of a step structure IZTO TFT. b) SEM image of the step area. c,d) Electrical characteristics of the corresponding TFT: c) transfer characteristic, d) output characteristic. a–d) Reproduced with permission.[72] Copyright 2019, American Chemical Society.

4. Memory Applications

Many different types of storage devices for various purposes are available in current computer architectures (von Neumann computing). The central processing unit (CPU) uses registers to execute or process information. Once the register has appropriate information, the load instruction is sent to look in the cache. If this instruction cannot be found in the cache, it will fire commands for the disk drive to check DRAM and read it from there. **Figure 34**a shows the latency and cost of each

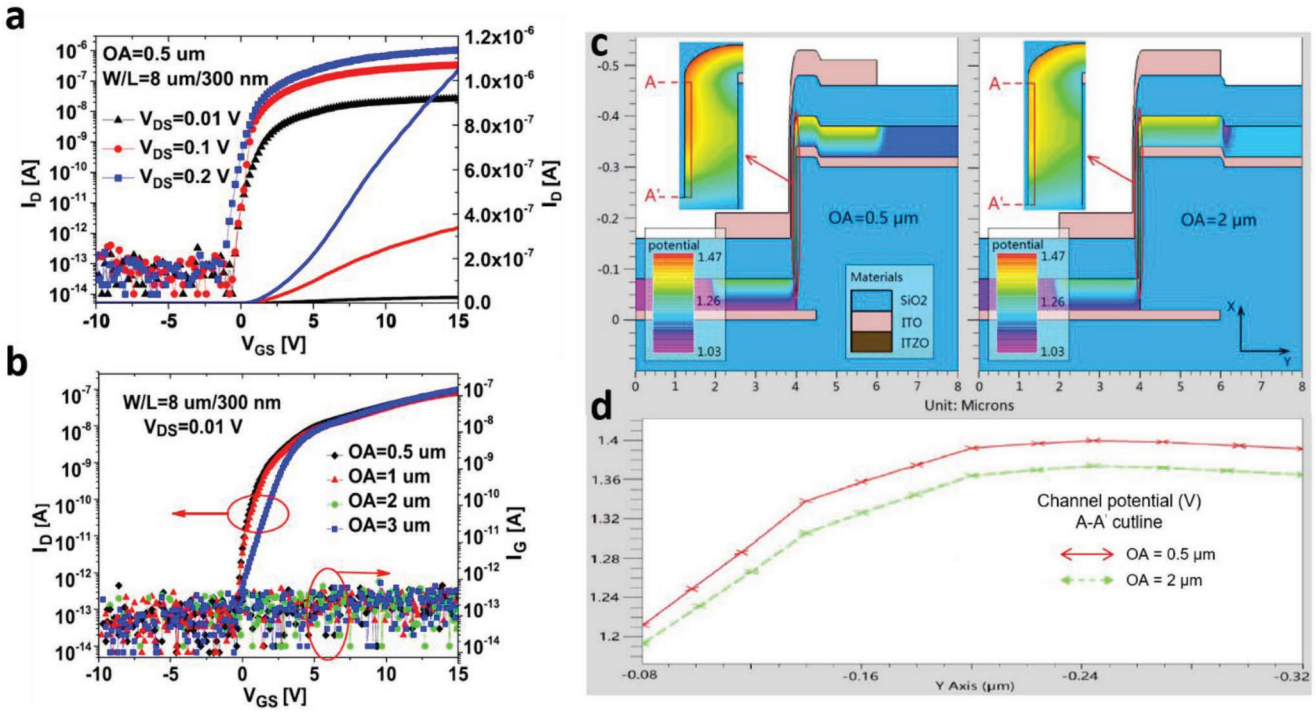
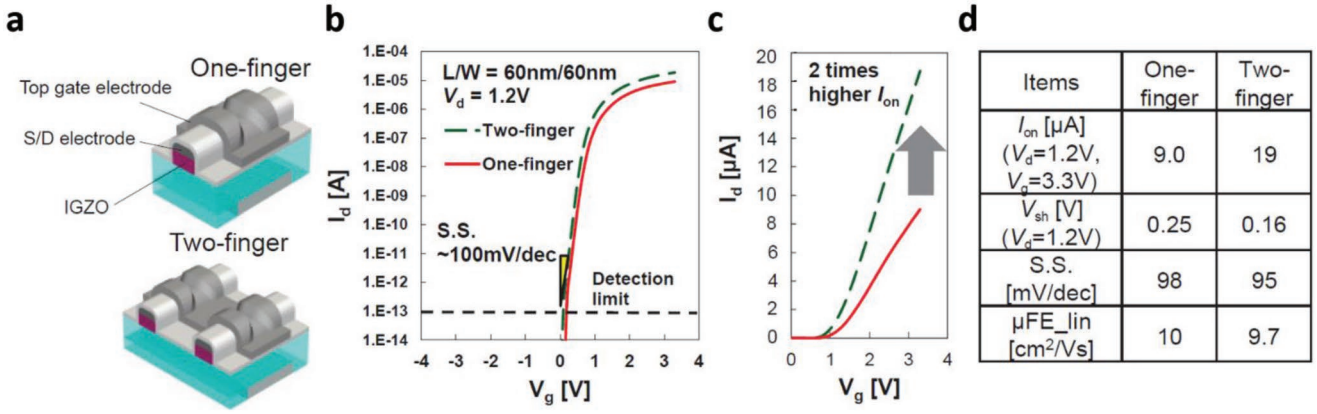


Figure 29. a) V_{DS}-dependent transfer characteristics of the step structure IZTO TFTs with OA of 0.5 µm. b) OA-dependent transfer characteristics at V_{DS} = 0.01 V. c,d) TCAD simulation for the step structure TFTs with OA of 0.5 and 2 µm: c) potential distribution in the active layer, d) potential in the line of A–A′. a–d) Reproduced with permission.[136] Copyright 2020, IEEE.

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Items	One-finger	Two-finger
I_{on} [μA] (V_d=1.2V, V_g=3.3V)	9.0	19
V_{sh} [V] (V_d=1.2V)	0.25	0.16
S.S. [mV/dec]	98	95
μFE_lin [cm²/Vs]	10	9.7

Figure 30. a) Schematics of single and two-finger structure IGZO TFTs. b) Transfer characteristics of the corresponding IGZO TFTs. c) Linear curve of (b). d) Comparison of key electrical parameters depending on the number of fingers. a–d) Reproduced with permission.[137] Copyright 2017, IEEE.

storage device. The memory hierarchy starts from the disk at the bottom, which has a remarkably large storage capacity to register a few kilobytes or megabytes. The register storage at the top level is close to the CPU. Because the wiring distance to the CPU is relatively close, the performance is higher and power can be reduced compared to other memories. A conventional 6-transistor static RAM (SRAM), the representative memory of the top level, has a large unit cell size. However, in the case of OS-based memory, the chip area can be reduced by placing it in BEOL or the size of the unit cell can be reduced by a stackable structure. Therefore, chip area and cost can be minimized.

Figure 34b compares the characteristics of a memory device including OS-based large-scale integration (LSI) with a conventional memory device including Si-based LSI and a nonvolatile memory (NVM) device including other materials.[142] Conventional memories, such as DRAM and SRAM, have high-speed writes and exhibit excellent endurance. However, data

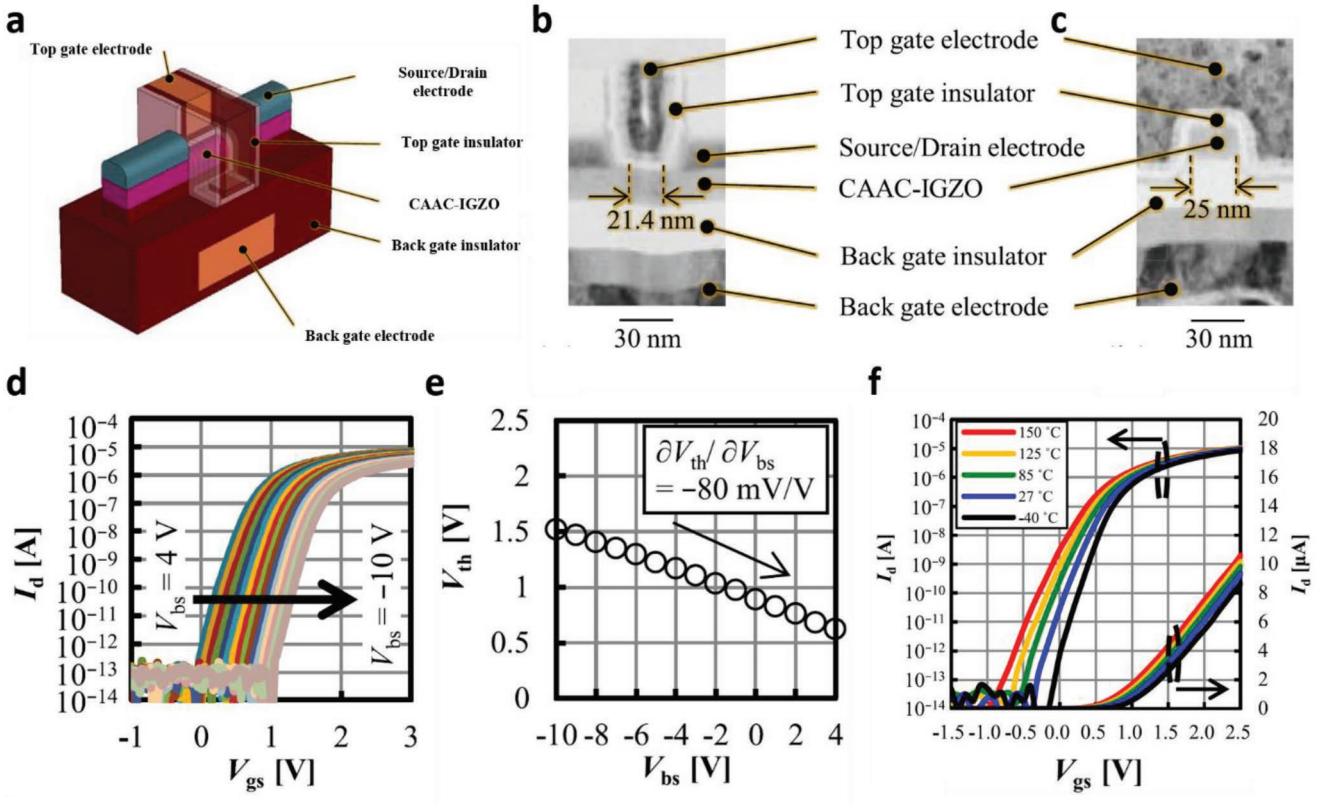


Figure 31. a) A schematic of a fin structure CAAC-IGZO transistor. b,c) Cross-sectional STEM images of a CAAC-IGZO transistor with different points of view: b) channel length direction, c) channel width direction. d) V_{BS}-dependent transfer characteristics. e) V_{BS}-dependent V_{TH} displacement. f) Temperature-dependent transfer characteristics. a–f) Reproduced with permission.[138] Copyright 2019, IEEE.

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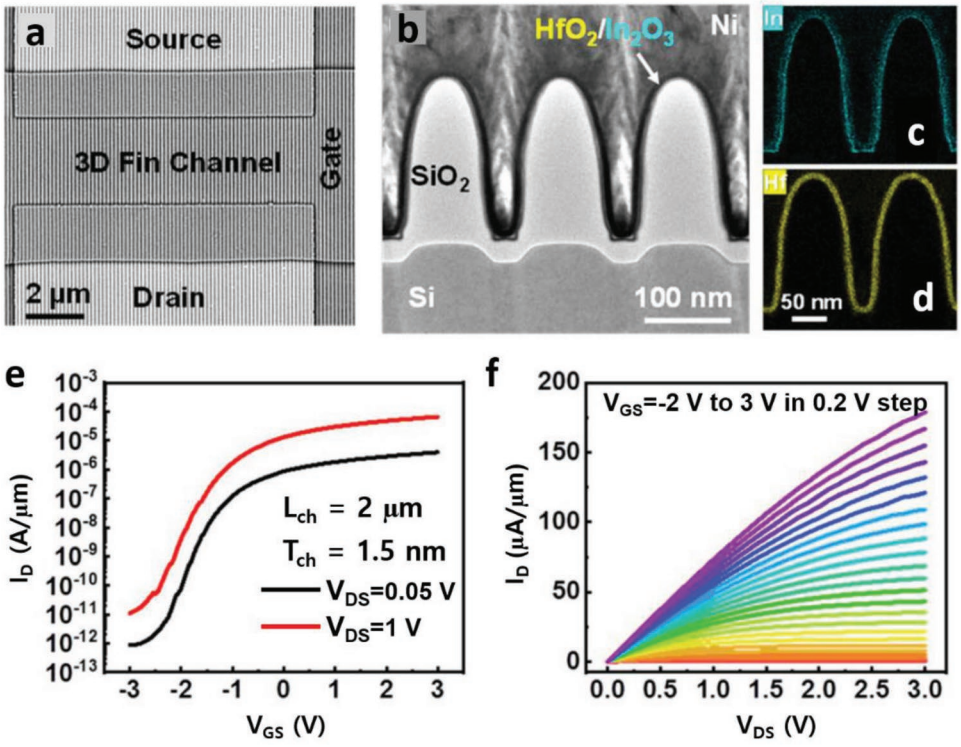


Figure 32. a) SEM image of an In_2O_3 fin transistor. b–d) Cross-sectional STEM image and EDS mappings of fin structure In_2O_3 thin film: b) HRTEM, c) In mapping, d) Hf mapping. e,f) Electrical characteristics of a 2 μm-long, 1.5 nm-thick In_2O_3 fin transistor: e) transfer characteristics, f) output characteristics. a–f) Reproduced with permission.[139] Copyright 2021, IEEE.

stored in volatile memory (VM) are lost when the device is powered off. By contrast, NOR and NAND flash memories, which are mainly used for NVM, have slow write speed, high write voltage, and limited write cycle. The proposed memory using OS like IGZO shows the advantages of VM and NVM. As previously mentioned, IGZO TFTs have an extremely low leakage current, which further helps realize VMs with a long retention time, saving the power in a standby state. NAND, a representative NVM, needs high mobility channel materials to overcome the worst-on-cell current. Ferroelectric FETs (FeFETs), another NVM, have poor write endurance, which can be improved by removing the interface layer (IL) when OS FETs are used.

4.1. Volatile Memory Applications

Constant progress in DRAM cell transistor structures, capacitors, and process densities are essential to continue scaling DRAM beyond 10 nm, as shown in Figure 34c.[8] For example, a vertical channel transistor can provide a compact unit cell area ($4F^2$). Alternatively, innovative multilayer cell structures can be developed, where bit cells are stacked on top of each other like the VNAND cell structure. However, several side effects like high subthreshold leakage current caused by the floating-body effect (FBE) should be minimized. Here, OS can be a remarkable solution to this phenomenon because the FBE is effectively suppressed due to a lack of hole carriers.

4.1.1. 1 Transistor, 1 Capacitor (1T1C) DRAM

DRAM is a VM, which makes it difficult to reduce power consumption when not being accessed, becoming a major drawback. OS FETs have a considerably low off-state leakage current. Thus, OS FETs can store the charge on the capacitors connected to these transistors for long periods, which is suitable for DRAM cell transistors. These transistors also suffer from various electric-field-induced leakages, such as gate-induced drain or junction leakage. Therefore, leakage should be suppressed strongly despite the short retention time (less than 64 ms). Thin capacitor dielectrics and high-κ materials should be developed to scale down to a sub-10 nm technology node to ensure the capacitance of cell capacitors. However, ensuring capacitance at small technology nodes is remarkably difficult.[8]

Yamazaki et al. reported CAAC-IGZO-based DRAM.[142] The peripheral circuit of this memory comprises Si FET, and an OS FET is used for 1T of the 1T1C memory cell. **Figure 35**a–c shows a cross-sectional view of the stacked structure of DRAM, an 8 kB chip microscopy image, and a memory cell schematic. OS FETs are stacked over Si FETs to reduce the number of cells per bit line, minimizing the load on each bit line and decreasing capacitance. DRAM uses OS FETs to enable a low I_{OFF} in memory cells, allowing data to be stored for long periods with a smaller capacitance than DRAM with Si memory cells. The horizontal dotted line in Figure 35d shows that three devices have achieved adequate memory characteristics of an operating frequency of more than 100 MHz and a refresh frequency of less than once h^{-1}. The cell storage capacitor only has 3.5 fF

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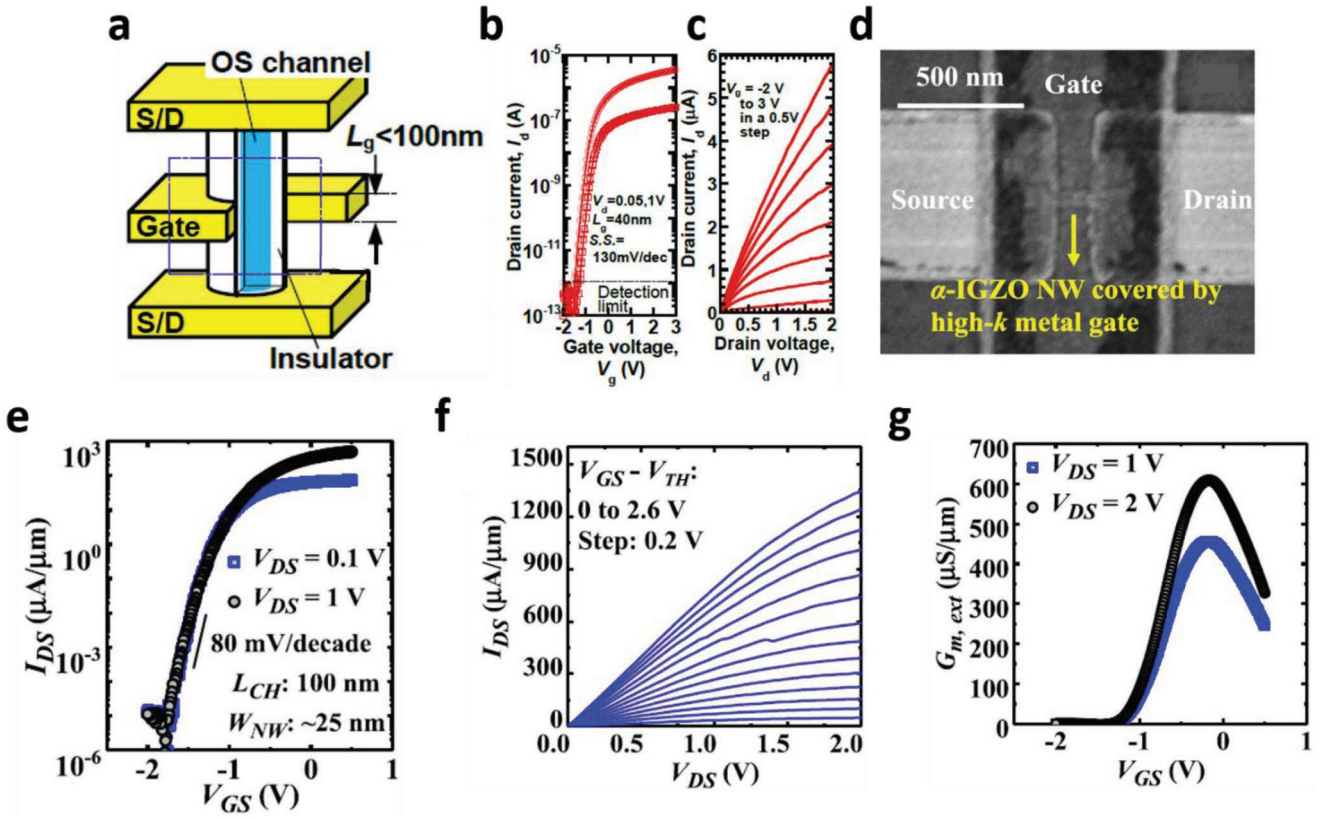


Figure 33. a) A schematic of structure of surrounding gate vertical-channel TFT. b,c) Electrical characteristics of the corresponding TFT: b) transfer characteristic, c) output characteristic. d) SEM images of IGZO NW TFT with a *W* of 20 nm. e–g) Electrical characteristics of the corresponding IGZO NW TFT: e) transfer characteristics, f) output characteristics, g) V_{GS}-dependent G_m extracted from transfer characteristics. a–c) Reproduced with permission.[140] Copyright 2020, IEEE. d–g) Reproduced with permission.[141] Copyright 2021, IEEE.

capacitance, which is substantially smaller than the currently commercialized standalone DRAM. This capacitor is expected to contribute to future DRAM scaling.

4.1.2. 1T1C DRAM in CPU

Total power consumption can be greatly reduced when IGZO-based DRAM and OS FETs are combined with the CPU. **Figure 36**a shows a significant reduction in power consumption for IoT chips by ≈94% and 70% for OS flip-flop and IGZO DRAM for memory, respectively.[137] Figure 36b shows a schematic of an analog circuit with an OS FET. OS FETs are ideal switches with a high hold time of 1 h. Thus, they can be connected to a capacitor to store a reference voltage and break the circuit, which can markedly reduce the power consumption of the voltage generator.

4.1.3. 2T DRAM

Capacitorless DRAMs are currently of interest because the storage capacitor in existing DRAM devices may limit the scalability of DRAMs. Belmonte et al. reported capacitorless DRAM cells with more than 1000 s retention time by integrating two IGZO TFTs in a two transistors and one capacitor (2T0C) configuration.[143]

Table 6. Summary of device performances of 3D structure oxide semiconductor TFTs.

Material [channel/GI]	I_{ON}	Mobility [$cm^2 V^{-1} s^{-1}$]	SS [$mV dec^{-1}$]	Structure	Ref.	Year
IZTO/Al_2O_3	–	10	240	Step	[72#72]	2019
IGO/SiO_2	–	14.28	340	Step	[59#59]	2020
IZTO/SiO_2	–	5.96	210	Step	[137#136]	2022
IGZO	19 μA	9.7	95	Fin	[138#137]	2017
CAAC-IGZO	4.4 μA	10.5	87	Fin	[139#138]	2019
In_2O_3/HfO_2	180 μA $μm^{-1}$	113	<100	Fin	[140#139]	2021
IAZO	5.8 μA	–	130	GAA	[141#140]	2020
IGZO/HfO_2	620 μA $μm^{-1}$	–	80	NW	[142#141]	2021

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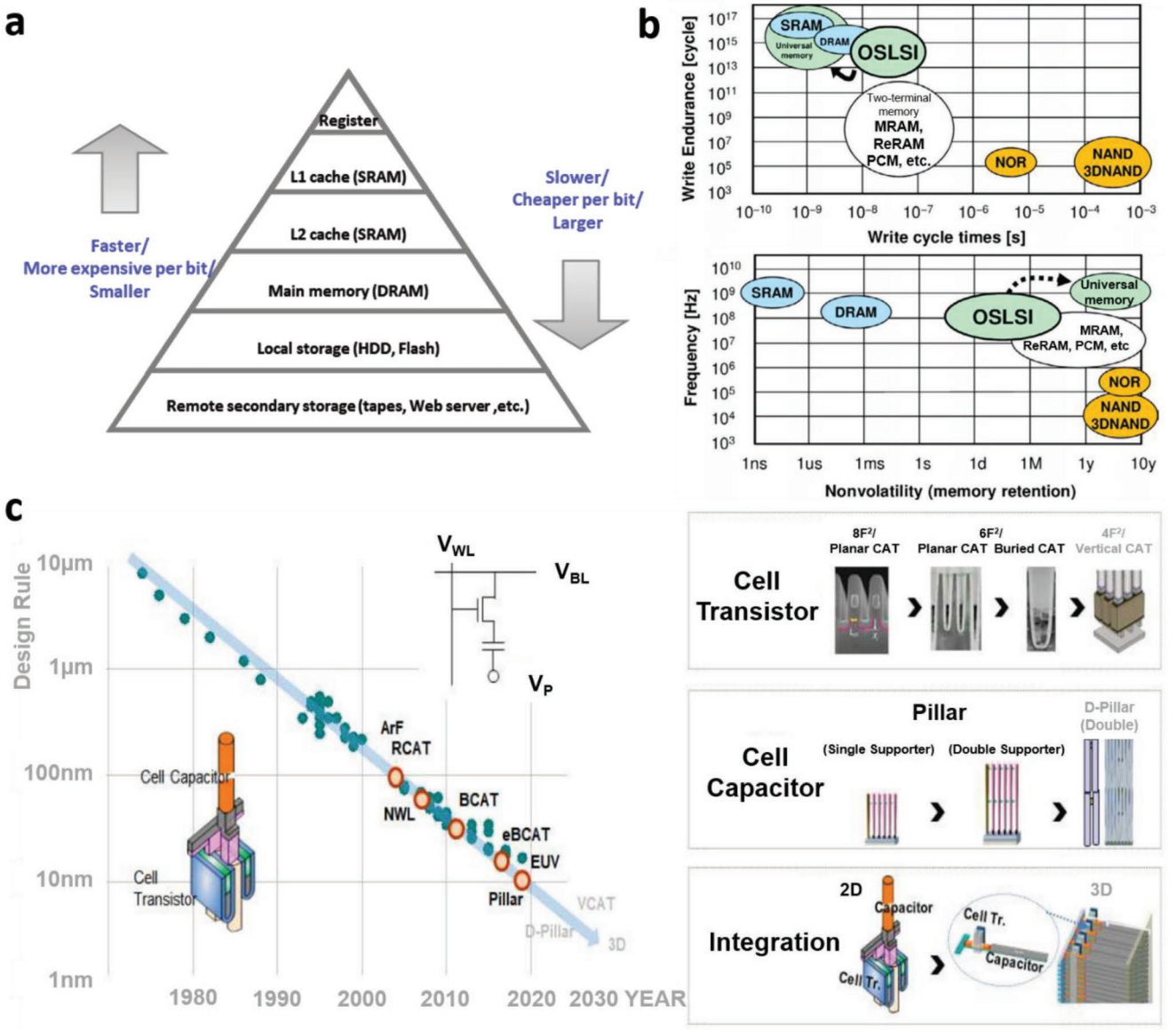


Figure 34. a) Memory hierarchy. b) Comparison of write endurance versus cycle time (top), and frequency versus nonvolatility (bottom). a,b) Reproduced with permission.[142] Copyright 2019, Wiley. c) Evolution of DRAM technology, and structure innovations. Reproduced with permission.[8] Copyright 2021, IEEE.

V_O and H can act as shallow donors in the IGZO bandgap.[144,145] The Si_3N_4 film was introduced below the IGZO channel layer, which acts as an oxygen blocker and aligns with the contacts to optimize the single IGZO-TFT performances. Meanwhile, SiO_2 is present only under the IGZO channel to enable local defect passivation without affecting the series resistance (Figure 36c). They achieved a long retention time due to the extremely low extracted IGZO TFT off-current, as shown in Figure 36d,e.

4.1.4. SRAM

Studies regarding the application of IGZO to SRAM, which requires the fastest speed, are currently available. Although the mobility of IGZO is smaller than that of Si and the formation of a p-channel FET is difficult,[142] the layout area can be substantially reduced compared with conventional SRAM. Hybrid inverters can be constructed by p-channel poly-Si and n-channel amorphous IGZO transistors.[146] Chang et al. demonstrated CMOS inverters and 6T-SRAM with n-type IGZO and p-type poly-Si channels, as shown in Figure 36f.[147] P-channel FET is implemented with poly-Si to overcome the current differences with a p-type Si transistor, and ≈628 mV static noise margin of SRAM can be achieved (Figure 36g). This finding is important in that Si and IGZO transistors can be fabricated simultaneously despite the challenges associated with fabrication temperature, H treatment, and contamination.

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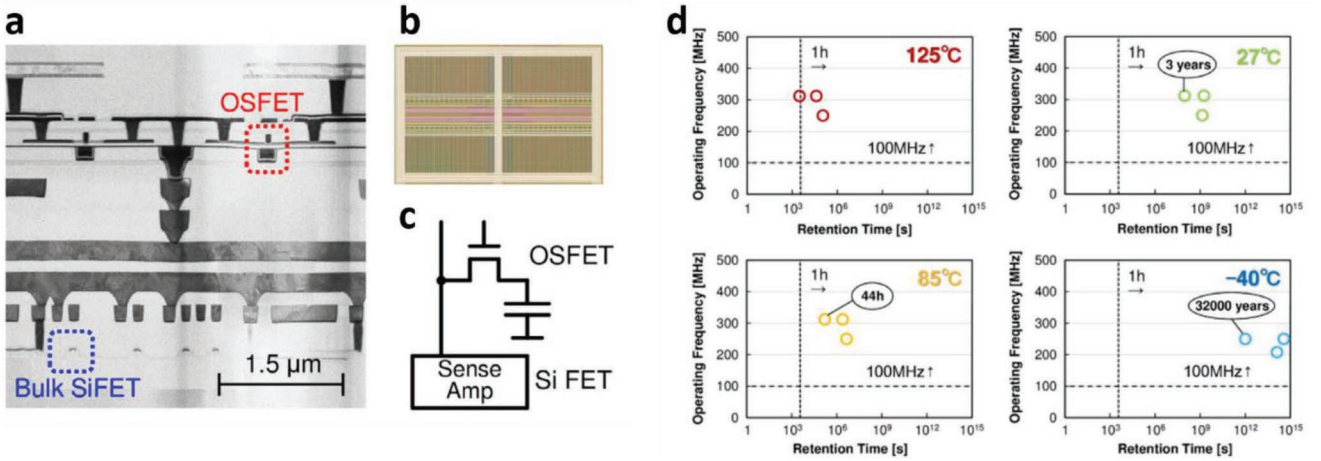


Figure 35. a) Cross-sectional SEM image of dynamic OS RAM (DOSRAM) with stacked OSFET and Si FET. b) Chip microscopy image. c) Circuit diagram of memory cell. d) Estimation results of operating frequency and retention time of nonvolatile OS RAM at different temperatures. a–d) Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[142] Copyright 2019, The Authors International Journal of Ceramic Engineering & Science published by Wiley Periodicals, Inc. on behalf of American Ceramic Society.

4.2. Nonvolatile Memory Applications

VNAND is being developed as a representative structure of NVM. The VNAND structure is successfully undergoing revolutionary development from 2D to 3D.[8] Thus, the state-of-the-art VNAND is a 3D structure. However, VNAND should be called 4D because it has one additional dimension, that is, each cell has multiple bits. Therefore, future VNAND may

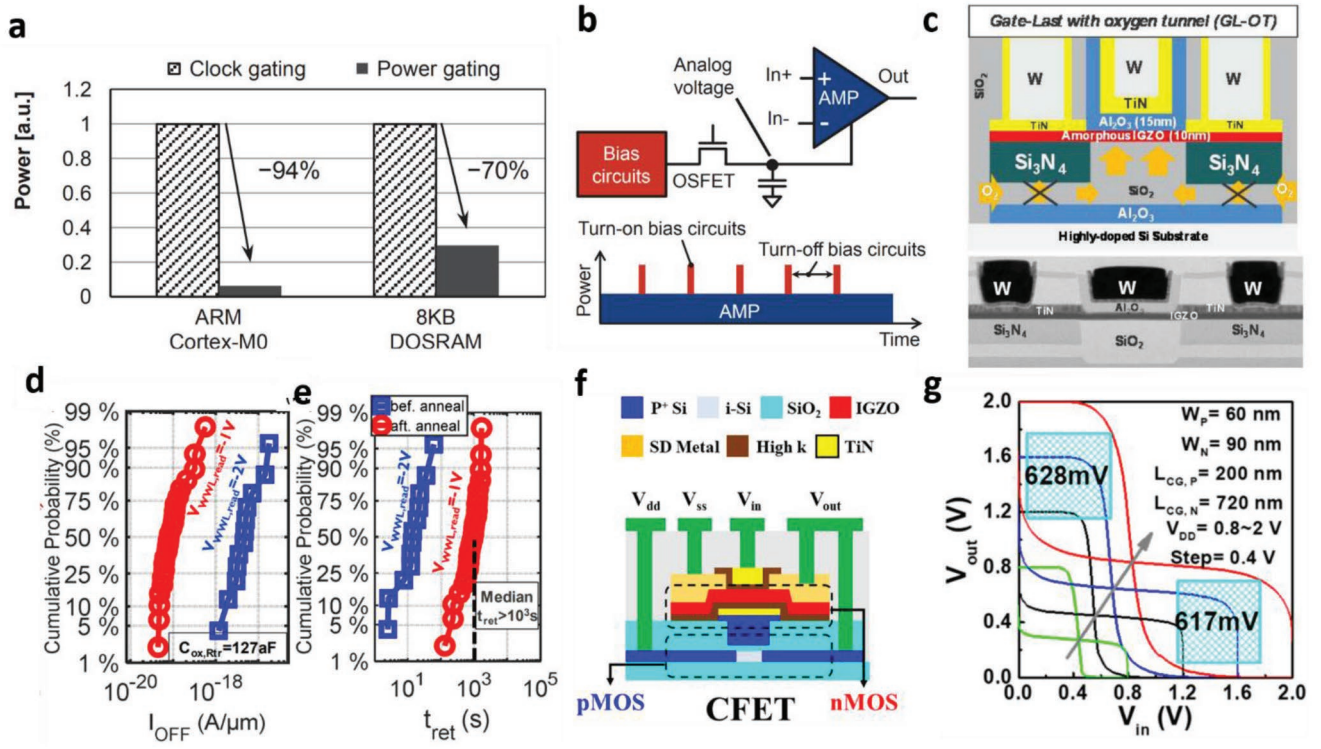


Figure 36. a) Power consumption for 100 MHz normally off CPU with clock gating and power gating with the assumption of sleep/active ratio = 1000:1, which shows it can be drastically reduced by 94% with OS flip-flop for ARM Cortex-M0 and 70% with DOSRAM for memory. b) Conceptual illustration of an analog circuit with an OSFET. a,b) Reproduced with permission.[137] Copyright 2017, IEEE. c) A schematic of the gate-last configuration with oxygen tunnel (top) and cross-sectional TEM image of one of the transistors implemented in the 2T0C configuration. d,e) Obtained electrical parameters: d) I_{OFF}, e) retention time. c–e) Reproduced with permission.[143] Copyright 2021, IEEE. f) Illustration of separate contacts connected to the top (nMOS) and bottom (pMOS) transistor source and drain electrodes. g) Voltage transfer characteristics from the 6T-SRAM with the n-type IGZO and p-type poly-Si channels. The inset shows noise margins at different V_{DD} values. f,g) Reproduced with permission.[147] Copyright 2022, IEEE.

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not deviate significantly from the current structure, which has multibit cells in a 3D vertical structure. Thus, solving the problem originating from an increase in the number of stacks is necessary, and OS can be proposed as one of the solutions.

4.2.1. VNAND

Polysilicon has been developed as a VNAND channel because it is suitable for high-temperature processes (>800 °C) and is most suitable for conventional processes. However, polysilicon presents numerous challenges, including reduced operating current, increased leakage current, and increased V_{TH} nonuniformity due to the presence of adverse grain boundaries.[148] Numerous alternative channel materials have been studied to solve the problem of polysilicon. In recent years, OS has been proposed as a possible candidate because its wide bandgap produces the desirable low leakage currents. Hwang reported a bilayer stacked integrated charge trap flash (CTF) in which $ZnSnO_3$ channels are grown by metal–organic chemical vapor deposition, as shown in **Figure 37**a.[149] Top and bottom CTF devices showed viable memory performance in the drain current, a gate voltage sweep mode with a sufficiently high memory window, and stability even at 85 °C with a guaranteed 10 years retention time. However, the estimated program times in pulse program/erase mode were impractically long (order of seconds), suggesting that significant improvements in material stacking or process conditions are necessary. Hole erase (the only erase method used in vertical channel NAND flash architectures) cannot be used because OS FETs have remarkably low hole concentrations and a large bandgap (E_G > 3 eV). Choi et al. proposed an IGZO–Oxide–P-filler (IOP) structure to overcome these problems and improve the poor erase performance of a VNAND flash structure using an IGZO channel, as shown in Figure 37b–d.[148] An IOP structure using a field-erasing method rather than a hole-erasing method was proposed. Though it is only a simulation, the erasing performance can be improved in accordance with the thickness of the macaroni oxide. These results indicate that IGZO can overcome the poor erase performance (due to poor hole concentration) while maintaining the desirable properties of IGZO.

4.2.2. 2T1C NVM

Oota et al. proposed a 2T1C structure nonvolatile oxide semiconductor RAM (NOSRAM) as a general-purpose memory (**Figure 38**a).[150] This structure is similar to that of 2T DRAM.[143] The retention time is relatively longer compared to 2T0C IGZO DRAM due to the additional capacitor, and a multibit structure can be implemented (3 bits per cell, as shown in Figure 38b). NOSRAM modules consist of OS FET and Si FET stacks. Damage caused by charge injection degrades the gate insulating film because the charge is injected and ejected from the floating gate through the gate insulating film in a conventional floating gate NVM, limiting the number of write cycles. A NOSRAM does not require charge injection through the gate insulating film. Charging and discharging of

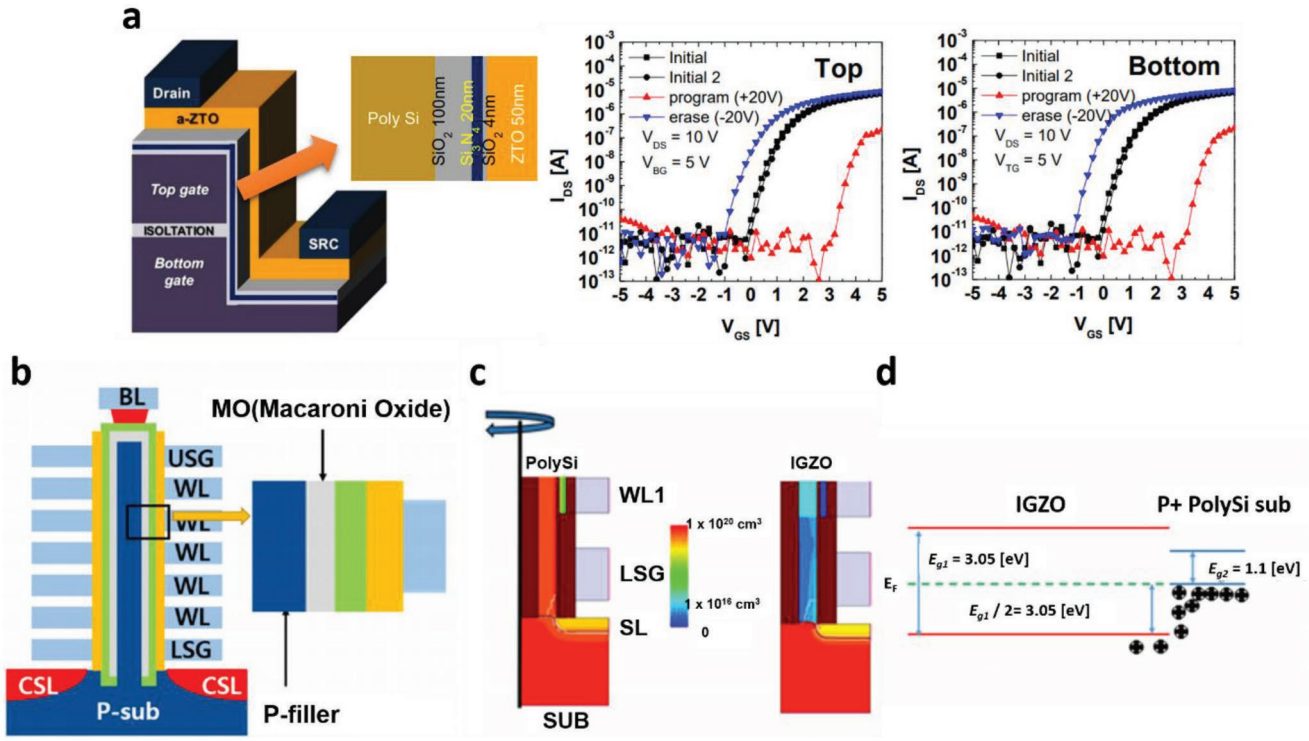


Figure 37. a) A schematic of double layer stacked CTF (left), and transfer characteristics of top (middle) and bottom (right) devices. Reproduced with permission.[149] Copyright 2017, ECI. b–d) Schematics of IOP structure: b) single string of IOP, c) hole density in each channel during erase, d) energy band diagram and hole movement in the IGZO and p+-poly Si sub during erase. Reproduced with permission.[148] Copyright 2019, IEICE.

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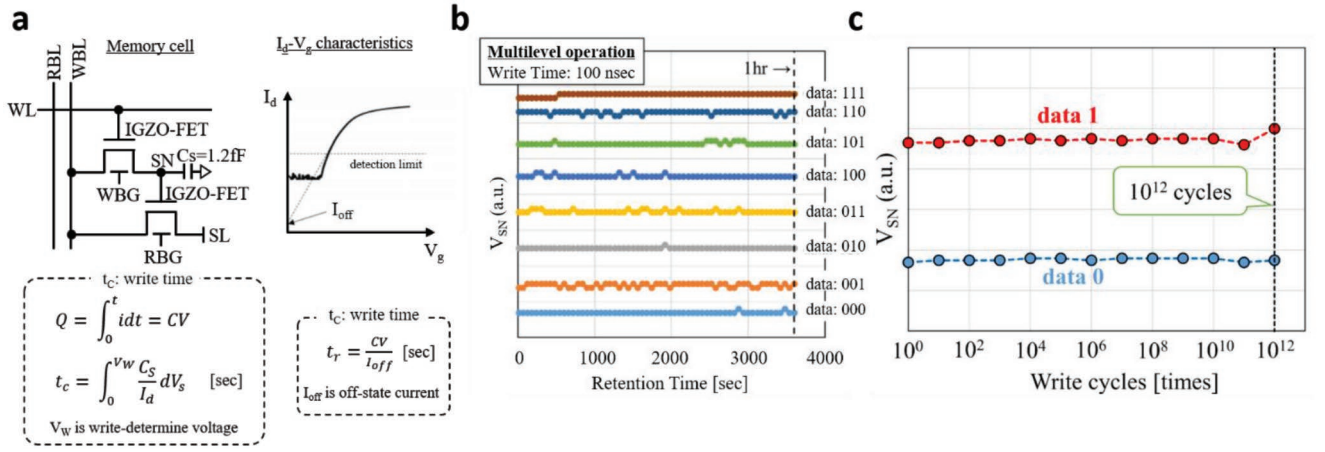


Figure 38. a) Configuration of IGZO NOSRAM cell and the formula of write and retention time. b) Data retention time of fabricated NOSRAM during multilevel operations. c) Endurance of NOSRAM during binary operations. a–c) Reproduced with permission.[150] Copyright 2019, IEEE.

the storage capacitor corresponding to writing are performed by turning the OS FET on and off, respectively. Therefore, the deterioration can be relatively suppressed (Figure 38c). In addition, the leakage current (I_{OFF}) of OS FETs is approximately yA μm^{-1}; thus, the charge accumulated in the floating node FN can remain nonvolatile.

4.2.3. Ferroelectric FET

HfO_2-based ferroelectric materials have been used to revive studies on a FeFET, which can be characterized by ultralow-power consumption, high write speed, and good scalability, making these devices promising for future embedded NVM (eNVM) applications.[151,152] However, many barriers must still be overcome when using ferroelectric materials in memory applications. Sufficient verification of variation in scaled dimensions and multibit implementation has not been provided yet. A 3D vertical FeFET with a structure similar to that of 3D NAND has recently been proposed.[153] Poly-Si is used as a channel material for vertical structures. However, some problems with poly-Si channels remain, such as low mobility and high thermal budget of substantially thin poly-Si channels. In addition, a low-κ IL is formed between the poly-Si and the ferroelectric gate oxide, which causes voltage loss and lowers the SS factor due to charge trapping. Poly-Si may also suffer from low endurance induced by interfacial layer degradation, which is typically less than 10^5 cycles.[109] The absence of an IL at the ferroelectric material/IGZO (or OS) interface reduces the depolarization field, mitigating retention degradation, and lowering the operating voltage of the device, which benefits eNVM applications.[154] IGZO has a high mobility for a remarkably thin body,[155,156] which makes IGZO a suitable material for 3D vertical FeFETs, as shown in **Figure 39**a. Figure 39b shows that the interface charge trapping in IGZO-based transistors may not be relevant compared with inversion mode transistors.[153] The voltage loss is substantially less due to the near-zero low-κ interfacial layer between the channel and gate oxide. Therefore, IGZO-channel-based FeFETs are promising candidates for high-density low-power memory applications.[157–159] Positive V_G induces accumulation charge in program mode. Spontaneous polarization can be flipped by the electric field and sustained by these charges. Meanwhile, a negative V_G induces a high density of depletion charge in erase mode. However, the IGZO channel has few hole carriers, which may be a problem with the polarization flip in erase mode. This drawback is encountered when using OS FETs, as shown in Figure 39c. Thus, Mo et al. suggested that thickness optimization is necessary, and 8 nm is proposed as an appropriate thickness.[153] V_{TH} increases and SS decreases with decreasing IGZO thickness. This phenomenon is due to the following: a thin IGZO involves a thin depletion layer; thus, a small negative voltage is necessary to turn off the channel as SS becomes steep.[155] Notably, the 5 nm thick IGZO FET has a significantly low drain current due to the nonuniformity of IGZO films and the mobility degradation. Figure 39d shows diagrams of energy bands for structures during write and read operations.[160] A high electric field stress across the IL, along with continuous charge trapping and trap generation at the interface, tends to limit the write endurance of conventional FeFETs. The presence of ILs also establishes a large charge detrapping time constant, which improves the read latency in conventional FeFETs. By contrast, the absence of IL lowers the write voltage and significantly reduces interfacial charge trapping, improving write endurance and read latency (**Table 7**).

5. Synaptic Device Applications

The human brain can recognize various objects, understand information, and learn on its own in complex real-world environments through complicated computations. Recently, artificial intelligence (AI) has attracted great attention and is being developed rapidly with the performance enhancement of the hardware-based computing power. Although there are many software-based advances in deep learning and other machine learning algorithms, most of the computations are still performed on the conventional von Neumann architecture. As the amount of computing data increases, computing systems based on von Neumann architectures are very inefficient in terms

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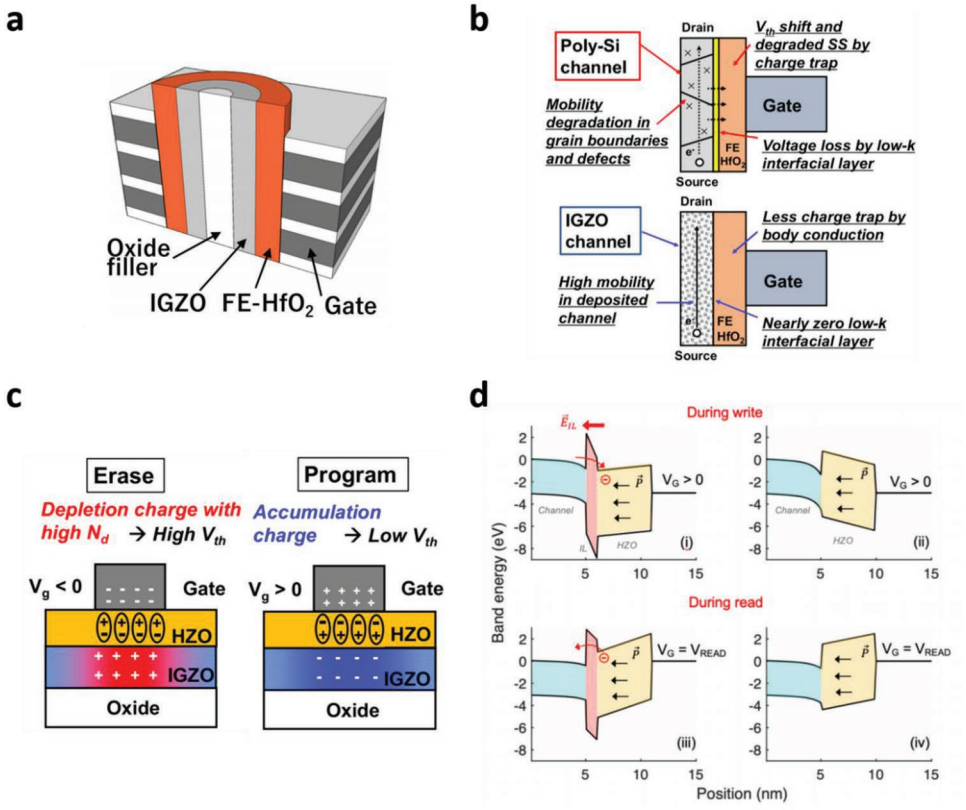


Figure 39. a) A schematic of 3D vertical FeFET of one pillar with an ultrathin IGZO channel layer. b) Difference between poly-Si (top) and IGZO (bottom) FeFET. c) Operation of an ultrathin IGZO FeFET in erase (left) and program mode (right). a–c) Reproduced with permission.[153] Copyright 2020, IEEE. d) Schematics of energy band diagrams for the two structures during write and read operations: IL FeFET (left) and IL-free FeFET (right). Reproduced under the terms of the CC-BY Creative Commons Attribution 4.0 International license (https://creativecommons.org/licenses/by/4.0).[160] Copyright 2021, The Authors, published in ArXiv.

of power consumption, the size of the hardware, and training times. In addition, the data transmission bottleneck worsens, and the energy consumption increases rapidly since data processing and storage units are separated in a von Neumann architecture.

A neuromorphic computing system that emulates the functions and information processing of the human brain has been proposed as an efficient solution for complex data processing. The human brain, which can be, in principle, mimicked as a neuromorphic chip, is a powerful and efficient biological computing system that can handle a variety of complex tasks with small volume occupation (≈1200 cm^3) and ultralow energy consumption (≈20 W). In the brain, information is processed in the form of the activity patterns of the neurons, which are connected to each other through synapses whose connection strength depends on recent activity. The connection strength between the two neurons linked by that synapse is known as synaptic plasticity, which plays an important role in achieving learning.

To realize neuromorphic computing that further utilizes the potential strengths and capabilities of the human brain, it is important both to develop a learning algorithm through spike-based information processing between synapses and neurons, and to develop artificial synapses that mimic biological synapses. Conventionally, Si-based CMOS analog circuits were fabricated to emulate synaptic functions. However, this approach requires a significant number of transistors (≈10 transistors per synapse). For example, a TrueNorth CMOS chip was made by IBM that integrated 5.4 billion transistors to simulate 1 million neurons and 256 million synapses.[161] Furthermore, the energy consumption of simple synaptic events in CMOS-based analog circuits can be much higher than that of biological synapses.

To overcome the limitations of CMOS-based analog circuits, a single electronic device such as two-terminal resistance change memory technology and synaptic transistors has been proposed to implement synaptic functions. Several two-terminal devices such as resistive RAM, phase change memory devices, and magnetoresistive RAM have been proposed to emulate synaptic functions. Although the two-terminal artificial synapse developed from relatively matured NVM technology has the advantages of simple device structure, small cell size, and low energy consumption, the efficient signal transmission and self-learning can be difficult to perform simultaneously in a two-terminal synaptic device. In addition, two-terminal synaptic devices require additional circuit components to select target cells and reduce cell-to-cell interference.

Recently, synaptic transistors with three terminals (source, drain, and gate electrodes) have attracted much attention because they can easily implement synaptic behavior within

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Table 7. Summary of performances of oxide memory devices.

Memory	Materials (channel)	Device performances	Ref.	Comment
1T1C-DOSRAM	CAAC-IGZO	– IOFF < 10–24 A μm–1 – Retention time > 10 year (@ 85 °C) – Operating frequency ≈ 100 MHz	[143#142]	VM
1T1C-DOSRAM	Crystalline-IGZO	– IOFF < 10–21 A – Power consumption reduction ≈ 70% – Operating frequency ≈ 100 MHz	[138#137]	
2T0C-DRAM	a-IGZO	– Write time > 10^{11} with t_{write} < 10 ns – Retention time > 10^3 s (for L_g = 50 nm)	[144#143]	
VNAND	a-ZTO	– Memory window > 4 V – Retention time > 10 year (@ 85 °C)	[150#149]	NVM
2T1C-NOSRAM	CAAC-IGZO	– I_{OFF} < 10^{-20} A μm^{-1} (@ 85 °C) – Write time < 100 ns, retention time > 1 h	[151#150]	
FeFET	a-IGZO	– 10^8 endurance and retention time > 1 h – Memory window ≈ 0.5 V	[154#153]	
FeFET	a-IGZO	– 10^{11} endurance – Memory window ≈ 1.6 V	[161#160]	

one device. The information transmission and process in the biological synapse proceed through the process in which the presynaptic pulse is transmitted as the current/voltage change of the postsynapse.[162] In synaptic transistors, the synaptic behavior is emulated by considering the gate electrode as a presynapse and the source/drain electrode as a postsynapse (**Figure 40**a).[163,164] Excitatory postsynaptic current (EPSC) occurs in the postsynaptic terminal (drain) when a stimulus in the form of a voltage pulse arrives at the presynaptic terminal (gate) (Figure 40b). Synaptic transistors can reliably implement spike generation and electrical plasticity and control synaptic weight more precisely through strong electrical coupling between two or more synaptic spikes. OSs are promising candidates for artificial synaptic transistors because of their high mobility, good uniformity, high visible light transparency, low processing temperature, and good compatibility with large-area processing technologies. In this subsection, recent developments in transistor-based artificial synapses using OS channels are reviewed and discussed. The working principles and major developments of electric-double-layer transistors, optoelectronic synaptic transistors, and ferroelectric gated synaptic transistors using OS channels are reviewed and discussed in this section.

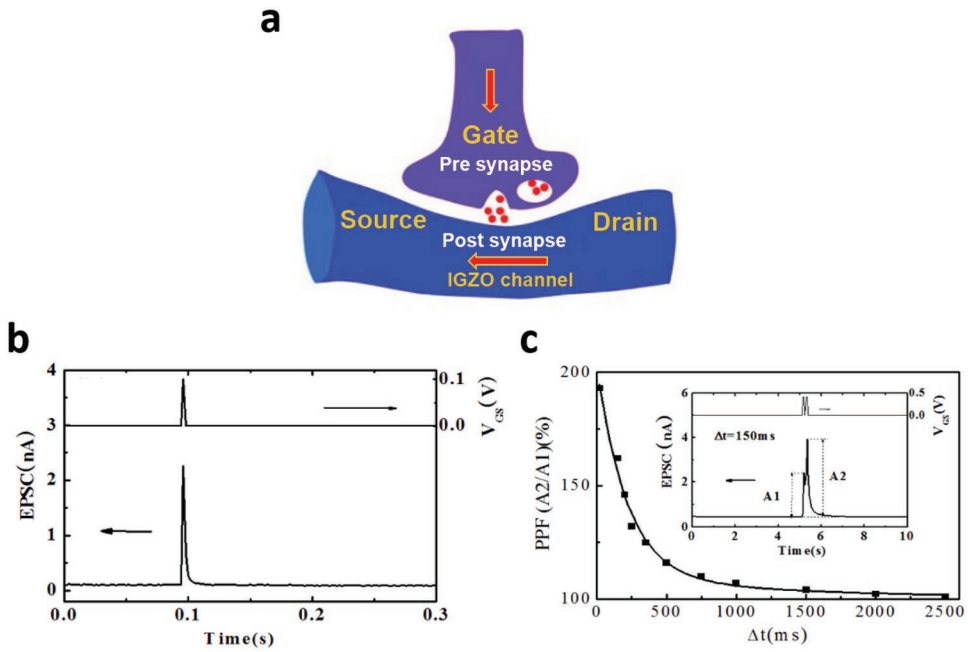


Figure 40. a) A schematic of an artificial synapse based on the synaptic transistor. b) EPSC of the synaptic transistor triggered by a gate spike. c) Paired-pulse facilitation ratio (A2/A1) as a function of the pulse interval (Δt). Inset shows EPSCs triggered by paired spikes with Δt = 150 ms. a–c) Reproduced with permission.[171] Copyright 2015, IEEE.

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5.1. Electric-Double-Layer Transistors

Recently, electric-double-layer transistors (EDLTs) using electrolyte dielectrics as a gate insulator have emerged to imitate biological synapses. Electrolytes can accumulate high density charges with low voltage and provide synaptic plasticity due to strong capacitance coupling between pulse intervals. When the presynaptic pulse arrives, ions in the electrolyte can move and accumulate at the interface between the semiconductor and the dielectric to form an EDL.[165] When the electric pulse is removed, the ions diffuse back into the first layer and return to equilibrium over a relaxation time. If the second spike is applied at an interval shorter than the relaxation time, some of the protons moved by the first spike do not diffuse back to the equilibrium position, so as the interval between pulses decreases, the degree of current increases by the second pulse increases. Figure 40c shows paired-pulse facilitation (PPF), one form of short-term plasticity (STP), which refers to the enhancement of postsynaptic current caused by paired presynaptic pulses.

In 2009, Wan and co-workers reported the IGZO-based EDLT using mesoporous SiO_2 films deposited by plasma-enhanced chemical vapor deposition (PECVD) at room temperature.[166] Despite the use of 4 μm thick SiO_2 as a gate insulator, these IGZO TFTs show a low operating voltage of about 1.0 V, a μ_{FE} of 28.5 cm^2 V^{-1} s^{-1}, and a SS of 110 mV dec^{-1} due to the high capacitance of EDL. Various electrolytes and dielectric films, including p-doped nanogranular SiO_2, chitosan proton conducting electrolyte film, phosphosilicate glass, methylcellulose, and 3-triethoxysilylpropylamine-modified graphene oxide (KH550-GO) were used as the gate dielectric.[167] The use of an appropriate electrolyte material can significantly lower the power consumption of the synaptic transistor, which is a key requirement of the neuromorphic system. The energy consumption per spike of a synaptic transistor with the p-doped nanogranular SiO_2 and IZO channel was about 160 pJ,[168] while the acetic-acid-doped chitosan could reduce the power consumption to 26 pJ due to its high proton conductivity.[169] Liu et al. showed that the self-assembled IZO synaptic transistor on a chitosan membrane could operate at a lower power of 3.9 pJ.[170] Since IGZO thin films have lower intrinsic electron concentration and lower channel current at a zero gate voltage than IZO, artificial synapses based on IGZO can potentially improve the energy efficiency. Zhou et al. reported that the energy consumption of IGZO synapse transistors with 600 nm thick p-doped nanogranular SiO_2 films was as low as 0.23 J per spike.[171]

According to the operation principle, EDL transistors can be classified into two types: electrostatic modulation and electrochemical doping. In the first case, the ions move only within the electrolyte dielectric under gate bias, and the channel conductivity is changed reversibly to mimic STP. In the second case, electrolyte ions penetrate the semiconductor and irreversibly change the conductivity of the channel by doping. The nonvolatile conductance change of postsynapse provides the basis for mimicking long-term plasticity (LTP). LTP was also mimicked in oxide semiconductor EDL transistors.[172] Yang et al. proposed that spike-rate-dependent plasticity (SRDP) could be emulated with an IZO-based EDL transistor with modified graphene oxide (KH550-GO), as shown in **Figure 41**a,b.[167] SRDP is one of the basic learning rules of long-term plasticity, in which the synaptic weight changes with the rate of presynaptic spikes. In addition to implementing typical synaptic function, dynamic logic and dendritic integration functions and a brain-inspired cognitive system have been emulated in oxide–semiconductor-based EDL transistors.[172,173]

5.2. Photonic Synaptic Transistors

Photonic synaptic transistors, which use light pulses as stimulus instead of gate pulses, have received attention because they have several advantages over electronic synapse transistors, such as wide bandwidth, fast signal transmission, low crosstalk, and low interconnection energy loss.[163,174] Various materials such as carbon nanotubes,[175,176] perovskite quantum dots,[177,178] 2D materials,[179,180] and OSs have been used in photon synaptic devices that mimic synaptic behavior using optical stimulation. Recently, OS has gained increasing attention to implement photonic synaptic transistors due to its inherent persistent photoconductivity (PPC) phenomenon that the light-induced channel conductance decays over time despite the removal of optical stimulus.[181] Among various point defects such as V_O, metal vacancies, interstitial oxygen, interstitial metal and peroxide, the *Vo* defects are generally believed

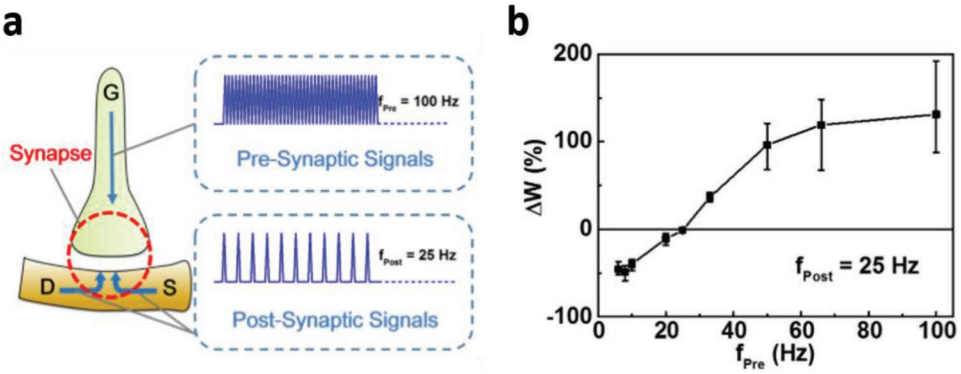


Figure 41. a) Schematic illustrating the SRDP learning rule in an artificial synapse when a pair of pre- and postsynaptic signals are applied. b) Distribution and mean values of synaptic weight change after experiencing SRDP learning rules. a,b) Reproduced with permission.[167] Copyright 2016, American Chemical Society.

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to be the origin of the PPC due to their low formation energy in most of the AOS.[181–184] When AOS is exposed to light with high energy, electrons are mainly generated by the ionization of V_O ($V_O \rightarrow V_O^{1+} + e^-$ or $V_O \rightarrow V_O^{2+} + 2e^-$) (**Figure 42**a). When the light is off, the conductance of the channel does not immediately drop to its initial value but decreases slowly over time since neutralization of V_O ions is a nonspontaneous process with high activation energy. These inherent PPC characteristics of OS yielded optoelectronic channel conductance switching behavior and nonvolatile synaptic plasticity functions with a simple TFT structure. For example, a UV pulse-stimulated synaptic transistor based on IGZO–Al_2O_3 was reported to emulate synaptic characteristics such as PPF and STP-to-LTP transition, as shown in **Figure 43**a–c.[185] In addition, a symmetric spike-timing-dependent plasticity (STDP) function was successfully implemented by connecting two a-IGZO-based photonic synaptic transistors in series (Figure 43d,e).[183] Like SRDP, STDP is an important synaptic modification rule involved in learning, in which synaptic weight is changed by the time difference between the signal from the presynaptic neuron and the signal from the postsynaptic neuron.[186] Wu et al. have demonstrated that the element composition ratio in the IGZO layer can effectively mediate the synaptic behaviors.[187] IGZO-based synaptic devices with In-rich and Ga-poor configurations exhibit higher photocurrent generation and slower decay rates compared to devices with Ga-rich and In-poor configurations due to a higher percentage of oxygen vacancies. Hence, the emulation of various synaptic functions can be enhanced by properly optimizing the content of the various In-, Zn-, and Sn-based oxide semiconductors. The application of a gate bias has been proposed to improve the photoconductive switching performance of the synapse.[183,186] The ionization and recombination reaction of V_O can be controlled by adjusting the activation energy of nonspontaneous reaction of V_O by applying gate bias to change the charge density at the interface between the oxide semiconductor and the dielectric layer. In 2020, a photon synaptic device capable of controlling the synaptic function using an oxide semiconductor and a ferroelectric layer was proposed.[188] Figure 42b,c shows that the polarization of ferroelectric HZO layer is used to control the amount by which conductance increase during PPF operation and to enhanced LTP.[188] However, compared with other types of synapses, the photonic synaptic devices using PPC have disadvantages of low energy efficiency and slow operation speed due to low photocarriers by photoionization and slow recombination dynamics of V_O. In addition, it is difficult to build an optoelectrical

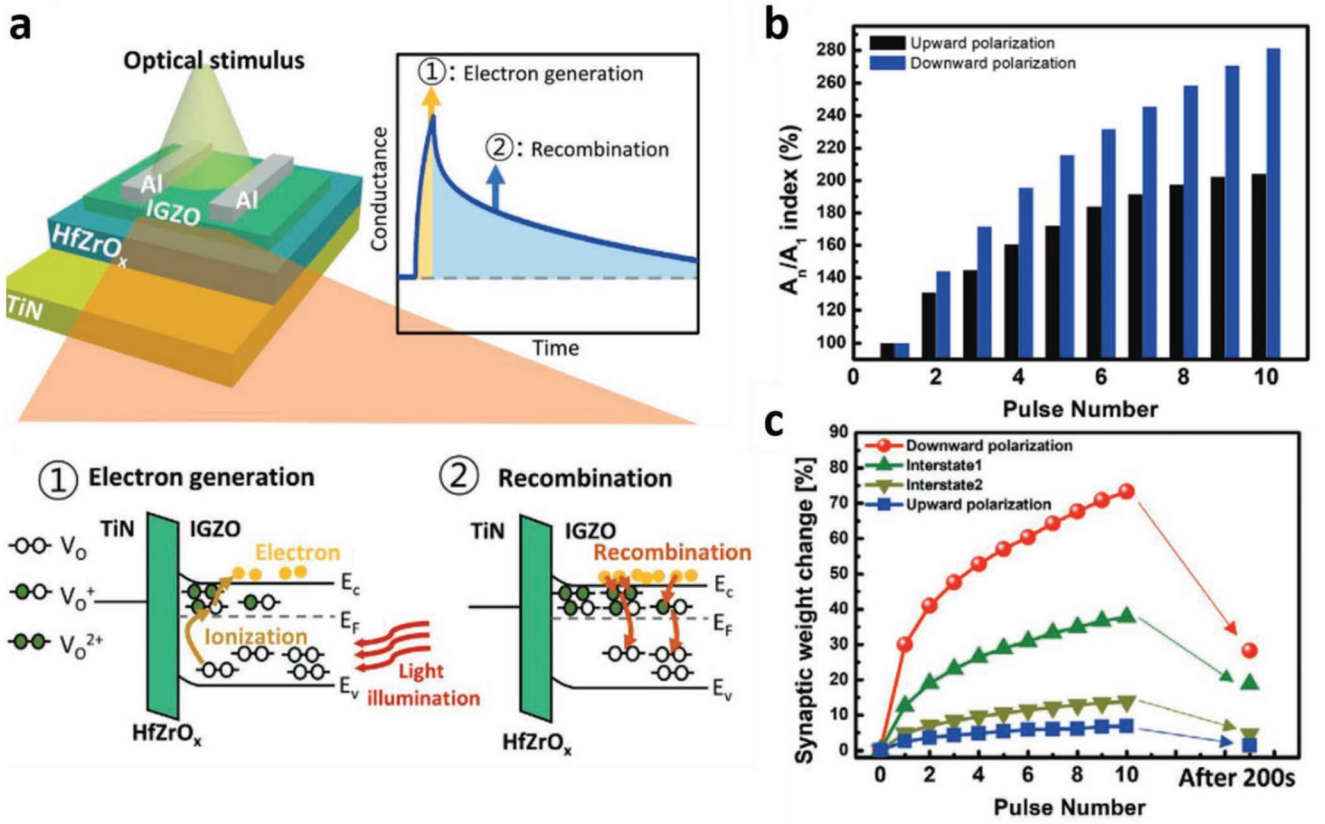


Figure 42. a) A schematic of photonic synapse and optical responses of IGZO-based transistors (top). Schematic mechanisms of optical responses in IGZO-based transistors, depicting the ionization and recombination processes of oxygen vacancies. b) A_n/A_1 index according to the number of applied optical pulses in upward and downward polarizations, wherein the A_n/A_1 index is defined as the ratio of the peak current amplitudes of optical responses after applying first (A_1) and *n*th (A_n) optical pulses. c) Synaptic weight changes of devices according to the number of applied optical pulses with different polarization states. Downward polarization state, interstate1, interstate2, and upward polarization state were induced by applying different gate bias pulses. a–c) Reproduced with permission.[188] Copyright 2020, Wiley-VCH.

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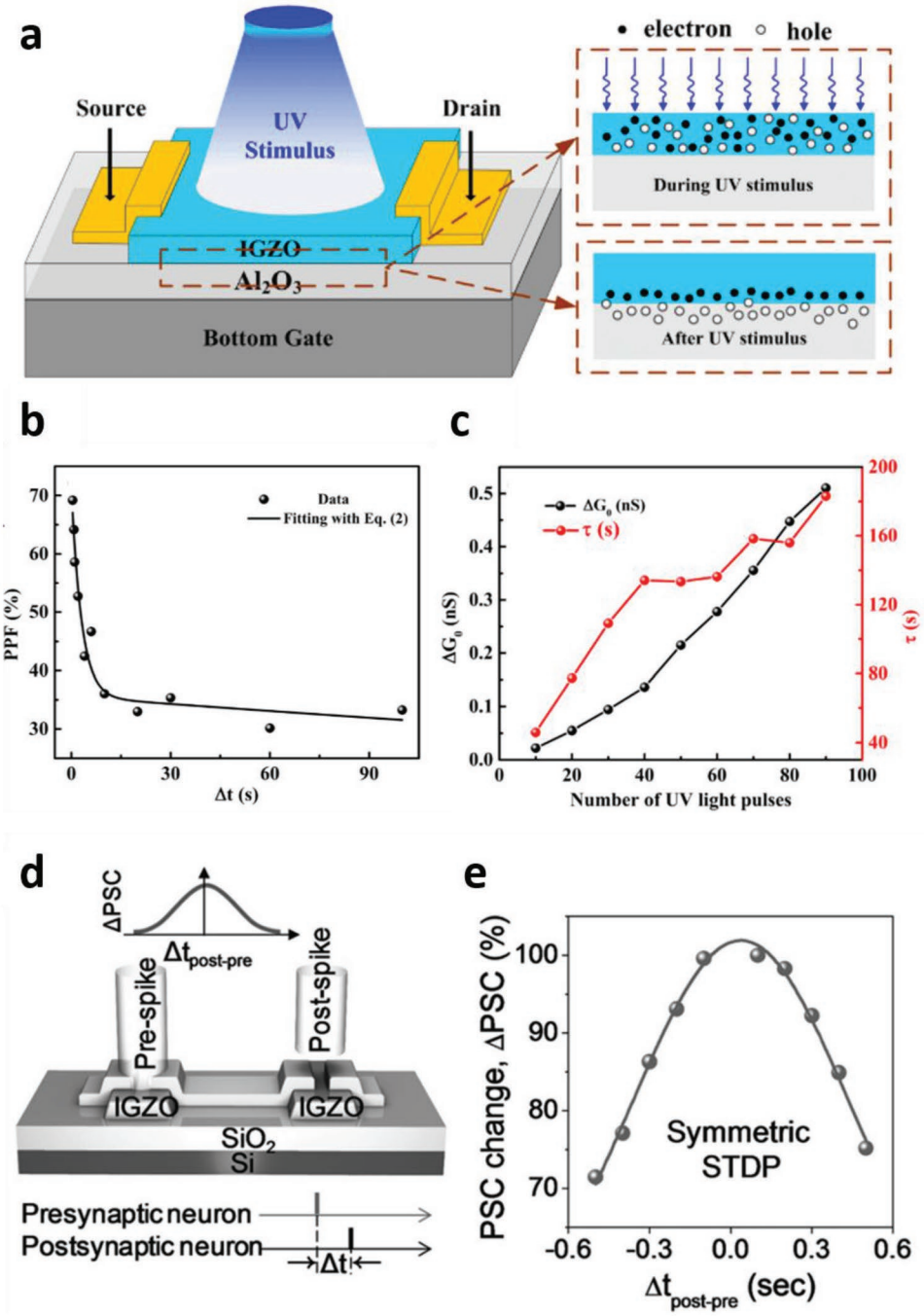


Figure 43. a) A schematic of a IGZO–Al_2O_3-based photonic synaptic transistor by UV light stimulus and the poststimulus distribution of the photogenerated electrons and holes. b) PPF as a function of the UV stimulus interval (Δt). c) ΔG_0 and τ as a function of the UV light pulse number. G_0 is the channel conductance recorded immediately after the last pulse of a pulse series, and τ is the characteristic relaxation time. a–c) Reproduced with permission.[185] Copyright 2016, American Institute of Physics. d) A schematic of two connected IGZO synaptic devices for the emulation of the STDP. e) Symmetric STDP realized in the two connected IGZO synaptic devices. d,e) Reproduced with permission.[183] Copyright 2017, Wiley-VCH.

synaptic system that can operate with long wavelength light because most photoionization reactions in AOS occur in the short-wavelength energy region (>2.8 eV). Therefore, it is necessary to develop innovative technologies that can increase the optical sensitivity of oxide semiconductors in a wide spectral region as well as improve device operational reliability and device to device uniformity.

5.3. Ferroelectric Synaptic Transistors

FeFETs with their nonvolatile properties, high switching speed, and ultralow energy dissipation have attracted considerable attention as promising platforms for artificial synapses. Ferroelectric materials provide multilevel channel conductance of FeFETs because they can achieve multistep residual

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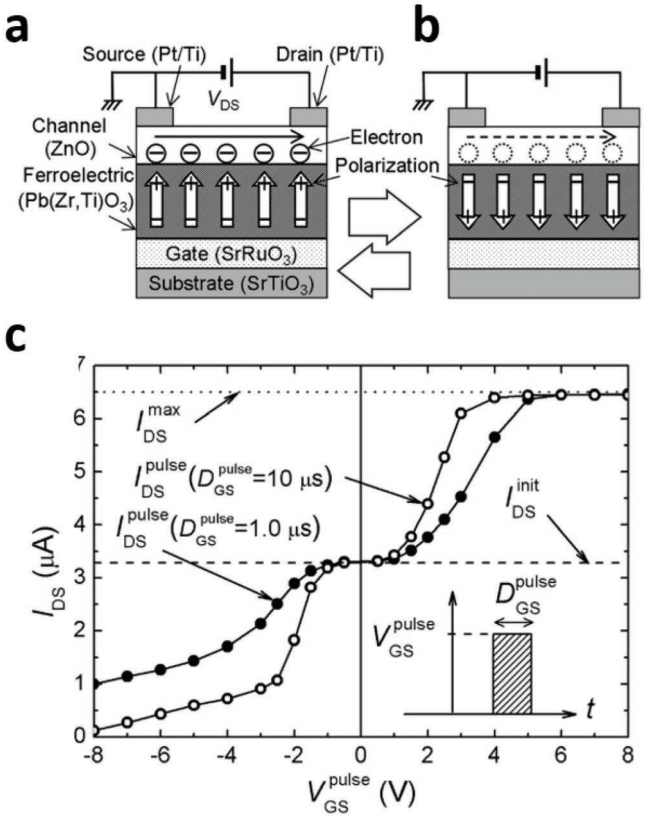


Figure 44. a,b) Schematics of FeFET with upward (a) and downward (b) polarization. c) Change of the I_{DS} before (I_{DS}^{init}) and after (I_{DS}^{pulse}) applying the pulsed gate voltage with various heights and widths. a–c) Reproduced with permission.[190] Copyright 2012, American Institute of Physics.

polarization states by adjusting the applied bias and maintain each state without the application of an electric field. Consequently, multilevel channel conductance of FeFETs can be used in synaptic weights.[189] Ferroelectric gated synaptic transistors (FGSTs) composed of a ZnO/Pb(Zr,Ti)O_3 (PZT) stack structure were fabricated, and multivalued conductance control was demonstrated by adjusting the height and the duration of applied pulse gate voltage (**Figure 44**a–c).[190] Utilizing these characteristics, symmetric and asymmetric STDP learning functions were successfully implemented in the FeFET-based synapse device by applying the nonlinear pulse gate voltage. Recently, synaptic devices based on HfO_2-based ferroelectric materials, such as HZO and Al-doped HfO_2, have been proposed because conventional perovskites ferroelectrics such as PZT have disadvantages such as poor CMOS compatibility, limited scalability, and complex configurations. FGSTs with HfO_2-based ferroelectric materials as the ferroelectric insulator layer and IGZO as channel were demonstrated.[191,192] As shown in **Figure 45**, IGZO/Al-doped HfO_2-based FeFETs successfully emulated the synaptic behavior such as PPF and STDP. In addition, an artificial neural network simulation with measured properties showed 90% recognition accuracy in recognizing handwritten digits on the Modified National Institute of Standards and Technology.

6. Conclusion and Perspectives

Over the past century, there have been remarkable achievements in science and technology, especially computing and communications. This trend has been accelerated by the emergence of coronavirus disease-19 (COVID-19), which led to huge progress in technologies enabling facile, contact-free transactions, such as big data, Internet of Things (IoT), and AI approaches. However, these advancements strongly require improved computational abilities with high power efficiency, ultimately demanding greater semiconductor device performance. In this regard, novel materials that can partly alter the conventional Si and thus facilitate the state-of-the-art M3D integration are increasingly important in the semiconductor industry, and OS can be a breakthrough in this digital age.

Here, fundamentals of AOS including a brief history, the functions of constituents and defects, transport mechanisms, and reliability issues were addressed first. Then, recent progress in OS transistors using diverse techniques such as cation composition control, crystallization, and introduction of multichannel layer and high-κ gate dielectrics was reviewed. Moreover, several impressive works demonstrating the potential of OS- for BEOL-compatible transistors in the M3D architecture were discussed, where their outstanding immunity against the notorious SCEs originating from device miniaturization and the feasibility of making transistors with complex 3D structures were investigated. Finally, articles where emerging applications using OS were implemented, such as memory and synaptic device applications, were introduced. More importantly, these substantial achievements have taken place in a very short period and are expected to accelerate. To do so, studies to improve device performances and investigate effects originating from further scaling and adopting advanced 3D structures must be performed simultaneously. The development of state-of-the-art advanced semiconductor devices involving OS could be assisted by the TCAD simulation. The in-depth understanding on carrier transport, contact, interface integrity, etc., in the nanoscaled structure such as OS-based FinFET and GAA has not been examined sufficiently, which indicates that there is still much room for being explored. In this respect, the modeling/simulation studies must be conducted more intensively in this field, which can greatly reduce the experimental R&D cost and contribute the overcome of technical challenges for the highly scaled, complex nanoscale OS devices.

It should be also noted that promising BEOL-compatible p-type candidates must be obtained to make the best use of the M3D integration. However, developing high-performance p-type semiconductors using CMOS-compatible routes at low temperature is very challenging. In general, VB edge, the hole conduction path, is considerably localized by the oxygen 2p orbitals in OS, which results in a high hole effective mass. Thus, p-type characteristics are not revealed in most of the OSs. To make matters worse, native hole killers like V_O are easily generated due to their low formation energy, which annihilates free hole carriers.[193] Notwithstanding these harsh conditions, some of the OSs based on copper (Cu) and Sn show p-type characteristics,[194–203] which are enabled by partial delocalization through Cu 3d and Sn 5s orbital overlapping, respectively, on the oxygen 2p orbitals. However, device performances of FETs

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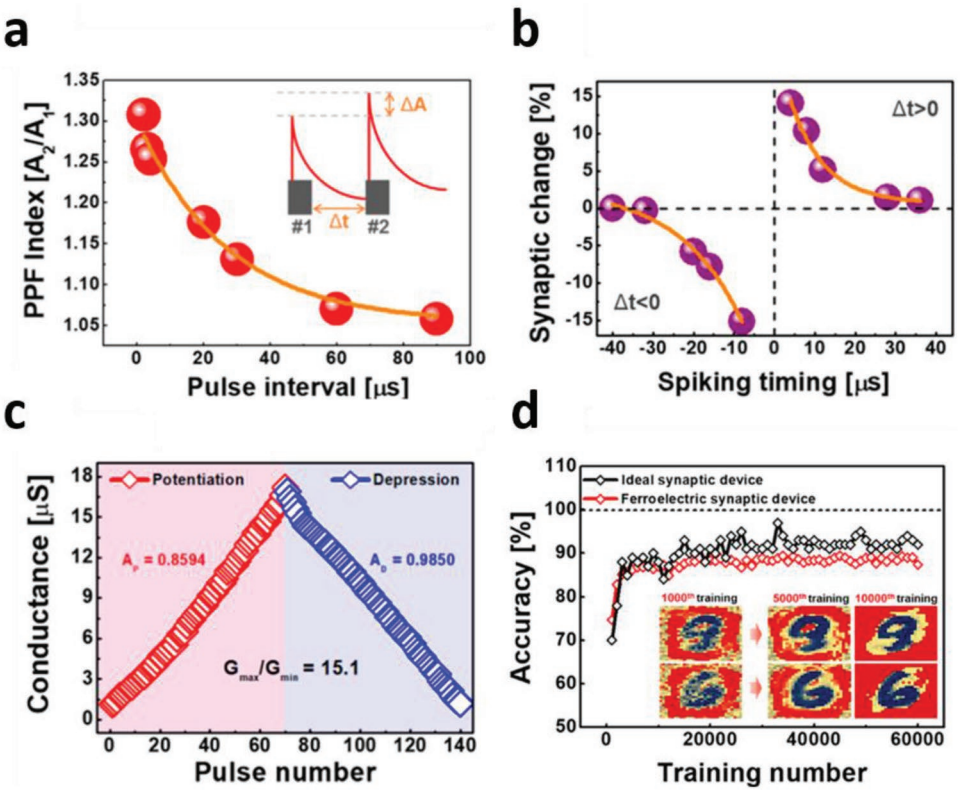


Figure 45. a,b) Synaptic characteristics of FeFETs: a) PPF, b) STDP. c) Linear potentiation/depression characteristic of FeFET with incremental pulse amplitude. d) Pattern recognition accuracy based on both the FeFET and ideal synaptic device. a–d) Reproduced with permission.[192] Copyright 2021, American Chemical Society.

using such p-type OSs as a channel layer have been inferior to those of their n-type counterparts. In the case of 2D materials such as phosphorene, MoS_2 and WSe_2, promising p-type electrical characteristics have been observed in their FETs.[204–209] However, their performance is drastically degraded when fabricated through CMOS-compatible fabrication methods or at low temperature, still hindering their practical use despite the outstanding properties. Recently, hexagonal tellurium has been resurrected as a promising p-type candidate because it reveals exceptional electrical characteristics even if fabricated through the practical routes at low temperature.[210–215] Nonetheless, there is still room for investigating their applications. As such, obtaining a BEOL-compatible p-type candidate is an important task but is still an open question at present.

Furthermore, multivalued logic (MVL) technology is increasingly investigated to implement logic circuits with considerably improved power efficiency and interconnect requirements. In principle, MVL devices have more than two states, e.g., a ternary logic device that has three states of 0, 1, and 2. Meanwhile, binary logic, the building block of current logical operating systems, consists of two logic states of 0 and 1. As the number of logic states increases, information density that the device can process increases significantly, which enables more information to be computed with fewer devices.[216] Therefore, it can process information more efficiently without further scaling of transistors. Furthermore, fewer interconnect lines are required to transmit information in MVL computing systems, which can simplify the IC design.[217] To date, various MVL devices have been researched mainly on basis of ternary logic, where negative differential resistance devices and quantum dot gate FETs are primarily implemented.[217–222] However, neither of them show a constant intermediate state, which makes it difficult to get a stable logic state, still hampering the demonstration of ternary logic devices. Here, OS can also be employed as a promising channel material for novel logic applications because their extremely low off-current provides stable intermediate states. Recently, OS-based MVL devices have been investigated using a layered quantization system with ZnO thin films,[223–225] where great electrical characteristics with reliable operation including adjustable multilevel states and no current fluctuations were exhibited. In this manner, a MVL application is another emerging potential technology of OS for next-generation electronics beyond the scaling era, requiring further studies on OS-based MVL applications.

In addition to memory/synaptic and MVL applications, there have been other emerging areas for OS devices to be adopted such as flexible optoelectronics,[20,226–228] power electronics,[228–230] and sensor applications, etc.[22,231–233] As such, the success of OS is now expected to be expanded to diverse areas of electronics beyond the display applications in the near future. Provided that the reports to date are fully understood and the remaining hurdles mentioned above are overcome, we believe that OSs can be the engine for the 4th industrial revolution.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords

3D devices, back-end-of-line transistors, field-effect transistors, memory devices, monolithic 3D integration, oxide semiconductors, synaptic devices

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